Circuit tools and semiconductor physics

Answers and worked solutions · Day 01 · Sat 24 Oct · pack E1

5. Practice

From your course files

WS1 Q1 (course question).

> **a) Negative.** More electrons than protons means a net negative charge.

WS1 Q2 (course question).

> **d) Right, Left.** Conventional current is the flow of positive charge, so i_1 (arrow pointing right) is positive when positive charges move right, or equivalently when negative charges move left.

WS1 Q3 (course question).

> **c) Negative.** A wrongly guessed polarity just gives a negative answer. The magnitude is still right.

WS1 Q4 (course question).

> **a) Lose, Positive, Negative, Absorbed.** A resistor always absorbs energy and dissipates it as heat, so carriers lose potential energy going from + to −.

WS1 Q5 (course question).

> **b) Always positive** (official answer). If $i < 0$ then $v < 0$ too, so $p = vi = i^2R$. Strictly $p \geq 0$, and it is zero when $i = 0$.

WS1 Q6 (course question).

> **b) Open circuit.** $i = C\,dv/dt = 0$ when v is constant, and zero current is an open circuit.

WS1 Q7 (course question).

> **a) Short circuit.** $v = L\,di/dt = 0$ when i is constant, and zero voltage is a short circuit.

WS1 Q8 (course question).

> **a) Low, High, Positive.** Active sign convention: current flows from the − end to the + end inside the source, and a source that supplies energy has positive $i(t)$.

WS1 Q9 (course question).

> **b) Positive, Positive.** In the active convention a supplying source has $v(t) > 0$, and positive current leaves the + end.

WS1 Q10 (course question).

> From left to right: **VCVS** ($v = k \cdot v_1$), **CCVS** ($v = k \cdot i_1$), **VCCS** ($i = k \cdot v_1$), **CCCS** ($i = k \cdot i_1$). Diamond with ± means a voltage source. Diamond with an arrow means a current source. The controlling variable is v_1 or i_1.

WS1 Q11 (course question).

> **a, c, f.** A short has zero voltage, zero resistance and infinite conductance. Its current is whatever the circuit sets.

WS1 Q12 (course question).

> **a, c.** A charged capacitor or inductor can give energy back. With the passive convention it delivers when $p < 0$ and absorbs when $p > 0$.

WS1 Q13 (course question).

> **c) $N = 3, B = 5$.** The curved wire joins the two top-left and top-right nodes of the 5 kΩ/10 kΩ pair, so those nodes are one node. The 20 kΩ then has both ends on the same node: it is shorted and is not a branch. That leaves three nodes (top, middle, bottom) and five elements: 12 V, 5 kΩ, 10 kΩ (top), 10 kΩ and 5 kΩ (vertical).

WS1 Q14 (course question).

Loop = 6, Mesh = 3 (official). Options a) and d) are printed identically on the sheet, so either letter carries this answer. Count meshes with $B - N + 1$; count loops by listing every closed path that uses no node twice.

WS1 Q15 (course question).

a). The horizontal 5 Ω is open at its right end, so $I_1 = 0$. KCL at the top-left node: $9 = 3 + I_2$, so $I_2 = 6$ A.

WS1 Q16 (course question).

a). Go round the blue loop in its (anticlockwise) direction and take each voltage's sign as you meet it: top resistor $-v_1$, A–B resistor $+v_4$, B–D resistor $-v_3$, D–C resistor $+v_2$. The loop does not pass through the 3 A source, so v_5 is not in it.

WS1 Q17 (course question).

c) 16 V. Everything is in series, so the sources can be collected. Taking the 9 V polarity as positive: $+9 + 3 - 1 + 5 = 16$ V. (The series resistance is $100 + 220 = 320\ \Omega$.)

WS1 Q18 (course question).

b) 4.8 A. The right-hand 2 Ω and 6 Ω are in series (8 Ω), in parallel with the middle 2 Ω. Current divider: $I_1 = 6 \times \dfrac{8}{8+2} = 4.8$ A.

WS1 Q19 (course question).

c) 7.2 V. The middle 2 Ω has $2 \times 4.8 = 9.6$ V across it, and so does the 8 Ω branch. Voltage divider: $V_2 = 9.6 \times \dfrac{6}{2+6} = 7.2$ V.

WS1 Q20 (course question).

c) 0.66 A.

- 2 A on, 3.5 V off (short). The short puts the 5 Ω out of circuit, so 2 A splits between the 7+3 = 10 Ω leg and the 15 Ω: $I_{x1} = 2 \times \dfrac{10}{10+15} = 0.8$ A.
- 3.5 V on, 2 A off (open). The 3.5 V sits across the 15 Ω + 10 Ω path, driving current against the I_x arrow: $I_{x2} = -\dfrac{3.5}{15+7+3} = -0.14$ A.

- Sum: $I_x = 0.8 - 0.14 = 0.66$ A.

WS1 Q21 (course question).

a) 45 V, 3 Ω. Remove the 6 Ω load. R_{Th}: short the 36 V, open the 3 A, leaving only the 3 Ω, so $R_{Th} = 3\ \Omega$. V_{Th}: with A–B open, the 3 A flows round through the 3 Ω, dropping $3 \times 3 = 9$ V. KVL round the outer loop: $V_{Th} = 36 + 9 = 45$ V.

WS1 Q22 (course question).

d) 15 A, 3 Ω. $R_N = R_{Th} = 3\ \Omega$. Short A–B: the 3 A source delivers 3 A into the short, and the 36 V drives $36/3 = 12$ A through the 3 Ω into the short. $I_N = 3 + 12 = 15$ A. Check: $I_N = V_{Th}/R_{Th} = 45/3 = 15$ A.

WS1 Q23 (course question).

b) 0.1923 mA. Transform 5 V + 5 kΩ into a Norton source: $5/5\ \mathrm{k} = 1$ mA in parallel with 5 kΩ. Now 1 mA + 1 mA = 2 mA feeds 5 kΩ ‖ 47 kΩ. Current divider:
$I_x = 2 \times \dfrac{5}{5 + 47} = 0.1923$ mA.

WS1 Q24 (course question).

c) $V_c = 0.2$ V. Full working is in Example 2 above: three KCLs (A, C, D) give $V_A = 0.3556$ V, $V_C = 0.2$ V, $V_D = 0.2444$ V.

WS1 Q25 (course question).

a) 0.3556 V (the options are labelled "V_c" on the sheet but mean V_{AB}). The 1 A source is on the outside of mesh 1, so $I_1 = 1$ A. KVL for meshes 2, 3, 4 (conductances, so each resistor drop is current/G):

$$-(1 - I_2) + \frac{I_2 - I_3}{2} + \frac{I_2 - I_4}{2} = 0, \quad \frac{I_3}{3} - \frac{I_4 - I_3}{2} - \frac{I_2 - I_3}{2} = 0, \quad -\frac{I_2 - I_4}{2} + \frac{I_4 - I_3}{2} + I_4 = 0$$

Simplified: $4I_2 - I_3 - I_4 = 2$, $-3I_2 + 8I_3 - 3I_4 = 0$, $-I_2 - I_3 + 4I_4 = 0$. So $I_2 = 0.6444$, $I_3 = 0.3333$, $I_4 = 0.2444$ A, and $V_{AB} = (I_1 - I_2)/1\ \mathrm{S} = 0.3556$ V. This matches V_A from Q24.

Note: the official solution prints the first simplified equation as "$= 0$". It should be "$= 2$" (multiply the first KVL by 2). Its numerical answers satisfy the "$= 2$" version.

WS2 Q1 (course question).

I **F** (it is an insulator at 0 K). II **T**. III **F** (intrinsic Si is not a conductor). IV **T**. V **T**. VI **T**. VII **F** (doped material is neutral: every free carrier is balanced by a fixed ionised dopant). VIII **T**. IX **T**. X **T**.

WS2 Q2 (course question).

$L = 0.8$ cm, $D = 0.1$ cm, $A = \pi D^2/4 = 7.854 \times 10^{-3}$ cm². $\rho = 1/\sigma = 1.77 \times 10^{-6}$ Ω-cm.

$$R = \frac{\rho L}{A} = \frac{1.77 \times 10^{-6} \times 0.8}{7.854 \times 10^{-3}} = \mathbf{1.80 \times 10^{-4}\ \Omega}$$

WS2 Q3 (course question).

(a) $n = p = n_i$, so
$\sigma = q(\mu_n + \mu_p) n_i = 1.6 \times 10^{-19}(1850)(1.5 \times 10^{10}) = 4.44 \times 10^{-6}\ (\Omega\text{-cm})^{-1}$ and $\rho = 2.252 \times 10^{5}\ \Omega$-cm. $R = \rho L/A = 2.252 \times 10^{5} \times 0.8/7.854 \times 10^{-3} =$ **22.9 MΩ** (the official 22.95 MΩ rounds A to 7.85×10^{-3}).

(b) $E = V/L = 5/0.8 = 6.25$ V/cm. $v_{dn} = \mu_n E = 8.44 \times 10^{3}$ **cm/s**, $v_{dp} = \mu_p E = 3.13 \times 10^{3}$ **cm/s**.

WS2 Q4 (course question).

(a) $N_D \gg n_i$, so $n_n \approx N_D = 10^{14}$ **cm⁻³** and $p_n = n_i^2/N_D = (1.5 \times 10^{10})^2/10^{14} = 2.25 \times 10^{6}$ **cm⁻³**.

(b) The hole term is negligible: $\sigma \approx q N_D \mu_n = 1.6 \times 10^{-19} \times 10^{14} \times 1350 = 0.0216\ (\Omega\text{-cm})^{-1}$.

WS2 Q5 (course question).

Worked in full in Example 4 above. Final answers: 300 K, $p \approx 2 \times 10^{15}$, $n = 1.125 \times 10^{5}$ cm⁻³; 405 K, $n = 5 \times 10^{10}$ cm⁻³. The exact equations change these only in the 5th significant figure.

WS2 Q6 (course question).

Worked in full in Example 3 above. Final answers: (a) $n_i = 7.88 \times 10^{10}$ cm⁻³; (b) $n = p = n_i$; (c) $\rho = 4.29 \times 10^{4}\ \Omega$-cm; (d) $R = 238$ MΩ; (e) 81 % lower than the 1.25 GΩ

at 300 K.

WS2 Q7 (course question).

Worked in full in Example 3, step 5. $\rho = 0.926\ \Omega$-cm and $R = 5.14$ kΩ. The official solution refers to "Problem 5" and works at 300 K. At 50 °C the minority term is still negligible ($p_n = 1.24 \times 10^6$), so $R = 5.14$ kΩ either way, assuming the mobilities do not change.

PT Q1 (practice test, 5 marks).

(I) Answer: a. Electrons and holes recombine near the junction and leave behind ionised dopants. (b) is wrong because the bulk still conducts. (c) is wrong because only carriers near the junction cross, and equilibrium stops further flow. (d) is wrong because a depletion region exists with no applied voltage at all.

(II) Answer: a. $r_d = \Delta V_d / \Delta I_d$ is the inverse slope of the I–V curve at the Q-point. This is covered in pack E2.

(III) Answer: b. Full-wave rectification uses both half-cycles.

(IV) Answer: a. A thin, lightly doped base means few injected electrons recombine, so most reach the collector (WS4 §3.2).

(V) Answer: b. The DC analysis fixes the Q-point. The AC analysis studies the small variations about it.

New questions

N1. [5 marks] Circle the correct answer (I to V).

I – b. $p = n_i^2 / N_D = (1.5 \times 10^{10})^2 / 10^{16} = 2.25 \times 10^4\ \text{cm}^{-3}$. Option (d) is n_i^2 itself, and (a) is n_i, which applies only to intrinsic material.

II – b. At equilibrium, the majority-carrier diffusion current I_D is balanced by the minority-carrier drift current I_S. Carriers do cross the junction, so (a) is wrong. The depletion region holds only bound ions, so (d) is wrong.

III – c. Charge balance gives $x_n N_D = x_p N_A$, so $x_n / x_p = N_A / N_D = 100$. The region extends into the lightly doped n side.

IV – b. Short V_{CC}. Both resistors then run from the base to ground, in parallel.

V – a. These are the two ideal-op-amp rules (WS4 §2).

N2. [6 marks]

(a) Since $N_A \gg n_i$: $p \approx N_A = 4 \times 10^{15}\ \text{cm}^{-3}$ and $n = n_i^2/N_A = (1.5 \times 10^{10})^2/(4 \times 10^{15}) = 5.625 \times 10^{4}\ \text{cm}^{-3}$.

(b) $\sigma = q(n\mu_n + p\mu_p) = 1.6 \times 10^{-19}\left(5.625 \times 10^{4} \cdot 1350 + 4 \times 10^{15} \cdot 500\right)$. The electron term is 3.8×10^{-11} times the hole term, so it is negligible. $\sigma = 0.320\ (\Omega\text{-cm})^{-1}$ and $\rho = 3.125\ \Omega$**-cm**.

(c) $L = 0.2$ cm and $A = 0.05 \times 0.02 = 1.0 \times 10^{-3}$ cm². $R = \rho L/A = 3.125 \times 0.2/10^{-3} =$ **625 Ω**, so $I = 3/625 =$ **4.8 mA**.

(d) $E = V/L = 3/0.2 = 15$ V/cm, so $v_{dp} = \mu_p E = 500 \times 15 = 7.5 \times 10^{3}$ **cm/s**. (Check: $J = qpv_{dp} = 4.8\ \text{A/cm}^2$, and $J \times A = 4.8$ mA, which agrees.)

N3. [7 marks] (chained)

(a) $V_{Th} = 15 \cdot \frac{12}{68+12} =$ **2.25 V**. $R_{Th} = 68 \parallel 12 = \frac{68 \times 12}{80} =$ **10.2 kΩ**.

(b) KVL around $V_{Th} \to R_{Th} \to 0.7\ \text{V} \to 101$ kΩ:

$$I_B = \frac{V_{Th} - 0.7}{R_{Th} + 101\ \text{k}} = \frac{1.55}{111.2\ \text{k}} = \mathbf{13.94\ \mu A}$$

$V_B = V_{Th} - I_B R_{Th} = 2.25 - 13.94\,\mu\text{A} \times 10.2\ \text{k} =$ **2.108 V**.

(c) Ignoring R_{Th}: $I_B = 1.55/101\ \text{k} = 15.35\ \mu$A. The error is $(15.35 - 13.94)/13.94 =$ **+10.1 %**. This is why the course asks for the Thevenin step (practice test Q7(a)).

N4. [7 marks] (chained)

(a) Short both supplies: $R_{Th} = 20 \parallel 30 \parallel 15$. Then $1/R_{Th} = 1/20 + 1/30 + 1/15 = 0.15\ \text{mS}$, so $R_{Th} = 6.667$ **kΩ**.

(b) KCL at X, open circuit: $\frac{12 - V_X}{20} + \frac{-6 - V_X}{30} + \frac{0 - V_X}{15} = 0$.

$$V_{Th} = \frac{12/20 - 6/30}{0.15} = \frac{0.6 - 0.2}{0.15} = \mathbf{2.667\ V}$$

Superposition:

- +12 V alone: $30 \parallel 15 = 10$ k, so the contribution is $12 \cdot \frac{10}{20+10} = 4.0$ V.
- −6 V alone: $20 \parallel 15 = 8.571$ k, so the contribution is $-6 \cdot \frac{8.571}{30+8.571} = -1.333$ V.
- Sum: 2.667 V. This agrees.

(c) $V_X = V_{Th}\frac{5}{5+6.667} =$ **1.143 V**, so $I_L = 1.143/5\ \text{k} =$ **0.229 mA**. (Check by direct nodal analysis with the 5 kΩ added: the same 1.143 V.)

(d) $I_N = V_{Th}/R_{Th} = 2.667/6.667\ \text{k} = \mathbf{0.400\ mA}$ (flowing into X when X is shorted to ground), in parallel with $R_N = 6.667$ kΩ.

N5. [6 marks]

(a) Currents leaving the node are positive.

$$\text{Node 1: } \frac{V_1 - 12}{2} + \frac{V_1}{4} + \frac{V_1 - V_2}{3} = 0$$
$$\text{Node 2: } \frac{V_2 - V_1}{3} + \frac{V_2}{6} = 1\ \text{mA}$$

(resistances in kΩ, currents in mA). In matrix form:

$$\begin{bmatrix} 1.0833 & -0.3333 \\ -0.3333 & 0.5 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 6 \\ 1 \end{bmatrix}$$

$V_1 = 7.742$ **V and** $V_2 = 7.161$ **V.** Check at node 2: $(7.742 - 7.161)/3 = 0.194$ mA plus 1 mA $= 1.194$ mA, and $7.161/6 = 1.194$ mA. KCL balances.

(b) Current leaves the + terminal of the source (active convention), so $P = V_2 \times 1\ \text{mA} =$ **7.16 mW, supplied**.

(c) Switch off both sources: $R_{Th} = \big((2 \parallel 4) + 3\big) \parallel 6 = (1.333 + 3) \parallel 6 = \mathbf{2.516\ k\Omega}$. (Check: $V_{oc} = V_2 = 7.161$ V and $I_{sc} = 2.846$ mA by direct calculation, so $7.161/2.846 = 2.516$ kΩ.)

N6. [8 marks] (hard, chained)

(a) $n \approx N_D = 10^{13}$ and $p = (1.5 \times 10^{10})^2/10^{13} = 2.25 \times 10^7$.
$\sigma = 1.6 \times 10^{-19}(10^{13} \cdot 1350) = 2.16 \times 10^{-3}\ (\Omega\text{-cm})^{-1}$, so ρ = 463 Ω-cm.

(b) $n_i = 7.3 \times 10^{15}(450)^{3/2} e^{-1.12/(2\times 8.62\times 10^{-5}\times 450)} = 3.744 \times 10^{13}\ \text{cm}^{-3}$. This is larger than N_D.

(c) The approximation gives $p = (3.744 \times 10^{13})^2/10^{13} = 1.40 \times 10^{14}$, which is bigger than the "majority" $n = 10^{13}$. A minority concentration cannot exceed the majority, so the approximation (which assumes $N_D \gg n_i$) fails.

(d) Exact:

$$n = \frac{N_D}{2} + \sqrt{\left(\frac{N_D}{2}\right)^2 + n_i^2} = 5 \times 10^{12} + \sqrt{(5 \times 10^{12})^2 + (3.744 \times 10^{13})^2} = \mathbf{4.277 \times 10^{13}}$$

$p = n_i^2/n = 3.277 \times 10^{13}$. Check: $n - p = 10^{13} = N_D$, so charge neutrality holds. The sample is now close to intrinsic.

(e) $\sigma = 1.6 \times 10^{-19}(4.277 \times 10^{13} \cdot 1350 + 3.277 \times 10^{13} \cdot 500) = 1.186 \times 10^{-2}\ (\Omega\text{-cm})^{-1}$. The ratio to 300 K is **5.49**. $V_T = \frac{1.38 \times 10^{-23} \times 450}{1.6 \times 10^{-19}} = \mathbf{38.8\ mV}$. (In reality mobility falls with temperature, so the true ratio would be smaller. That is why we stated the assumption.)

6. Quick recall check

Question

1. $V_T = kT/q = 1.38 \times 10^{-23} \times 300 / 1.6 \times 10^{-19} = 25.875$ mV ≈ 26 mV.
2. $n_i^2 = 2.25 \times 10^{20}\ \text{cm}^{-6}$, so $p = 2.25 \times 10^{5}\ \text{cm}^{-3}$.
3. A voltage source becomes a short. A current source becomes an open. Dependent sources stay in.
4. The lightly doped side: $x_n / x_p = N_A / N_D$.
5. $V_{Th} = V_{CC} R_2 / (R_1 + R_2)$ and $R_{Th} = R_1 R_2 / (R_1 + R_2)$.

Diode models

Answers and worked solutions · Day 03 · Mon 26 Oct · pack E2

5. Practice

From your course files

WS2 Q8 (course question).

> The official solution uses $V_T = 25$ mV (its "V_T at 20 °C ≈ 25 mV") and $n = 1$. At mA currents $I_D \gg I_S$, so $V_D = nV_T \ln(I_D/I_S)$. (The full equation with the "−1" gives the same 0.5583 V at 1 mA, which confirms the diode is fully conducting.)
>
> - **i.** $V_D = 0.025 \ln \dfrac{10^{-3}}{0.2 \times 10^{-12}} = 0.025 \times 22.33 =$ **0.5583 V**
> - **ii.** $V_D = 0.025 \ln(5 \times 10^{10}) =$ **0.6159 V**
> - **iii.** $V_D = 0.025 \ln(5 \times 10^{11}) =$ **0.6734 V**
> - **iv.** Each decade adds $\Delta V = nV_T \ln 10 = 2.303 \times 25\ \text{mV} =$ **57.6 mV**. With $V_T = 25.9$ mV (27 °C) it is about 60 mV per decade.

WS2 Q9 (course question).

> Worked in full in Example 1 above (redo it closed-book). Dividing the two operating points removes I_S: $0.1 = e^{-0.1/(n \times 0.025)}$, so $n = 1.737$, and then $I_S = 10^{-3} e^{-0.7/(1.737 \times 0.025)} =$ **0.1 nA**.

WS2 Q10 (course question).

> Identical diodes share I equally, so each now carries $I/2$. From $I = I_S e^{V_1/nV_T}$ and $I/2 = I_S e^{V_2/nV_T}$:
>
> $$\frac{1}{2} = e^{(V_2 - V_1)/nV_T} \;\Rightarrow\; V_2 - V_1 = -nV_T \ln 2$$
>
> The voltage **falls** by $nV_T \ln 2$, about 17–18 mV for $n = 1$.

WS2 Q11 (course question).

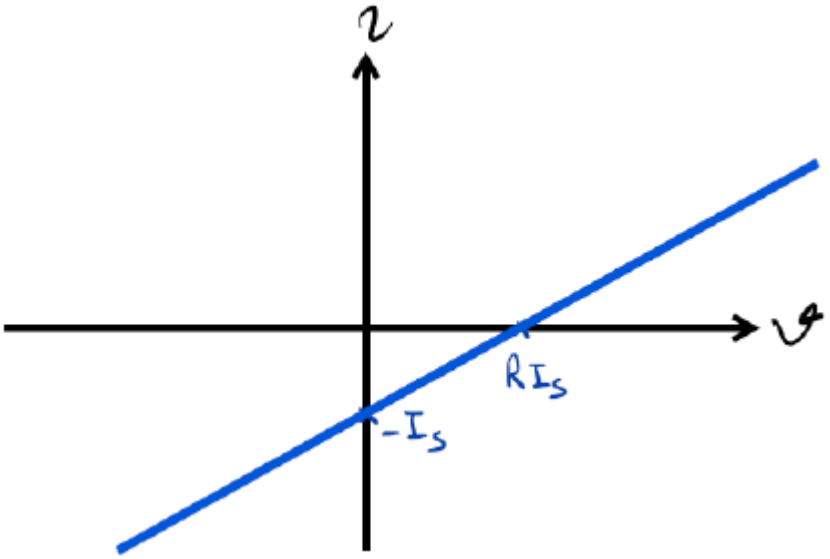
> [A] **anode**, [B] **p-type**, [C] **boron** (or another trivalent, Group III atom), [D] **holes**, [E] **electrons**.

WS2 Q12 (course question).

Yes, both are linear. Each contains only linear elements (a resistor and an independent source), so by superposition the terminal i–v relation is a straight line. Here I_S is an independent current source, not a diode's saturation current.

WS2 Q13 (course question).

Circuit A: KCL at the top node, $i + I_S = I_R = v/R$, so $i = \dfrac{v}{R} - I_S$. Intercepts: $i = 0$ at $v = RI_S$; $v = 0$ at $i = -I_S$.



WS2 Q13 official sketch for circuit A

Circuit B: KCL, $I_S = i + v/R$, so $i = I_S - \dfrac{v}{R}$. Intercepts: $i = 0$ at $v = RI_S$; $v = 0$ at $i = +I_S$. A straight line of slope $-1/R$.

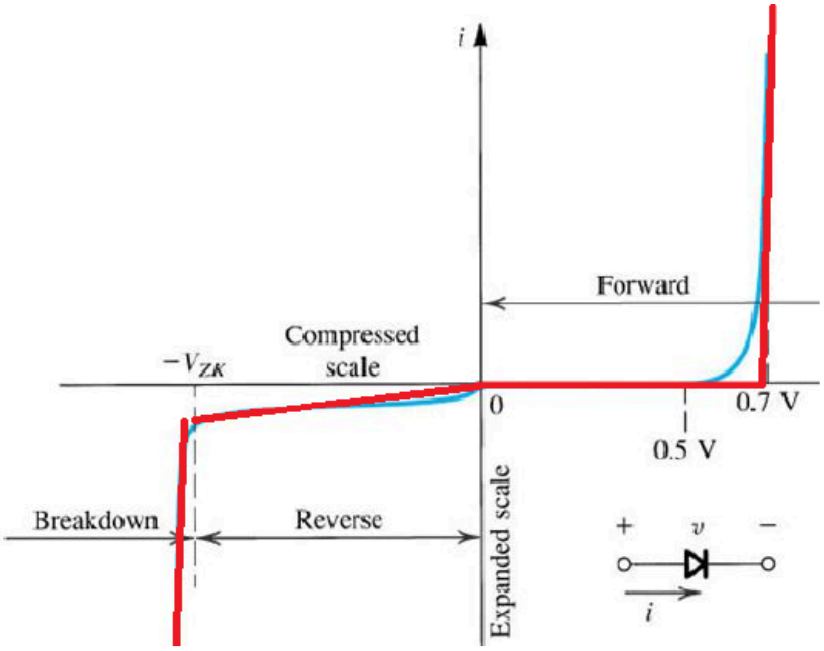
Both are straight lines, confirming Q12.

WS2 Q14 (course question).

- Region A: **breakdown**. Region B: **reverse bias** (only I_S flows). Region C: **partial conduction**. Region D: **full conduction**.
- Voltage A: **Zener knee / breakdown voltage**. Voltage B: **cut-in voltage**. Voltage C: **forward voltage drop**.

WS2 Q15 (course question).

Four linear pieces, one per region (breakdown, reverse, partial conduction, full conduction), because each region behaves differently and no single straight line fits two of them.



WS2 Q15 official sketch: four straight pieces (red) over the real curve

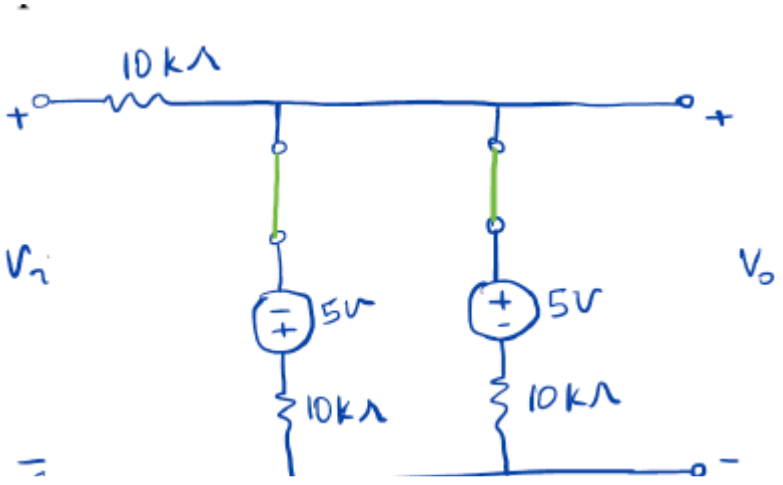
WS2 Q16 (course question).

The ohmmeter's own source is **reverse biasing** the diode, so it reads the very large reverse resistance (almost an open circuit). **Nothing is wrong.** Rotate the diode and it is forward biased, so you expect a **very low** reading (a few to a few tens of ohms). This is a quick way to find a diode's polarity, but the forward reading is at one unknown operating point, so it is not the diode's forward resistance r_D.

WS2 Q17 (course question).

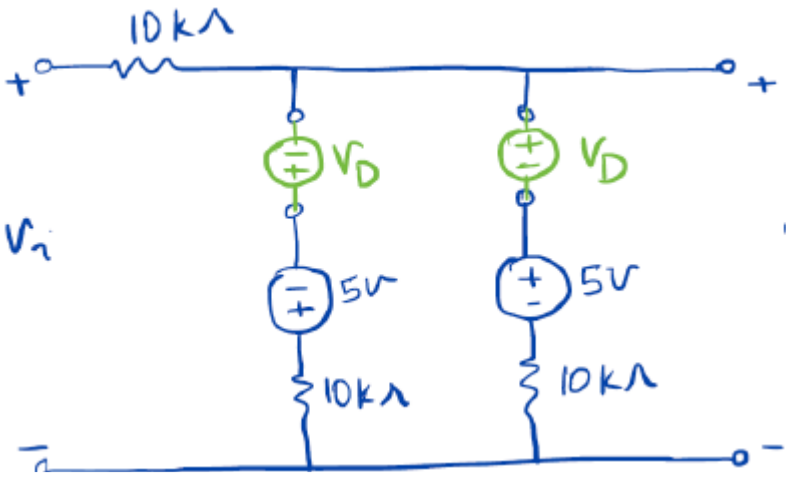
Redraw only. Both diodes are assumed ON, so each is replaced in turn by the forward model.

1. Ideal: each diode becomes a short circuit (green).



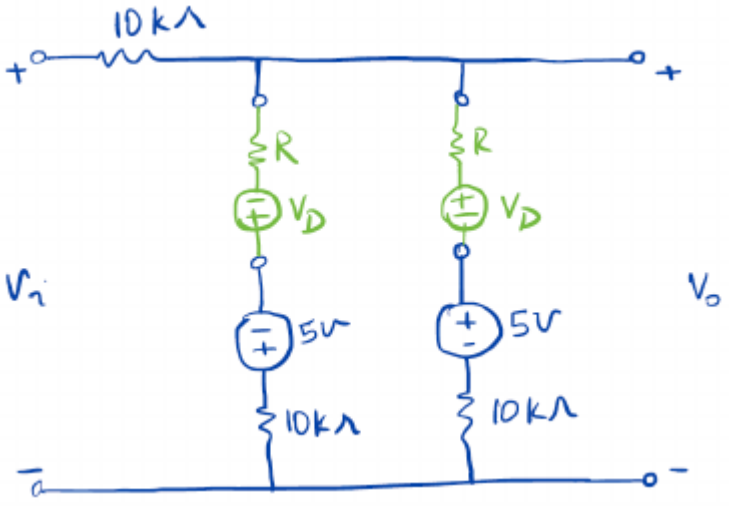
WS2 Q17 official redraw, ideal model

2. Voltage drop: each diode becomes a V_D source with + on the anode side.



WS2 Q17 official redraw, voltage-drop model

3. $V_D + R$**:** each diode becomes V_D in series with R (that is, r_D).



WS2 Q17 official redraw, voltage drop with resistor

In reverse bias every piecewise-linear model is just an open circuit.

WS3 Q1 (course question).

$T = 298.15$ K, so $V_T = \dfrac{1.38 \times 10^{-23} \times 298.15}{1.6 \times 10^{-19}} = 25.72$ mV. With $I_D \gg I_S$:

$$V = nV_T \ln \frac{I_D}{I_S} = 25.72\ \text{mV} \times \ln \frac{100 \times 10^{-3}}{10^{-6}} = 25.72 \times 11.51 = \mathbf{296\ mV}$$

(official 296.11 mV; see the note above).

WS3 Q2 (course question).

The diode is reverse biased, so the current round the loop is I_S and $V = I_S \times 1\ \text{M}\Omega$. At 20 °C: $I_S = 1/10^6 = 1\ \mu$A.

- 40 °C is +20 °C, two doublings: $I_S = 4\ \mu$A, so $V = 4$ **V**.
- 0 °C is −20 °C, two halvings: $I_S = 0.25\ \mu$A, so $V = 0.25$ **V**.

WS3 Q3 (course question).

Each diode must drop $2.4/3 = 0.8$ V, which is 0.1 V above 0.7 V, so one decade above 1 mA: $I_D = 10$ mA.

$$R = \frac{10 - 2.4}{10\ \text{mA}} = \mathbf{760\ \Omega}$$

WS3 Q4 (course question).

$T = 293$ K: $V_T = \dfrac{1.38 \times 10^{-23} \times 293}{1.6 \times 10^{-19}} =$ **25.27 mV**. Full equation:

$$I_D = I_S\left(e^{V_D/nV_T} - 1\right) = 40 \times 10^{-9}\left(e^{0.5/(2\times 0.02527)} - 1\right) = \mathbf{0.79\ mA}$$

(official 0.789 mA; I get 0.791 mA).

WS3 Q5 (course question).

Use the knee voltage for the source and the chord between two points on the steep part for $r_{av} = \Delta V_d / \Delta I_d$ (the official solution uses 4 mA and 30 mA).

- **Ge:** $V_K = 0.3$ V, $r_{av} = \dfrac{0.42 - 0.3}{(30 - 4)\ \text{mA}} = 4.62\ \Omega$. Model: ideal diode + **0.3 V** + **4.62 Ω**.
- **GaAs:** $V_K = 1.2$ V, $r_{av} = \dfrac{1.32 - 1.2}{(30 - 4)\ \text{mA}} = 4.62\ \Omega$. Model: ideal diode + **1.2 V** + **4.62 Ω**.

WS3 Q6 (course question).

(a) At 4 mA the graph gives $V_D \approx 0.7$ V, so $R_D = V_D / I_D = 0.7/4\ \text{mA} =$ **175 Ω**.

(b) Draw the tangent at 10 mA and read ΔV_D and ΔI_D off it. The official answer is $r_d \approx 4$ **Ω** (see the note at the top of this section about its printed ΔI_D).

WS3 Q7 (course question).

(a) Both diodes conduct (Si 0.7 V, Ge 0.3 V). Divider on what is left:

$$V_o = \frac{10}{10+2}(12 - 0.7 - 0.3) = \mathbf{9.167\ V}$$

(b) The top Si diode and the parallel pair (which drops only one 0.7 V) are on. Current through 4.7 kΩ:

$$I = \frac{16 - 0.7 - 0.7 - (-4)}{4.7\ \text{k}} = 3.96\ \text{mA}, \qquad V_o = -4 + 4.7\ \text{k} \times 3.96\ \text{mA} = \mathbf{14.6\ V}$$

WS3 Q8 (course question).

Assume the diode is **OFF**. Then both sides are plain dividers:

$$V_A = 5 \cdot \frac{2}{2+3} = \mathbf{2\ V}, \qquad V_B = 5 \cdot \frac{2}{2+2} = \mathbf{2.5\ V}$$

Check: $v_d = V_A - V_B = -0.5$ V, which is below the 0.7 V threshold (it is reverse biased), so **OFF** is correct.

PT Q2 (practice test, 5 marks).

Step 1. Thermal voltage. $T_K = 273 + 27 = 300$ K.

$$V_T = \frac{kT_K}{q} = \frac{1.38 \times 10^{-23} \times 300}{1.6 \times 10^{-19}} = 25.875\ \text{mV}$$

Step 2. Ratio form. Both currents are much larger than I_S, so $I_D \simeq I_S e^{V_D/nV_T}$. Divide the two operating points:

$$\frac{I_2}{I_1} = e^{(V_2 - V_1)/nV_T} \;\Rightarrow\; \Delta V_D = nV_T \ln \frac{I_2}{I_1}$$

I_S cancels, which is why the question does not give it.

Step 3. Substitute.

$$\Delta V_D = 1.5 \times 25.875 \times 10^{-3} \times \ln \frac{10}{0.1} = 1.5 \times 0.025875 \times 4.605 = \boxed{0.179\ \text{V}\ (179\ \text{mV})}$$

The voltage **increases** by 179 mV.

> **SANITY CHECK:** one decade costs $2.303 \times 1.5 \times 25.875 = 89.4$ mV, and two decades cost 179 mV ✓.

PT Q3 (practice test, 8 marks).

READING THE FIGURE. The 10 V source feeds a 5 kΩ series resistor to node A. A 5 kΩ resistor runs from A to ground. The three diodes run from A to node B, all pointing from A to B. From B, 10 kΩ goes down to the + terminal of a 3 V source whose − terminal is grounded. V_o is measured from B to ground.

(a) Redraw. Replace each diode with an ideal diode in series with a 0.6 V source (+ on the anode side, so it opposes forward current) and a 0.2 kΩ resistor. Draw all three separately, because the question says not to simplify:

```
10 V o--[5k]--+--|>|-(+0.6V-)-[0.2k]--|>|-(+0.6V-)-[0.2k]--|>|-(+0.6V-)-[0.2k]--+--o
              A          D1                        D2                       D3      B
              |                                                                    |
            [5k]                                                                [10k]
              |                                                                    |
             GND                                                          (+) 3 V (-)
                                                                                   |
                                                                                  GND
```

Practice test Q3(a) redrawn: each diode replaced by an ideal diode, a 0.6 V source (+ on the anode side) and 0.2 kΩ.

Label the chain current I, flowing from A to B (anode to cathode).

(b) Verify. Work in V, mA and kΩ.

1. Model: 3rd model, all three ON (as the question tells you to assume).
2. KCL at A: $\dfrac{10 - V_A}{5} = \dfrac{V_A}{5} + I \;\Rightarrow\; V_A = 5 - 2.5I$.
3. Branch B to ground: $V_B = 3 + 10I$.
4. Diode chain: $V_A - V_B = 3(0.6) + 3(0.2)I = 1.8 + 0.6I$.
5. Substitute: $(5 - 2.5I) - (3 + 10I) = 1.8 + 0.6I \;\Rightarrow\; 0.2 = 13.1I$

$$I = 0.01527\ \text{mA} = 15.3\ \mu\text{A}$$

The cross-check by Thévenin is quicker. The source, the 5 k and the 5 k seen from A give $V_{Th} = 5$ V and $R_{Th} = 2.5$ kΩ. Then $I = \dfrac{5 - 1.8 - 3}{2.5 + 0.6 + 10} = \dfrac{0.2}{13.1} = 15.3$ μA ✓.

CHECK: $I = 15.3$ μA > 0 in the anode-to-cathode direction. Because the diodes are in series, the same positive current flows through each one, so **all three diodes are conducting** ✓. The node voltages are $V_A = 4.962$ V and $V_B = 3.153$ V. The chain drops 1.809 V, which equals $1.8 + 0.6(0.0153)$ ✓.

(c) Output voltage.

$$V_o = V_B = 3 + 10\ \text{k}\Omega \times 15.3\ \mu\text{A} = \boxed{3.15\ \text{V}}$$

Comment worth writing: the margin is tiny. Only 0.2 V is left after the three 0.6 V drops and the 3 V source. With the 2nd model at 0.7 V per diode, $5 - 2.1 - 3 = -0.1$ V would mean the diodes are OFF and $V_o = 3$ V. That is why the verification part is worth 4 marks.

New questions

N1 [4 marks]

a. $T_K = 323$ K, so $V_T = 1.38 \times 10^{-23} \times 323 / 1.6 \times 10^{-19} = \mathbf{27.86\ mV}$.

b. $\Delta V_D = nV_T \ln(I_2/I_1) = 1.8 \times 0.02786 \times \ln(0.1) = -0.1155$ V. So $V_D = 0.62 - 0.1155 = \mathbf{0.505\ V}$. One decade costs $2.303 nV_T = 115$ mV ✓.

c. $r_d = nV_T / I_D = 1.8 \times 27.86\ \text{mV} / 2\ \text{mA} = \mathbf{25.1\ \Omega}$.

N2 [6 marks]

a. $V_T = 25.875$ mV. Dividing the two operating points gives

$$n = \frac{0.69 - 0.60}{0.025875 \ln 10} = \mathbf{1.511}$$

Then $I_S = 0.5 \times 10^{-3} e^{-0.60/(1.511 \times 0.025875)} = \mathbf{1.08 \times 10^{-10}\ A}$ (0.108 nA). The second point gives the same value ✓.

b. $V_D = 0.69 + nV_T \ln(20/5) = 0.69 + 1.511 \times 0.025875 \times 1.386 = \mathbf{0.744\ V}$.

c. $R_D = 0.69/5\ \text{mA} = \mathbf{138\ \Omega}$. $r_d = 1.511 \times 25.875\ \text{mV}/5\ \text{mA} = \mathbf{7.82\ \Omega}$. A 2 mV signal is below about 5 mV, so it sees r_d.

N3 [4 marks]

a. $V_D \approx 0.65 - 2.5\ \text{mV} \times 50 = \mathbf{0.525\ V}$.

b. At 75 °C: $10\ \text{nA} \times 2^{50/10} = 10 \times 32 = \mathbf{320\ nA}$. At 0 °C: $10 \times 2^{-25/10} = \mathbf{1.77\ nA}$.

c. $V_T(25\ ^\circ\text{C}) = 25.70$ mV, so $r_d = 25.7\ \Omega$. $V_T(75\ ^\circ\text{C}) = 30.02$ mV, so $r_d = 30.0\ \Omega$. (V_T here uses $T_K = T + 273$.)

N4 [7 marks] (multi-diode, iterate)

a. **Assume D1 ON and D2 ON.** Write KCL at A, in mA with kΩ:

$$\frac{10 - V_A}{1} = \frac{V_A - 0.7}{2} + \frac{V_A - 0.7 - 8}{1} \;\Rightarrow\; 19.05 = 2.5V_A \;\Rightarrow\; V_A = 7.62 \text{ V}$$

This gives $i_{d1} = (7.62 - 0.7)/2 = 3.46$ mA > 0 ✓, but $i_{d2} = (7.62 - 8.7)/1 = -1.08$ mA < 0 ×. **D2 cannot be ON.** The 8 V source pushes back harder than node A can push forward.

b. **Assume D1 ON and D2 OFF.** D2 is now an open circuit.

$$I = \frac{10 - 0.7}{1 + 2} = 3.10 \text{ mA}, \quad V_A = 10 - 3.10 = 6.90 \text{ V}, \quad V_B = 6.90 - 0.7 = 6.20 \text{ V}$$

No current flows in the 1 kΩ of D2's branch, so D2's cathode sits at 8 V. That gives $v_{d2} = 6.90 - 8 = -1.10$ V.

Check: $i_{d1} = 3.10$ mA > 0 ✓ and $v_{d2} = -1.10$ V < 0.7 V ✓.

$$\boxed{V_o = 6.20 \text{ V}} \quad (\text{D1 ON, D2 OFF})$$

N5 [8 marks] (two diodes, two input polarities, chained)

a. **Assume D1 ON and D2 OFF.** D1's anode faces $+6$ V. D2's cathode is at the positive node. Branch 1 is 0.7 V + 2 V in series with $0.1 + 0.5 = 0.6$ kΩ, so its current is $(V_A - 2.7)/0.6$. KCL at A:

$$\frac{6 - V_A}{1} = \frac{V_A}{4} + \frac{V_A - 2.7}{0.6} \;\Rightarrow\; V_A = 3.60 \text{ V}$$

That gives $i_{d1} = (3.60 - 2.7)/0.6 = 1.50$ mA > 0 ✓. D2's anode is at 0 V (no current in its 1 kΩ) and its cathode is at 3.60 V, so $v_{d2} = -3.60$ V < 0.7 ✓. As a KCL check, $2.40 = 0.90 + 1.50$ mA ✓. $V_o = 3.60$ **V.**

b. **Assume D2 ON and D1 OFF.** Current flows up from ground through 1 kΩ and D2 into A, then through R to the -6 V source. The current into A through D2 is $(0 - 0.7 - V_A)/1.1$. KCL at A (current in through R = current out through the 4 k + current out through branch 2, which is negative here):

$$\frac{-6 - V_A}{1} = \frac{V_A}{4} - \frac{-0.7 - V_A}{1.1} \;\Rightarrow\; V_A = -3.07 \text{ V}$$

That gives $i_{d2} = (-0.7 + 3.074)/1.1 = 2.16$ mA > 0 ✓. D1's cathode is held at +2 V by the battery (no current in 0.5 kΩ), so $v_{d1} = -3.07 - 2 = -5.07$ V < 0.7 ✓. $V_o = -3.07$ **V.**

c. With both diodes OFF, $V_A = V_i \times \dfrac{4}{1 + 4} = 0.8V_i$.

- D1 turns on when $v_{d1} = V_A - 2 = 0.7$, that is $V_A = 2.7$ V, so $V_i = 3.375$ V.
- D2 turns on when $v_{d2} = 0 - V_A = 0.7$, that is $V_A = -0.7$ V, so $V_i = -0.875$ V.

Both diodes are OFF for $-0.875\ \text{V} < V_i < 3.375\ \text{V}$. Parts (a) and (b) lie outside this range, which is consistent.

N6 [8 marks] (OR gate with unequal inputs: the near-miss)

Each ON branch is a source $V_k - 0.7$ in series with $1 + 0.1 = 1.1$ kΩ.

a. **Assume both ON.** KCL at V_o:

$$\frac{4.3 - V_o}{1.1} + \frac{3.3 - V_o}{1.1} = \frac{V_o}{4} \;\Rightarrow\; V_o = 3.341\ \text{V}$$

That gives $i_{d1} = 0.872$ mA > 0 ✓, but $i_{d2} = (3.3 - 3.341)/1.1 = -0.037$ mA < 0 ×. **D2 is not ON.** The failure is small, so it is easy to miss if you round early.

b. **Assume D1 ON and D2 OFF.**

$$i_{d1} = \frac{4.3}{1.1 + 4} = 0.843\ \text{mA}, \qquad V_o = 4 \times 0.843 = 3.373\ \text{V}$$

No current flows in D2's 1 kΩ, so its anode is at 4 V and $v_{d2} = 4 - 3.373 = 0.627$ V.

Check: $i_{d1} > 0$ ✓ and $v_{d2} = 0.627$ V < 0.7 V ✓. $V_o = 3.37$ **V.** D2 is forward biased, but by less than V_D, so in this model it is OFF.

c. D2 starts to conduct when $v_{d2} = 0.7$ V with zero current, that is $V_2 = 3.373 + 0.7 = \mathbf{4.07\ V}$.

d. $V_2 = 5$ V is symmetric with V_1, so both are ON with equal currents: $2(4.3 - V_o)/1.1 = V_o/4$. That gives $V_o = \mathbf{3.78\ V}$, with $i_{d1} = i_{d2} = 0.473$ mA > 0 ✓.

6. Quick recall check

Question

1. $V_T \ln 10 = 57.6$ mV. With $n = 2$ it is 115 mV.
2. It falls, by $nV_T \ln 2 \approx 18$ mV for $n = 1$.
3. An open circuit in all three models.
4. $26/13 = 2\ \Omega$.
5. $5 \times 2^2 = 20$ nA. The forward drop falls by about $2.5 \times 20 = 50$ mV at a fixed current.

Rectifiers, Zener regulators, clippers

Answers and worked solutions · Day 05 · Wed 28 Oct · pack E3

5. Practice

From your course files

WS3 Q9 (course question).

(a) $V_m = \sqrt{2} \times 12 = 16.97$ V and $V_P = V_m - V_D = 16.17$ V. Half-wave, so $V_{rpp} = V_P/(fCR_L)$:

$$C = \frac{V_P}{V_{rpp} f R_L} = \frac{16.17}{1 \times 50 \times 100} = \mathbf{3.23\ mF}$$

(b) $V_{Odc} = V_P - \frac{1}{2}V_{rpp} = 16.17 - 0.5 =$ **15.67 V**. $I_L = 15.67/100 =$ **156.7 mA**. PIV $\approx 2V_m =$ **33.94 V** (official; see the note above for 33.14 V).

WS3 Q10 (course question).

Zener model: $V_O = V_{zk} + I_Z r_z$. KCL at the output node, $I_s = I_Z + I_L$:

$$\frac{15 - V_O}{100} = \frac{V_O - 10}{20} + \frac{V_O}{500} \;\Rightarrow\; 6.2V_O = 65 \;\Rightarrow\; V_O = \mathbf{10.48\ V}$$

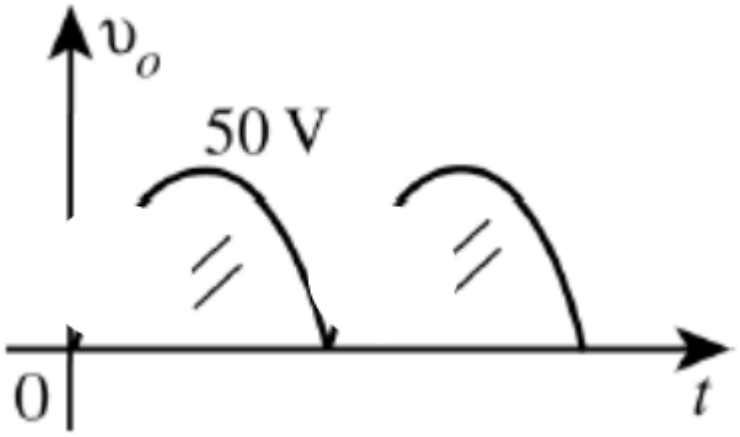
$I_s = (15 - 10.48)/100 =$ **45.2 mA**, $I_L = 10.48/500 =$ **20.97 mA**, $I_Z = (10.48 - 10)/20 =$ **24.2 mA**, $P_Z = V_O I_Z =$ **254 mW** (official 251.5 mW after rounding I_Z to 24 mA).

WS3 Q11 (course question).

Ideal diodes. In each half-cycle one diode conducts, and the circuit reduces to one 2.2 kΩ bridge resistor in series with the 2.2 kΩ load, so it is a 2.2/2.2 divider:

$$V_{o,peak} = \frac{2.2}{2.2 + 2.2} \times 100 = \mathbf{50\ V}$$

The load current has the same direction in both half-cycles, so v_o is a full-wave rectified sine of peak 50 V.



WS3 Q11 official output sketch

$$V_{dc} = 0.636 \times 50 = \mathbf{31.8\ V}$$

WS3 Q12 (course question).

Worked in full in Worked example 4 above (this is also practice test Q5). $v_i = +20$ V: diode reverse biased, $v_o = 0$ **V**. $v_i = -5$ V: diode on, $+5 - 4 + v_o = 0$, so $v_o = -1$ **V**. v_o is a square wave between 0 and −1 V.

WS3 Q13 (course question).

Worked in full in Worked example 3 above (this is also practice test Q4). $V_Z = 12$ **V.** At full load $R_L = 12/0.2 = 60\ \Omega$ and $I_Z \to 0$, so $\dfrac{60 \times 16}{60 + R_S} = 12$ gives $R_S = 20\ \Omega$. At no load the Zener takes 200 mA, so it needs $P_Z \geq 2.4$ W.

WS3 Q14 (course question).

Worked in full in Worked example 1 above. $C = V_P/(V_r f R) =$ **83.3 µF**.
$\omega \Delta t = \sqrt{2V_r/V_P} = 0.2$ rad, so the diode conducts for **3.18 %** of the cycle.
$i_{Dav} = I_L(1 + \pi\sqrt{2V_P/V_r}) =$ **324 mA** and $i_{Dmax} = I_L(1 + 2\pi\sqrt{2V_P/V_r}) =$ **638 mA**.

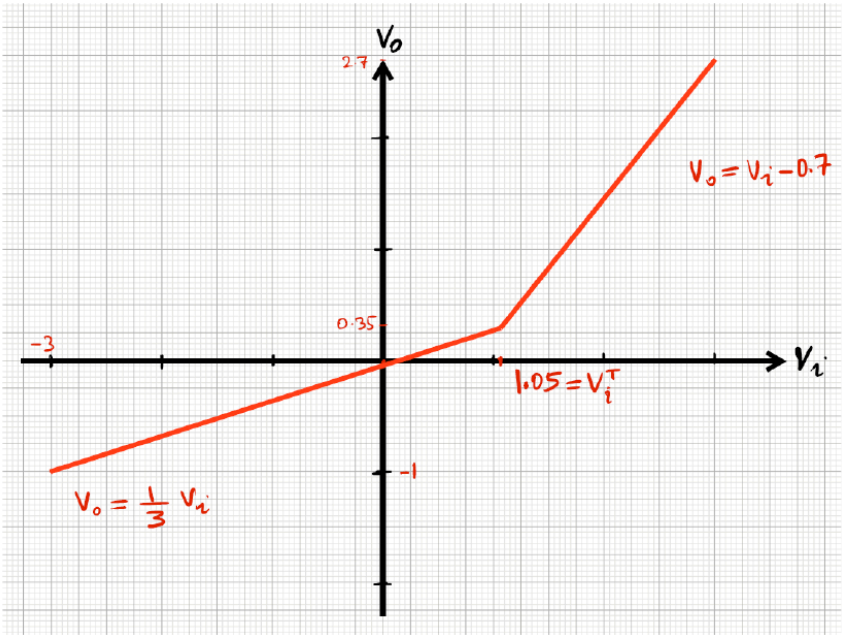
WS3 Q18 (course question).

The diode sits across R_1, so while it is off $v_d = v_{R1} = V_i \dfrac{R_1}{R_1 + R_2} = \dfrac{V_i}{1.5}$. It turns on when this reaches 0.7 V:

$$\frac{v_{iT}}{1.5} = 0.7 \ \Rightarrow\ v_{iT} = \mathbf{1.05\ V}$$

- $V_i < 1.05$ V (diode off): divider, $V_o = V_i \dfrac{0.5}{1.5} = \dfrac{V_i}{3}$.
- $V_i > 1.05$ V (diode on, 0.7 V source): KVL, $V_o = V_i - 0.7$.

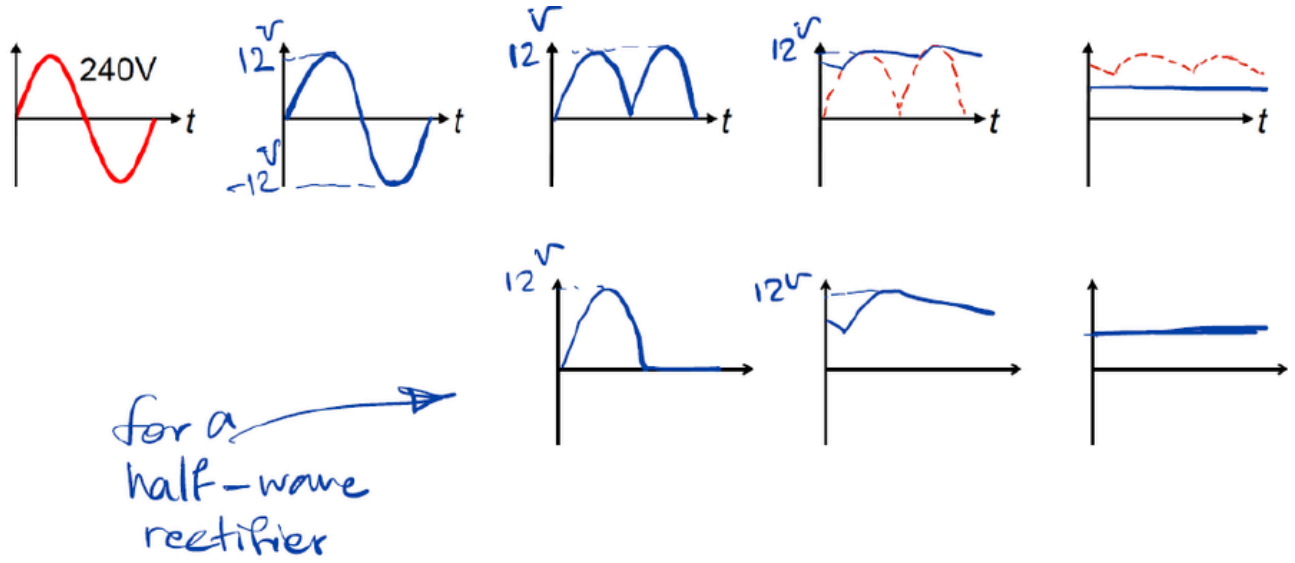
Endpoints: $V_o = -1.67$ V at $V_i = -5$ V, $V_o = 0.35$ V at the break, $V_o = 4.3$ V at $V_i = +5$ V.



WS3 Q18 official plot (drawn over a narrower range than the question asks; extend both lines to ±5 V)

WS3 Q19 (course question).

Sketches only. 240 V ac sine → transformer: 12 V peak sine → rectifier: pulsating dc (bridge: a hump every half-cycle, at $2f$; half-wave: one hump per cycle, at f) → capacitor: dc with a sawtooth ripple (the half-wave ripple is larger because it refills only once per cycle) → regulator: flat 5 V dc.



WS3 Q19 official sketches: top row bridge, bottom row half-wave

WS3 Q20 (course question).

Half-wave: current flows only for $0 < \omega t < \pi$.

1. $I_m = \dfrac{V_m}{r_f + R} = \dfrac{50}{820} =$ **61.0 mA**
2. $I_{dc} = \dfrac{1}{2\pi}\displaystyle\int_0^{\pi} I_m \sin\omega t \, d(\omega t) = \dfrac{I_m}{\pi} =$ **19.4 mA**
3. $V_{m,L} = 50 \times \dfrac{800}{820} =$ **48.8 V**
4. $V_{DC,L} = RI_{dc} =$ **15.5 V**
5. $I_{rms} = \sqrt{\dfrac{I_m^2}{2\pi}\displaystyle\int_0^{\pi} \sin^2 \omega t \, d(\omega t)} = \dfrac{I_m}{2} =$ **30.5 mA**
6. $V_{L,rms} = V_{m,L}/2 =$ **24.4 V**
7. The diode sees $r_f I_m \sin\omega t$ forward and $V_m \sin\omega t$ reverse, so $V_{D,DC} = \dfrac{1}{2\pi}\left[\displaystyle\int_0^{\pi} r_f I_m \sin\omega t \, d(\omega t) + \int_{\pi}^{2\pi} V_m \sin\omega t \, d(\omega t)\right] = \dfrac{r_f I_m - V_m}{\pi} = -$ **−15.5 V** (equal and opposite to the load's dc voltage, as KVL requires).
8. $P_{AC,i} = I_{rms}^2 (r_f + R) =$ **0.762 W**
9. $P_{DC,o} = I_{dc}^2 R =$ **0.301 W**
10. $\eta = P_{DC,o}/P_{AC,i} =$ **39.5 %**

WS3 Q21 (course question).

Two diodes conduct in series at any moment, so the path resistance is $2r_f + R$.

1. $I_m = \dfrac{240}{2(20) + 460} =$ **0.48 A**
2. $I_{dc} = \dfrac{2I_m}{\pi} =$ **0.306 A**

3. $V_{m,L} = 240 \times \dfrac{460}{500} =$ **220.8 V**
4. $V_{DC,L} = RI_{dc} =$ **140.6 V**
5. PIV $= V_m =$ **240 V**
6. Two output pulses per input cycle: $\omega_o = 2 \times 314 = 628$ rad/s, so $f_o = 628/2\pi =$ **99.95 Hz**
7. Each diode conducts half the cycle, so its $I_{rms} = I_m/2 = 0.24$ A and $P_D = I_{rms}^2 r_f =$ **1.15 W**

WS3 Q22 (course question).

$I_R = (V_s - V_Z)/R_s$: $I_{R,max} = (11 - 4.8)/1 = 6.2$ mA and $I_{R,min} = (9 - 4.8)/1 = 4.2$ mA. KCL: $I_L = I_R - I_Z$.

- **Smallest load current.** Even with no load the Zener takes at most $I_{R,max} = 6.2$ mA, which is below $I_{Z,max} = 7$ mA. So $I_{L,min} = 0$ is allowed and $R_{L,max} = \infty$.
- **Largest load current.** It must work at the lowest supply with the Zener still at $I_{Z,min}$: $I_{L,max} = I_{R,min} - I_{Z,min} = 4.2 - 0.2 = 4$ mA. So $R_{L,min} = 4.8/4\,\text{mA} = 1.2$ **kΩ**.

New questions

N1 (easy) [4 marks].

a. $V_m = 15\sqrt{2} = 21.21$ V. Two diodes are in the path, so $V_P = 21.21 - 1.4 = \mathbf{19.81\ V}$.

b. Full-wave, so use $2f$:

$$V_r = \frac{V_P}{2fCR} = \frac{19.81}{2 \times 50 \times 1000 \times 10^{-6} \times 470} = \mathbf{0.422\ V}$$

The ripple frequency is $2f = \mathbf{100\ Hz}$.

c. $V_{O(dc)} = V_P - V_r/2 = 19.81 - 0.21 = \mathbf{19.60\ V}$. Bridge PIV $= V_S - V_D = 21.21 - 0.7 = \mathbf{20.5\ V}$, so choose a diode rated for at least about 31 V.

(If you used f instead of $2f$, you got 0.843 V. That is the trap.)

N2 (medium) [6 marks].

a. $V_m = 12\sqrt{2} = 16.97$ V and $V_P = 16.97 - 0.7 = 16.27$ V. Full-wave with a known load current:

$$C = \frac{I_L}{2fV_r} = \frac{0.5}{2 \times 50 \times 1} = \mathbf{5000\ \mu F}\ \text{(choose the next standard value up)}$$

b. $V_{O(dc)} = 16.27 - 0.5 = \mathbf{15.77\ V}$. Centre-tap PIV $= 2V_S - V_D = 2(16.97) - 0.7 = \mathbf{33.2\ V}$.

c. $\omega\Delta t = \sqrt{2V_r/V_P} = \sqrt{2/16.27} = \mathbf{0.351\ rad}$ (20.1°), so $\Delta t = 0.351/(2\pi \cdot 50) = 1.12$ ms each half-cycle. For full-wave, each diode only has to replace the charge lost in $T/2$:

$$i_{Dmax} = I_L\left(1 + 2\pi\sqrt{\frac{V_P}{2V_r}}\right) = 0.5\left(1 + 2\pi\sqrt{8.135}\right) = \mathbf{9.46\ A}$$

($i_{Dav} = 0.5(1 + \pi\sqrt{8.135}) = 4.98$ A.) Using the half-wave form from the formula sheet would give $i_{Dav} = 9.46$ A, which is twice too large. If you use it in a test, say so.

N3 (medium) [6 marks].

a. KCL at the output node:

$$\frac{V_O - 6}{5} + \frac{V_O}{1000} = \frac{12 - V_O}{150}$$

This gives $\mathbf{V_O = 6.164\ V}$. Then $I_Z = (6.164 - 6)/5 = \mathbf{32.7\ mA}$, $I_L = 6.16$ mA, $I_S = (12 - 6.164)/150 = 38.9$ mA (check: 32.7 + 6.2 = 38.9 ✓).
$P_Z = 6.164 \times 32.7\ \text{mA} = \mathbf{202\ mW}$.

b. Repeat with 13 V: $V_O = 6.196$ V, so $\Delta V_O = 32.1$ mV for $\Delta V_s = 1$ V. **Line regulation = 32.1 mV/V.** Check with the formula: $\dfrac{r_z \parallel R_L}{R_S + r_z \parallel R_L} = \dfrac{4.975}{154.975} = 0.0321$ ✓.

c. No load: $\dfrac{V_O - 6}{5} = \dfrac{12 - V_O}{150}$, which gives $V_{NL} = 6.194$ V. Then

$$\%V.R. = \frac{6.194 - 6.164}{6.164} \times 100 = \mathbf{0.48\%}$$

Compare this with the 33 % of your unregulated Prac 1 supply.

N4 (hard, chained) [10 marks].

a. $V_m = 18\sqrt{2} = 25.46$ V. $V_P = 25.46 - 1.4 = \mathbf{24.06\ V}$. The valley is $V_P - V_r = \mathbf{22.06\ V}$.

b. The worst case is the valley with full load:

$$R_S \le \frac{22.06 - 9.1}{0.050 + 0.005} = \mathbf{235.6\ \Omega}$$

Choose **220 Ω**, the next standard value below. Check: at the valley, $(22.06 - 9.1)/220 - 5\ \text{mA} = 53.9$ mA ≥ 50 mA ✓.

c. While the Zener is in breakdown, $V_O = 9.1$ V is fixed, so $I_S = (v_C - 9.1)/R_S$ depends only on the capacitor voltage. The load only changes how I_S splits between I_Z and I_L. This is the same thing your Prac 1 showed. With $V_{C,avg} = V_P - V_r/2 = 23.06$ V:

$$I_S = \frac{23.06 - 9.1}{220} = 63.4\ \text{mA}$$

The capacitor discharges into this current, and it is full-wave:

$$C = \frac{I_S}{2fV_r} = \frac{0.0634}{2 \times 50 \times 2} = 317\ \mu\text{F} \Rightarrow \mathbf{choose\ 330\ \mu F}$$

That gives $V_r = 1.92$ V ✓.

d. The worst case for the Zener is no load, where it takes all of I_S. At the average capacitor voltage, $P_Z = 9.1 \times 63.4\ \text{mA} = 0.577$ W. At the ripple peak, $I_S = (24.06 - 9.1)/220 = 68.0$ mA, so $P_{Z,max} = \mathbf{0.62\ W}$. Specify a 1 W Zener. Bridge PIV $= V_m - V_D = \mathbf{24.8\ V}$, so choose diodes rated at least 37 V (for example, 50 V parts).

e. With $V_r = 1.92$ V, using the full-wave form:

$$i_{Dmax} \approx I_S \left(1 + 2\pi\sqrt{\frac{V_P}{2V_r}}\right) = 0.0634\left(1 + 2\pi\sqrt{6.26}\right) \approx \mathbf{1.06\ A}$$

N5 (medium) [5 marks].

a. With both diodes off, no current flows in the 1 kΩ, so $v_o = v_i$.

- D_1 turns on when $v_o = 3 + 0.7 = 3.7$ V, so $v_{iT1} = +3.7$ V. Above it, $\mathbf{v_o = 3.7\ V}$ (clipped).
- D_2 turns on when $v_o = -2 - 0.7 = -2.7$ V, so $v_{iT2} = -2.7$ V. Below it, $\mathbf{v_o = -2.7\ V}$.
- For $-2.7 < v_i < 3.7$ V, $\mathbf{v_o = v_i}$.

Input and output of the double clipper in N5.

b. v_o is the sine with its top flattened at +3.7 V and its bottom flattened at −2.7 V. The current in the 1 kΩ at the positive peak is $(10 - 3.7)/1\text{k} = 6.3$ mA. At the negative peak it is $(10 - 2.7)/1\text{k} = 7.3$ mA in the other direction. The largest current is therefore **7.3 mA**, at the negative peak.

N6 (easy) [4 marks].

a. When the input is −10 V, the diode conducts and C charges to 10 V. Thereafter the output is $v_i + 10$, so it switches between $\mathbf{0\ V}$ and $\mathbf{+20\ V}$. The shape is unchanged and it is still 20 V peak-to-peak.

b. $V_m = 12\sqrt{2} = 16.97$ V. The ideal output is $2V_m = \mathbf{33.9\ V}$. With two 0.7 V drops, it is $2(16.97 - 0.7) = \mathbf{32.5\ V}$. Each diode must block $2V_m = \mathbf{33.9\ V}$.

6. Quick recall check

Question

1. Bridge : half-wave = 1 : 2. The bridge refills every $T/2$, so $V_r = V_P/(2fCR)$.
2. $C = I_L/(fV_r)$, or $V_P/(fRV_r)$.
3. Size R_S at maximum load and minimum supply: $R_S = (V_{s,min} - V_Z)/(I_{L,max} + I_{Z,min})$. Check P_Z at no load and maximum supply.
4. $2V_S - V_D$.
5. It decreases, because a wider depletion layer means a larger plate separation.

BJT operation and DC bias I

Answers and worked solutions · Day 06 · Thu 29 Oct · pack E4

5. Practice

From your course files

WS4 Q6 (course question).

- **6.1 (A).** $i_C = I_S e^{v_{BE}/V_T}$: the base–emitter voltage controls the collector current. A PNP has p-type emitter and collector, and a BJT has two junctions, not three.
- **6.2 (C).** In cut-off both junctions are reverse biased. Option C names the emitter-base junction. (A is the same junction written as “base-emitter”, but the official answer is C.)
- **6.3 (D, official).** The official solution reads D as “electrons flow from emitter to base”, which is the correct physics: the forward-biased B–E junction injects electrons from the n-type emitter into the base. On the question sheet, options B and D are both printed as “holes flow from emitter to base”, so D as printed is wrong for an NPN. Answer with the physics: electrons flow from emitter into the base (and only *indirectly* on to the collector, so C is wrong).

WS4 Q7 (course question).

$\beta = \dfrac{\alpha}{1-\alpha}$: $\dfrac{0.5}{0.5} =$ **1**, $\dfrac{0.8}{0.2} =$ **4**, $\dfrac{0.98}{0.02} =$ **49**, $\dfrac{0.999}{0.001} =$ **999**. β grows very fast as α approaches 1.

WS4 Q8 (course question).

1. $I_B = I_C/\beta = 4/50 =$ **0.08 mA**, $I_E = I_C + I_B =$ **4.08 mA**.
2. $\alpha = \beta/(1+\beta) = 50/51 = 0.98$, $I_C = \alpha I_E =$ **3.92 mA**, $I_B = I_E - I_C =$ **0.08 mA**.
3. $I_C = \beta I_B = 50 \times 0.1 =$ **5 mA**, $I_E = I_C + I_B =$ **5.1 mA**.

WS4 Q9 (course question).

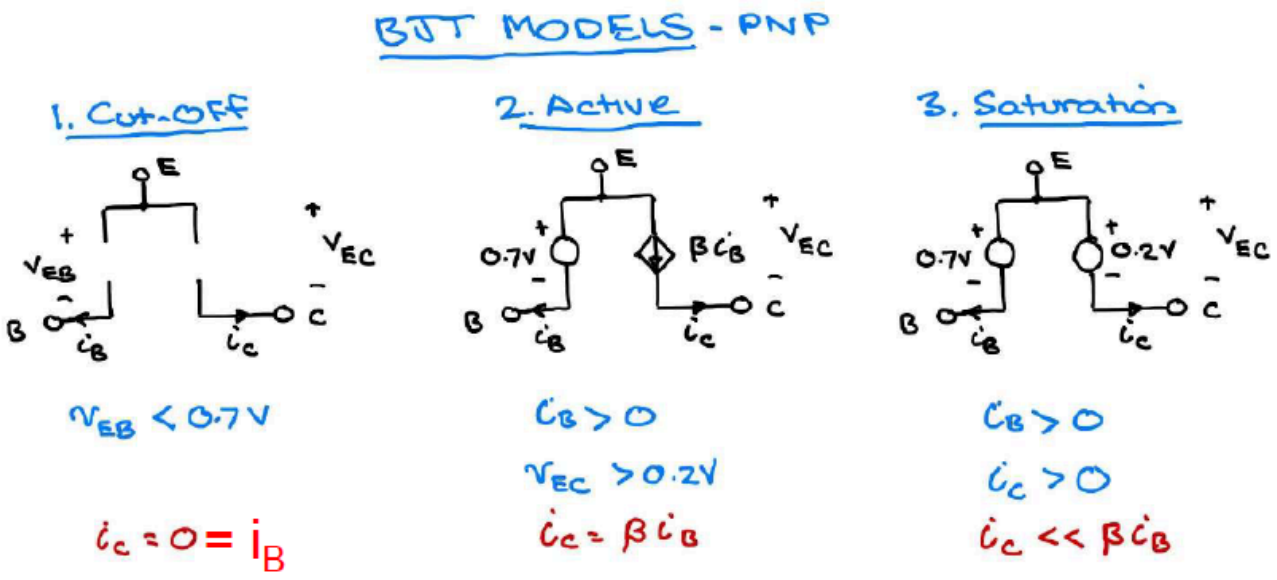
1. $\beta = I_C/I_B = 8/0.1 =$ **80**, $\alpha = \beta/(1+\beta) =$ **0.988**, $I_E = I_C + I_B =$ **8.1 mA**.
2. Part 2 is Worked example 1 above: $I_C =$ **4.9 mA**, α = **0.98**, β = **49**.
3. $I_B = I_E - I_C =$ **0.1 mA**, $\alpha = 10/10.1 =$ **0.9901**, $\beta = 10/0.1 =$ **100**.

WS4 Q10 (course question).

Worked in Worked example 1 above (the table): **cut-off, active, saturation, active.** Compare I_C with βI_B: equal means active, smaller means saturation, and $I_B = 0$ means cut-off.

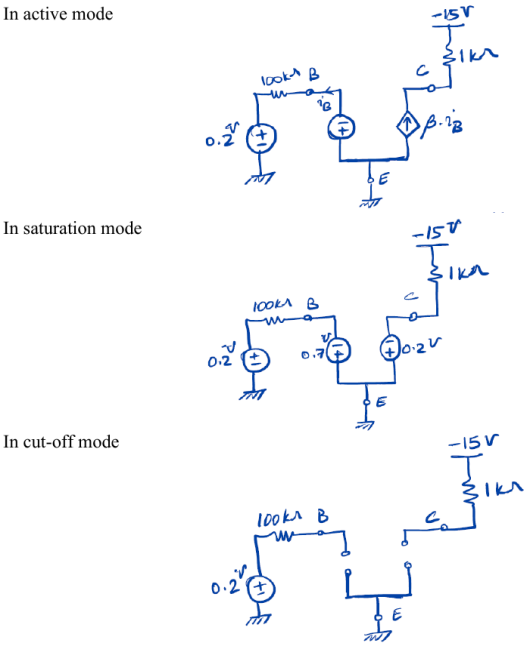
WS4 Q11 (course question).

Use the PNP models from the sheet (Figure 5): active is $V_{EB} = 0.7$ V between E and B plus βi_B from E to C; saturation is 0.7 V (E–B) plus 0.2 V (E–C); cut-off is open between all terminals.



WS4 Figure 5: PNP large-signal models (from the workshop sheet)

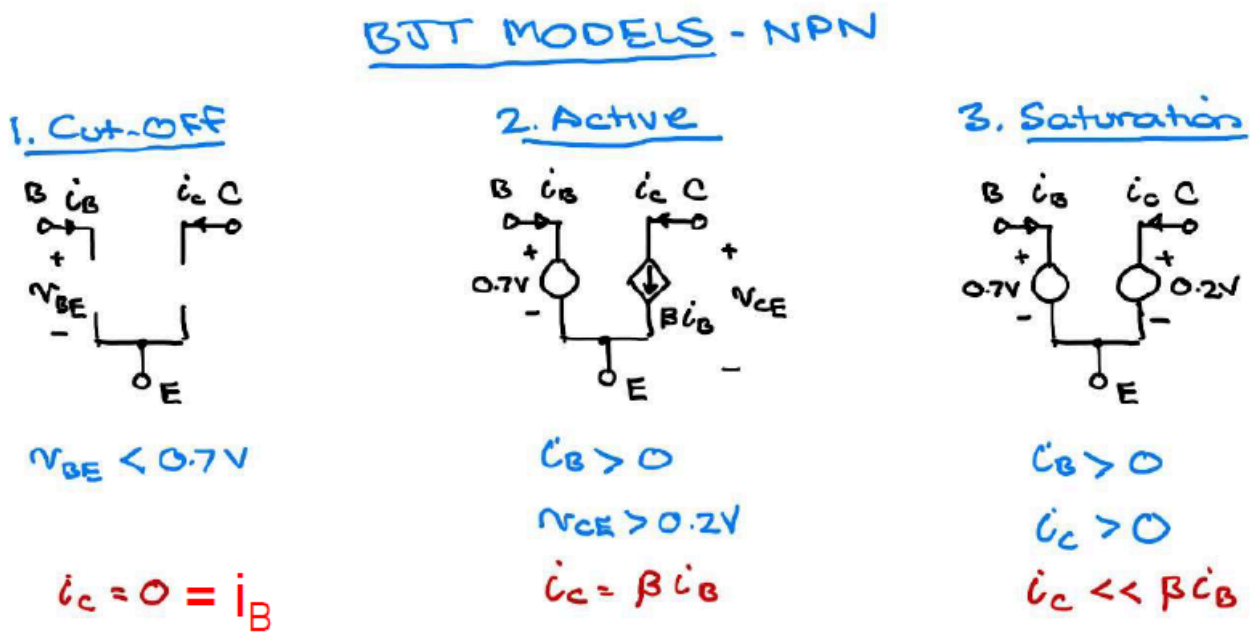
Applied to this circuit (emitter to ground, base through 100 kΩ to 0.2 V, collector through 1 kΩ to −15 V):



WS4 Q11 official redraws in the three modes

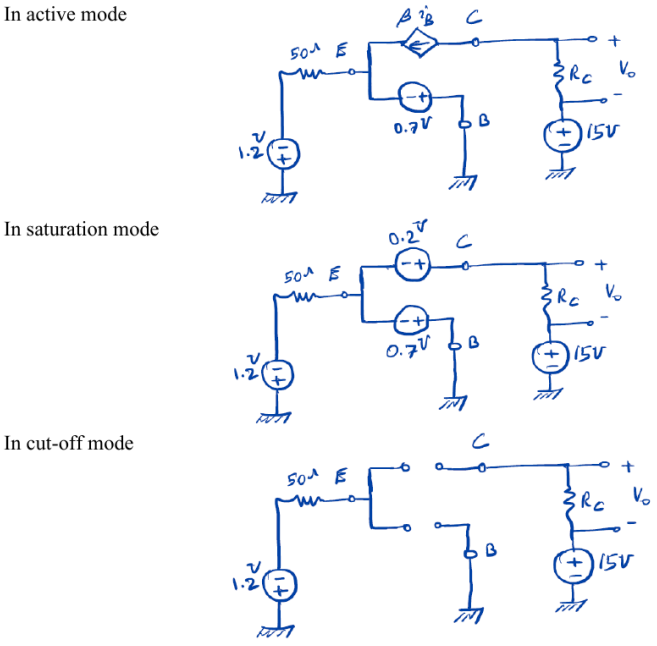
WS4 Q12 (course question).

Use the NPN models from the sheet (Figure 4): active is $V_{BE} = 0.7$ V between B and E plus βi_B from C to E; saturation is 0.7 V (B–E) plus 0.2 V (C–E); cut-off is open.



WS4 Figure 4: NPN large-signal models (from the workshop sheet)

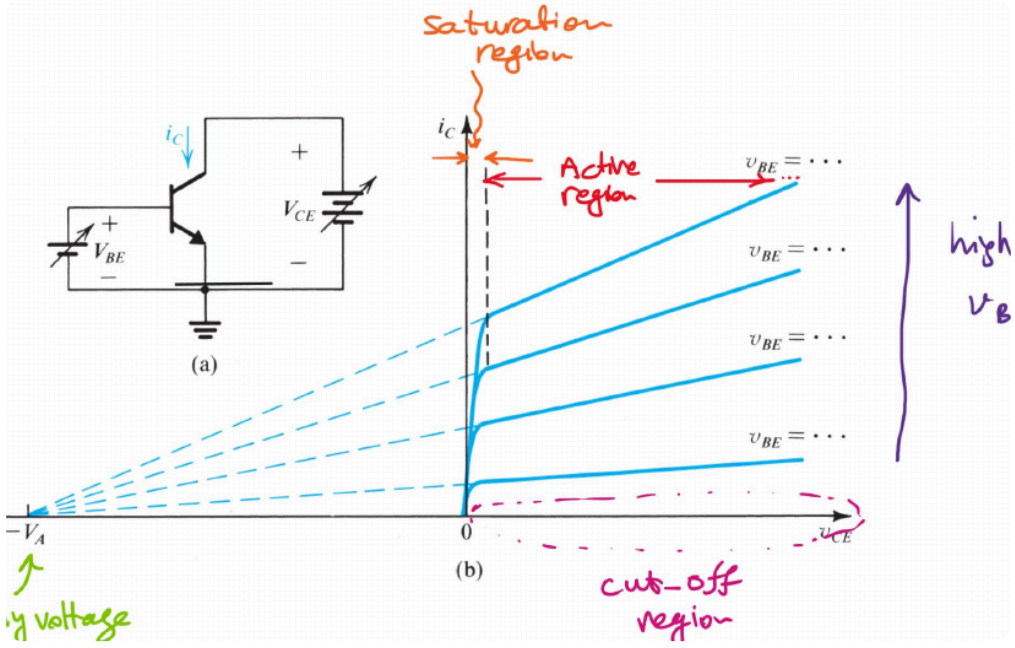
Applied to this common-base circuit (emitter through 50 Ω to 1.2 V, base grounded, collector through R_C to 15 V):



WS4 Q12 official redraws in the three modes

WS5 Q1 (course question).

- **Saturation:** the steep part near $v_{CE} \approx 0$, left of the dashed line. **Active:** the sloped, nearly flat part to the right. **Cut-off:** along the v_{CE} axis, $i_C \approx 0$. **Early voltage:** the point $-V_A$ where the extended active-region lines meet on the v_{CE} axis.
- The non-zero slope means i_C still rises with v_{CE} at fixed v_{BE} (the Early effect, modelled by a resistor r_o in the model).
- Higher curves need higher v_{BE}: v_{BE} increases as i_C increases.



WS5 Q1 official annotated answer

(The question says $i_c - v_{CB}$, but the axes on the figure are i_C against v_{CE}.)

WS5 Q3 (course question).

Hint: no α or β is given, so get I_B from the KVL round the loop that contains the 50 kΩ.

1. $I_E = \dfrac{3 - V_E}{5} =$ **0.4 mA**. Loop +3 V → 5 kΩ → E → B → 50 kΩ → ground: $-3 + 5I_E + 0.7 + 50I_B = 0$, so $I_B = \dfrac{3 - 0.7 - 2}{50} =$ **6 µA**. $I_C = I_E - I_B =$ **0.394 mA**.
2. $V_B = 50I_B =$ **0.3 V** (check: $V_E - V_B = 0.7$ V). $V_C = -3 + 5I_C =$ **−1.03 V**.
3. $\alpha = I_C/I_E =$ **0.985**, $\beta = \alpha/(1-\alpha) =$ **65.7**.

WS5 Q4 (course question).

$R_C = \dfrac{1.5 - V_C}{I_C} = \dfrac{1.5 - 0.5}{0.2} =$ **5 kΩ**. The base is grounded, so $V_E = -0.7$ V.
$\alpha = 100/101 = 0.9901$ and $I_E = I_C/\alpha = 0.202$ mA.

$$R_E = \frac{V_E - (-1.5)}{I_E} = \frac{0.8}{0.202} = \mathbf{3.96\ k\Omega}$$

WS5 Q5 (course question).

1. Emitter loop: $-2.7 + 0.2I_E + 0.7 = 0$, so $I_E = \mathbf{10\ mA}$. $I_C = \alpha I_E = \mathbf{9\ mA}$, $I_B = I_E - I_C = \mathbf{1\ mA}$.
2. Outer loop: $-2.7 + 0.2I_E + V_{EC} + R_C I_C - 20 = 0$, so $R_C = \dfrac{20.7 - V_{EC}}{9}$ (kΩ, mA). The smallest R_C goes with the largest V_{EC}: $R_{C,min} = \dfrac{20.7 - 20}{9} = \mathbf{77.8\ \Omega}$.

WS5 Q6 (course question).

1. Emitter loop: $+0.7 + 0.05I_E - 1.2 = 0$, so $I_E = 10$ mA and $I_C = 0.9 \times 10 = 9$ mA. $R_C = |V_o|/I_C = 4.5/9 = \mathbf{0.5\ k\Omega}$.
2. Outer loop: $-15 + R_C I_C + V_{CE} + 0.05I_E - 1.2 = 0$, so $V_{CE} = 15 - 4.5 - 0.5 + 1.2 = \mathbf{11.2\ V}$.

WS5 Q7 (course question).

Worked in full in Worked example 3 above. The $\beta R_E \geq 10R_2$ test fails (54.4 k < 91 k), so use the exact Thévenin method: (a) **21.4 µA**, (b) **1.71 mA**, (c) **8.15 V**, (d) **9.33 V**, (e) **1.18 V**, (f) **1.88 V**.

WS5 Q8 (course question).

Test: $\beta R_E = 120 \times 1 = 120$ kΩ $\geq 10R_2 = 82$ kΩ, so the approximate approach is allowed.

- e. $V_B = 18 \times \dfrac{8.2}{39 + 8.2} = \mathbf{3.13\ V}$
- d. $V_E = V_B - 0.7 = \mathbf{2.43\ V}$
- a. $I_C \approx I_E = V_E/R_E = \mathbf{2.43\ mA}$
- b. $V_{CE} = 18 - 2.43(3.3 + 1) = \mathbf{7.55\ V}$ (7.56 V unrounded; the official 7.55 V rounds I_C first)
- c. $I_B = I_C/\beta = \mathbf{20.2\ \mu A}$ (official 20.25 µA)

WS5 Q14 (course question).

Thévenin first (this is the carry-on in pack E1, Example 1): $V_{Th} = 20 \times \dfrac{10}{100} = 2$ V, $R_{Th} = 90 \parallel 10 = 9$ kΩ.

Assume active. Base loop: $-2 + 9I_B + 0.7 + 1 \cdot (\beta+1)I_B = 0$, so $I_B = \dfrac{1.3}{9+51} =$ **21.7 µA** and $I_C = 50 I_B =$ **1.08 mA**. $I_E = I_C + I_B = 1.107$ mA.

Collector loop: $V_{CE} = 20 - 5I_C - 1 \cdot I_E = 20 - 5.42 - 1.11 =$ **13.5 V**.

> **CHECK:** $I_B > 0$ and $V_{CE} = 13.5 > 0.2$ V, so the transistor is **active**.

PT Q6 (practice test, 6 marks).

This is emitter-stabilised bias: one base resistor from V_{CC} plus an emitter resistor.

Load line. Output loop: $V_{CE} = V_{CC} - I_C(R_C + R_E)$ with $R_C + R_E = 0.47 + 2.2 = 2.67$ kΩ.

- $I_C = 0$ gives $V_{CE} = 20$ V.
- $V_{CE} = 0$ gives $I_{C,sat} = 20/2.67\ \text{k} = 7.49$ mA.

Join (20 V, 0) to (0, 7.49 mA) with a ruler.

Base current (this decides which curve to use):

$$I_{BQ} = \frac{V_{CC} - V_{BE}}{R_B + (\beta+1)R_E} = \frac{20 - 0.7}{270 + 126(2.2)} = \frac{19.3}{547.2\ \text{k}} = 35.3\ \mu\text{A}$$

(a) Q-point. 35.3 µA lies between the 30 µA and 40 µA curves, about half-way. Near $V_{CE} \approx 8$ V the printed curves sit at about 3.9 mA (30 µA) and 5.05 mA (40 µA). Interpolating gives $I_C \approx 3.9 + 0.53(1.15) \approx 4.5$ mA. The load line passes through 4.5 mA at $V_{CE} = 20 - 4.5(2.67) = 8.0$ V. So read:

$$I_{CQ} \approx 4.5\ \text{mA}, \qquad V_{CEQ} \approx 8.0\ \text{V}$$

Analytic cross-check with β = 125: $I_{CQ} = 125 \times 35.3\ \mu\text{A} = 4.41$ mA, $I_E = 4.44$ mA and $V_{CEQ} = 20 - 4.41(0.47) - 4.44(2.2) = 8.15$ V. These agree with the graph to within reading accuracy. Quote the graph values, because the question asks for the Q-point on the characteristics.

(b) β at the Q-point.

$$\beta_{dc} = \frac{I_{CQ}}{I_{BQ}} \approx \frac{4.5\ \text{mA}}{35.3\ \mu\text{A}} \approx 127$$

This is consistent with the printed β = 125. Write the two numbers you read and then the division.

Cross-check with your Test 1 Reference. The reference has the same load line, I_{BQ} and analytic Q-point (4.41 mA, 8.15 V). For the graph it reads I_{CQ} as roughly 4.2–4.4 mA and gets β ≈ 120. I measured the printed curves from the PDF image: the 30 µA and 40 µA curves sit

at about 3.86 mA and 5.05 mA near 8 V, which puts the crossing nearer 4.5 mA and gives β ≈ 127. Either answer, in the range of about 120–130, should earn the marks if you show the readings. The spacing of the curves, $\Delta I_C/\Delta I_B \approx 1.19\ \text{mA}/10\ \mu\text{A} \approx 119$, is β_{ac}, not what part (b) asks for. The Practice Test pack (EPT) reads the same graph slightly differently (about 4.6 mA, 7.6 V, β ≈ 130). Treat **4.5–4.6 mA, 7.6–8.0 V, β ≈ 127–130** as the graph answer; any reading in that band, with the readings shown, is fine.

PT Q7 (practice test, 10 marks).

(a) Thévenin equivalent seen from the base.

Resistance. Short both sources to ground. R_1 and R_2 are then in parallel:

$$R_{Th} = R_1 \parallel R_2 = \frac{15 \times 30}{45} = 10\ \text{k}\Omega$$

Voltage. With the base open there are two sources, so use superposition:

$$V_{Th} = 9 \cdot \frac{30}{15+30} + (-12) \cdot \frac{15}{15+30} = 6 - 4 = 2\ \text{V}$$

Check by inspection: the base node sits $30/45$ of the way from −12 V up to +9 V, so $-12 + 21 \times \frac{2}{3} = 2$ V. ✓

$V_{Th} = 2$ **V and** $R_{Th} = 10$ **kΩ.**

(b) Redraw. Draw a 2 V source in series with 10 kΩ feeding the base. Between base and emitter, draw a 0.7 V source (+ at the base). Between collector and emitter, draw a current source $\beta I_B = 50 I_B$ pointing from collector to emitter. Connect the emitter to ground (the common node). Connect the collector to 12 V through 1.5 kΩ, and label V_C at the collector and I_B and I_C with arrows. For the marks, label every element value and the current directions.

Practice Test Q7(b) redrawn: Thévenin base source and the active-mode DC model.

(c) Currents and collector voltage. The emitter is grounded, so there is no $(\beta+1)R_E$ term:

$$I_B = \frac{V_{Th} - V_{BE}}{R_{Th}} = \frac{2 - 0.7}{10\ \text{k}} = 130\ \mu\text{A}, \qquad I_C = \beta I_B = 50 \times 130\ \mu\text{A} = \mathbf{6.5}\ \text{mA}$$

$$V_C = V_{CC} - I_C R_C = 12 - 6.5(1.5) = \mathbf{2.25}\ \text{V}$$

Check: $V_{CE} = V_C = 2.25 > 0.2$ V and $I_B > 0$, so the active assumption holds.

This matches your Test 1 Reference exactly. The method is the same as WS5 Q14. The only new step is the superposition for V_{Th}.

New questions

New question 1

a. $\alpha = \beta/(\beta+1) = 99/100 = \mathbf{0.99}$.

b. $I_{CEO} = I_{CBO}/(1-\alpha) = 10\ \text{nA}/0.01 = \mathbf{1}\ \boldsymbol{\mu}\mathbf{A}$, which equals $(\beta+1)I_{CBO}$.

c. $I_C = \beta I_B + I_{CEO} = 99(20\ \mu\text{A}) + 1\ \mu\text{A} = 1.980 + 0.001 = \mathbf{1.981}\ \text{mA}$.
$I_E = I_C + I_B = \mathbf{2.001}\ \text{mA}$. The leakage is 0.05 % of I_C, so it is negligible at room temperature. That is why we normally write $I_C = \beta I_B$. It becomes important at high temperature, because I_{CBO} roughly doubles every 10 °C.

New question 2

a. $I_B = (15 - 0.7)/470\ \text{k} = \mathbf{30.4}\ \boldsymbol{\mu}\mathbf{A}$, $I_C = 100 \times 30.4\ \mu\text{A} = \mathbf{3.04}\ \text{mA}$, and $V_{CE} = 15 - 3.04(3.3) = \mathbf{4.96}\ \text{V}$.

b. The load-line intercepts are $V_{CE} = 15$ V (cut-off) and $I_{C,sat} = 15/3.3\ \text{k} = \mathbf{4.55}\ \text{mA}$. $V_B = 0.7$ V and $V_C = 4.96$ V, so $V_{BC} = 0.7 - 4.96 = -4.26$ V. The CBJ is reverse-biased and the EBJ is forward-biased, so the transistor is **active**. ✓ The Q-point sits at 3.04/4.55 = 67 % of the way up the load line, towards saturation.

New question 3

a. Base loop: $I_B = \dfrac{18 - 0.7}{510 + 101(1.5)} = \dfrac{17.3}{661.5\ \text{k}} = \mathbf{26.2}\ \boldsymbol{\mu}\mathbf{A}$. $I_C = \beta I_B = \mathbf{2.62}\ \text{mA}$ and $I_E = 101 I_B = 2.64$ mA.
$V_{CE} = 18 - 2.615(2.4) - 2.641(1.5) = 18 - 6.28 - 3.96 = \mathbf{7.76}\ \text{V}$.

b. $V_E = I_E R_E = \mathbf{3.96}$ V, $V_B = V_E + 0.7 = \mathbf{4.66}$ V, $V_C = 18 - 6.28 = \mathbf{11.72}$ V, and $V_{BC} = 4.66 - 11.72 = \mathbf{-7.06}$ V. The CBJ is reverse-biased, so the transistor is active. ✓

c. $I_{C,sat} = V_{CC}/(R_C + R_E) = 18/3.9\ \text{k} = \mathbf{4.62}\ \text{mA}$.

Note that the reflected emitter resistance $(\beta+1)R_E = 151.5$ kΩ is a significant part of the 661.5 kΩ in the base loop.

New question 4

a. $\beta R_E = 150 \times 1.2 = 180$ kΩ and $10R_2 = 100$ kΩ. Since 180 ≥ 100, the approximation is **valid**.

b. $V_B = 15 \times 10/57 = 2.632$ V and $V_E = 1.932$ V.
$I_C \approx I_E = 1.932/1.2 = \mathbf{1.61}$ mA. $V_{CE} \approx 15 - 1.61(2.7 + 1.2) = \mathbf{8.72}$ V.
$I_B = 1.61/150 = \mathbf{10.7}\ \boldsymbol{\mu}\mathbf{A}$.

c. $R_{Th} = 47 \parallel 10 = 8.25$ kΩ and $E_{Th} = 2.632$ V.
$I_B = \dfrac{2.632 - 0.7}{8.25 + 151(1.2)} = \dfrac{1.932}{189.5\ \text{k}} = \mathbf{10.2}\ \boldsymbol{\mu}\mathbf{A}$. $I_C = 150 I_B = \mathbf{1.53}$ mA and $I_E = 1.54$ mA. $V_{CE} = 15 - 1.529(2.7) - 1.539(1.2) = \mathbf{9.02}$ V. The approximation over-estimates I_C by $(1.610 - 1.529)/1.529 = \mathbf{5.2}\ \%$. Even when the test passes, expect a few percent of error.

New question 5

a. $\beta R_E = 60 \times 0.56 = 33.6$ kΩ and $10 R_2 = 120$ kΩ. Since 33.6 < 120, the approximation is **not valid**.

b. Approximate: $V_B = 12 \times 12/45 = 3.2$ V, $V_E = 2.5$ V, and $I_C \approx 2.5/0.56 = 4.46$ mA. Then $V_{CE} = 12 - 4.46(2.76) = -0.32$ V. A negative V_{CE} is impossible in active mode. The student would wrongly conclude that the transistor is saturated. Also, 4.46 mA exceeds $I_{C,sat}$ (part d), which no active transistor can do. The base current is not negligible here, so the divider is heavily loaded and V_B is really much lower than 3.2 V.

c. Exact: $R_{Th} = 33 \parallel 12 = 8.8$ kΩ and $E_{Th} = 3.2$ V.

$$I_B = \frac{3.2 - 0.7}{8.8 + 61(0.56)} = \frac{2.5}{42.96\ \text{k}} = \mathbf{58.2}\ \boldsymbol{\mu}\mathbf{A}$$

$I_C = 60 I_B = \mathbf{3.49}$ mA and $I_E = 61 I_B = 3.55$ mA.
$V_{CE} = 12 - 3.492(2.2) - 3.550(0.56) = \mathbf{2.33}$ V. $V_E = \mathbf{1.99}$ V, $V_B = \mathbf{2.69}$ V (compare 3.2 V unloaded), and $V_C = 12 - 7.68 = \mathbf{4.32}$ V. Mode: $V_{CE} = 2.33 > 0.2$ V and $V_{BC} = 2.69 - 4.32 = -1.63$ V, so the transistor is **active**. The approximation was 28 % high on I_C.

d. $I_{C,sat} = 12/(2.2 + 0.56) = \mathbf{4.35}$ mA. The Q-point is at $I_C/I_{C,sat} = 3.49/4.35 = 80\ \%$ of the way up the load line, with $V_{CE} = 2.33$ V out of 12 V. It is well towards saturation, which is a poor bias point for symmetrical signal swing.

New question 6

a. $V_{CC} = 24$ V (the cut-off intercept). $R_C + R_E = V_{CC}/I_{C,sat} = 24/8$ mA $= \mathbf{3\ k\Omega}$, so $R_E = 3 - 2.4 = \mathbf{0.6\ k\Omega}$.

b. $V_{CEQ} = 24 - 4.2(3) = \mathbf{11.4}$ V and $\beta_{dc} = 4.2$ mA$/30\ \mu$A $= \mathbf{140}$.

c. $I_E = (\beta + 1) I_B = 141 \times 30\ \mu\text{A} = 4.23$ mA, so $V_E = 4.23(0.6) = 2.54$ V. Then

$$R_B = \frac{V_{CC} - V_{BE} - V_E}{I_B} = \frac{24 - 0.7 - 2.54}{30\ \mu\text{A}} = \mathbf{692\ k\Omega}$$

The nearest E12 value is **680 kΩ**. Then $I_B = \dfrac{23.3}{680 + 141(0.6)} = \dfrac{23.3}{764.6\ \text{k}} = \mathbf{30.5\ \mu A}$, which is 1.6 % above the target. That is fine.

6. Quick recall check

Question

1. $\beta = 0.995/0.005 = 199$ and $\alpha = 150/151 = 0.9934$.
2. $V_{BE} = 0.7$ V, so the EBJ is forward-biased. $V_{CE} = 0.2$ V, which means $V_{BC} = +0.5$ V and the CBJ is also forward-biased. Both forward means **saturation**.
3. $I_B = \dfrac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E}$.
4. $\beta R_E \geq 10 R_2$. The resistance looking into the base (about βR_E) is then large compared with R_2, so the base current does not load the divider and $V_B \approx R_2 V_{CC}/(R_1 + R_2)$.
5. $(V_{CE}, I_C) = (V_{CC},\ 0)$ and $(0,\ V_{CC}/(R_C + R_E))$.

BJT DC bias II

Answers and worked solutions · Day 08 · Sat 31 Oct · pack E5

5. Practice

From your course files

WS5 Q12 (course question).

Part 1. The base current is fixed by the source: I_B = 0.01 mA.

Guess active. $I_C = \beta I_B = 150(0.01) = 1.5$ mA and $I_E = I_C + I_B = 1.51$ mA. Collector loop:

$$V_{CE} = 12 - 9I_C - 1 \cdot I_E = 12 - 13.5 - 1.51 = -3.01 \text{ V}$$

Negative, so not active.

Guess saturation. V_{CE} = 0.2 V, and I_B is still 0.01 mA, so

$$I_E - I_C = 0.01 \quad (A), \qquad 9I_C + 1 \cdot I_E = 11.8 \quad (B)$$

Solving: I_C = **1.179 mA**, I_E = **1.189 mA**.

Check: $I_B, I_C > 0$ and $I_C = 1.179 < \beta I_B = 1.5$ mA (the solution writes this as $I_B = 0.01 > I_C/\beta = 0.00786$ mA). **Saturated**, forced β = 117.9.

Part 2. At the edge of saturation $I_C = \beta I_B$ and $V_{CE} = 0.2$ V, which gives $I_B[\beta(9 + R_E) + R_E] \geq 11.8$. The worst case is $\beta_{\min}$ = 100: $101R_E \geq 1180 - 900$, so $R_E \geq$ **2.77 kΩ**. The full argument, and why the reverse question ("always active") has no answer, is in Example 4 above.

WS5 Q13 (course question).

The emitter is grounded, so in saturation $V_C = V_{CE,sat} = 0.2$ V and

$$I_{C,sat} = \frac{5 - 0.2}{1 \text{ k}\Omega} = 4.8 \text{ mA}$$

At the edge of saturation $I_B = I_{C,sat}/\beta$.

Course answer (uses $\beta_{\max}$ = 200): $I_{B,\min} = 4.8/200$ = **24 µA**, and the base loop gives $V_{BB} = 0.7 + 100(0.024)$ = **3.1 V**.

Read this carefully. 24 µA is the current at which a β = 200 device *starts* to saturate. A β = 100 device needs $4.8/100$ = **48 µA**, which is $V_{BB} = 0.7 + 100(0.048)$ = **5.5 V**. To

guarantee saturation for every device in the range, give 48 µA and 5.5 V. In a test, give the course's number and add this line, saying which one you mean.

WS5 Q17 (course question).

V_{CE} = 1 V > 0.2 V, so the BJT is active.

Base loop (10 V, 80 kΩ, V_{BE}, 5 kΩ): the 80 kΩ goes straight from the 10 V rail to the base, so the base loop does not involve R_C:

$$I_B = \frac{10 - 0.7}{80 + 5(101)} = 15.9\ \mu\text{A}, \qquad I_E = 101 I_B = 1.606\ \text{mA}, \qquad I_C = 100 I_B = 1.590\ \text{mA}$$

Collector side. Thevenin the collector network (R_C from 10 V, 100 kΩ to ground): $V_{Th} = 10 \cdot \dfrac{100}{100 + R_C}$ and $R_{Th} = \dfrac{100 R_C}{100 + R_C}$. With $V_C = V_{CE} + 5 I_E = 1 + 8.03 = 9.03$ V:

$$V_{Th} - R_{Th} I_C = 9.03 \ \Rightarrow\ 1000 - 158.97 R_C = 9.03(100 + R_C) \ \Rightarrow\ R_C = \frac{97.2}{168.0} = \mathbf{0.58\ k\Omega}$$

The course solution prints **0.59 kΩ**, because it rounds V_C = 9.03 V to 9 V in the last line ($100 = 9 R_C + 158.9 R_C$). Either is fine; 0.58 kΩ is the unrounded value.

WS5 Q18 (course question).

V_{CE} = 6 V, so the BJT is active. $I_E = I_C/\alpha = 2.030$ mA and $I_B = I_E - I_C$ = 0.030 mA (β = α/(1−α) = 65.7).

$$V_B = V_{BE} + 2 I_E = 0.3 + 2(2.030) = 4.36\ \text{V}$$

Course method (divider, base current ignored): $4.36 = 16 \cdot \dfrac{20}{20 + R_1}$, so R_1 **= 53.4 kΩ**.

Collector loop: $16 = 2 R_C + 6 + 2 I_E$, so $R_C = (16 - 6 - 4.06)/2$ = **2.97 kΩ**.

Why the course's R_1 is only approximate. βR_E = 65.7 × 2 kΩ = 131 kΩ, which is less than $10 R_2$ = 200 kΩ, so the "ignore I_B" approximation fails its own test. Exactly, R_1 carries $V_B/20\text{k} + I_B$ = 0.218 + 0.030 = 0.248 mA, so $R_1 = (16 - 4.36)/0.248$ = **46.8 kΩ**. If you are asked this in a test, give the exact value and say why.

WS5 Q19 (course question).

V_{CE} = 2 V, so active: $I_B = I_C/\beta$ = 0.01 mA and I_E = 1.01 mA.

Collector feedback: R_C carries $I_C + I_B = I_E$. The emitter is grounded, so $V_C = V_{CE}$ = 2 V:

$$R_C = \frac{15 - 2}{1.01} = 12.87\ \text{k}\Omega$$

(The question asks only for R_B, but you need R_C first. The course solution finds both.)

Base path: $15 - R_C I_E - R_B I_B - V_{BE} = 0$, so

$$R_B = \frac{15 - 0.3 - 12.87(1.01)}{0.01} = \mathbf{170\ k\Omega}$$

Quicker: $R_B = (V_C - V_{BE})/I_B = (2 - 0.3)/0.01$ = 170 kΩ.

WS5 Q9 (course question, *).

The collector is tied to +20 V, so V_C = 20 V. Guess active.

Base loop: $-20 + 10I_B + 0.7 + 1(\beta + 1)I_B = 0$, so

$$I_B = \frac{19.3}{10 + 101} = \mathbf{0.174\ mA}$$

$V_B = 20 - 10(0.174) = 18.26$ V and $V_E = V_B - 0.7 = 17.56$ V, so $V_{CE} = 20 - 17.56$ = **2.44 V**.

Check: $I_B > 0$ and V_{CE} > 0.2 V, so **active**. (I_C = 17.4 mA, I_E = 17.56 mA.)

WS5 Q11 (course question, *).

Guess active. Base-emitter loop: $V_B - 0.7 - 1 \cdot I_E = 0$, so $I_E = (0 - 0.7)/1 = -0.7$ mA. With very high β, $I_C \approx I_E = -0.7$ mA, so I_B would also be negative. (The collector loop gives $V_{CE} = 3 - 1(-0.7) - 1(-0.7) = 4.4$ V, but that does not rescue it.) $I_B < 0$, so not active.

Guess cut-off. $I_B = I_C = I_E = 0$, so V_E = 0 and $V_{BE} = 0 - 0 = 0$ V < 0.7 V. **Cut-off.** (V_C = 3 V.)

WS5 Q14 (course question, *).

Thevenin the base: $V_{Th} = 20 \cdot \frac{10}{100}$ = 2 V and R_{Th} = 90 ∥ 10 = 9 kΩ. Guess active:

$$I_B = \frac{2 - 0.7}{9 + 51(1)} = \mathbf{21.7\ \mu A}, \qquad I_C = 50 I_B = \mathbf{1.08\ mA}, \qquad I_E = 1.105\ \text{mA}$$

$$V_{CE} = 20 - 5I_C - 1 \cdot I_E = 20 - 5.42 - 1.11 = \mathbf{13.5\ V}$$

$I_B > 0$ and V_{CE} > 0.2 V, so **active**.

WS5 Q15 (course question, *).

Two Thevenins.

- Base: $V_{Th,b} = V_I \cdot \dfrac{40}{400} = 0.1V_I$ = 1.5 V, and $R_{Th,b}$ = 360 ∥ 40 = 36 kΩ.
- Collector (superposition of +5 V and −10 V): $V_{Th,c} = 5 \cdot \dfrac{27}{30} - 10 \cdot \dfrac{3}{30}$ = 3.5 V, and $R_{Th,c}$ = 3 ∥ 27 = 2.7 kΩ.

Guess active (emitter grounded):

$$I_B = \frac{1.5 - 0.7}{36} = 22.2\ \mu\text{A}, \qquad I_C = 40I_B = 0.889\ \text{mA}$$

$$V_o = V_C = 3.5 - 2.7(0.889) = \mathbf{1.10\ V}$$

$V_{CE} = V_o$ = 1.10 V > 0.2 V and $I_B > 0$, so **active**.

The course solution gives 1.124 V because it rounds I_B to 22 µA (I_C = 0.88 mA). Either is fine.

PT Q8 (practice test, 6 marks).

READING THE FIGURE. Q_1 is an npn whose base and collector are joined at the 2.5 V terminal. Its emitter connects directly to the emitter of Q_2, a pnp. Q_2's base goes through 10 kΩ to the 1 V terminal, and its collector is the output V_{C2}, with 1 kΩ to ground.

Step 1, Q_1. With base tied to collector, $V_{CB1} = 0$, so Q_1 is diode-connected and always active. It drops one V_{BE}:

$$V_{E1} = 2.5 - 0.7 = 1.8\ \text{V} = V_{E2}$$

Step 2, Q_2 base (pnp). Emitter above base:

$$V_{B2} = V_{E2} - V_{EB2} = 1.8 - 0.7 = 1.1\ \text{V}$$

Step 3, base current. For a pnp the base current flows out of the base, from 1.1 V through 10 kΩ to the 1 V terminal:

$$I_{B2} = \frac{1.1 - 1.0}{10\ \text{k}\Omega} = 10\ \mu\text{A}$$

Step 4, collector. $I_{C2} = \beta_2 I_{B2} = 50 \times 10\ \mu\text{A}$ = 0.5 mA. It flows out of the collector, down through 1 kΩ to ground:

$$V_{C2} = 0.5\ \text{mA} \times 1\ \text{k}\Omega = \mathbf{0.5\ V}$$

Step 5, checks. $V_{EC2} = 1.8 - 0.5 = 1.3$ V > 0.2 V and I_{B2} > 0, so Q_2 is active. $I_{E2} = 51 \times 10\ \mu\text{A}$ = 0.51 mA. That current flows out of Q_1's emitter, so I_{B1} = 0.51/101 =

5.05 µA and I_{C1} = 0.505 mA, both drawn from the 2.5 V terminal. These are consistent but play no part in V_{C2}. **β1 does not affect the answer**, because Q_1 only fixes the 1.8 V level. Say this in one line. It agrees with the Test 1 Reference.

PT Q7 (practice test, 10 marks).

(a) With the base disconnected, the divider runs from +9 V to −12 V:

$$V_{Th} = 9 \cdot \frac{30}{45} + (-12) \cdot \frac{15}{45} = 6 - 4 = \mathbf{2\ V}, \qquad R_{Th} = 15 \parallel 30 = \mathbf{10\ k\Omega}$$

(b) The drawing: V_{Th} = 2 V in series with R_{Th} = 10 kΩ into the base; a 0.7 V source from base to the grounded emitter; a current source βI_B from collector to emitter; R_C = 1.5 kΩ from V_{CC} = 12 V to the collector.

(c) Emitter grounded, so

$$I_B = \frac{2 - 0.7}{10} = 0.130\ \text{mA}, \qquad I_C = 50 I_B = \mathbf{6.5\ mA}, \qquad V_C = 12 - 1.5(6.5) = \mathbf{2.25\ V}$$

$V_{CE} = 2.25$ V > 0.2 V, so the active assumption holds. This matches the Test 1 Reference.

The approximate method, compared with the exact one.

Step 1, validity test. The approximate method needs $\beta R_E \geq 10 R_2$. Here $R_E = 0$, so $\beta R_E = 0 < 10 \times 30$ kΩ = 300 kΩ. **The test fails completely.**

Step 2, what happens if you try anyway. The approximate method takes $V_B = V_{Th}$ = 2 V, so $V_E = 2 - 0.7 = 1.3$ V. But the emitter is grounded, so V_E must be 0. With no R_E, the "1.3 V across R_E" step would give $I_E = 1.3/0 \to \infty$. The method has nothing to work with. It relies on R_E to set the current, and this circuit has none.

Step 3, why, with numbers from the exact solution. The base is clamped at $V_B = V_{BE}$ = 0.7 V, not 2 V:

- current in R_1: $(9 - 0.7)/15$ = 0.553 mA
- current in R_2: $(0.7 + 12)/30$ = 0.423 mA
- base current: 0.553 − 0.423 = **0.130 mA**, which agrees with the exact method.

The base current is 24 % of the divider current, so it is far from negligible. That is exactly what the $\beta R_E \geq 10 R_2$ test is designed to detect.

Comparison where the approximate method does give a number (WS5 Q7, the divider-bias circuit from E4): βR_E = 54.4 kΩ < $10 R_2$ = 91 kΩ, so the test fails, but only mildly.

	Exact	Approximate	Error
I_C	1.71 mA	1.98 mA ($I_E = (2.048 - 0.7)/0.68$)	+16 %
V_{CE}	8.15 V	6.92 V	−15 %

For the lecture example (22 V, 39 kΩ / 3.9 kΩ, R_E = 1.5 kΩ, β = 100), where the test passes, the approximate method gives I_C = 0.867 mA and V_{CE} = 12.03 V. The exact method gives 0.838 mA and 12.36 V (slide 33 prints 0.84 mA and 12.34 V). The error is about 3 %.

Conclusion to write: the approximate method is only as good as $\beta R_E/R_2$ is large. It cannot be used at all without an emitter resistor, so practice test Q7 must be done with Thevenin.

New questions

New Q1 [5 marks].

(a) Active.

$$I_B = \frac{5 - 0.7}{47 + 121(0.5)} = \frac{4.3}{107.5} = 40.0\ \mu\text{A}, \qquad I_C = 120 I_B = 4.80\ \text{mA}, \qquad I_E = 4.84\ \text{mA}$$

$$V_{CE} = 9 - 4.80(2.2) - 4.84(0.5) = -3.98\ \text{V}$$

This is negative, so the transistor is not active.

(b) Saturation. Set V_{CE} = 0.2 V and $I_E = I_B + I_C$. Because R_E is shared, the two loops are coupled:

- Base loop: $5 = 47 I_B + 0.7 + 0.5(I_B + I_C)$, so $47.5 I_B + 0.5 I_C = 4.3$.
- Collector loop: $9 = 2.2 I_C + 0.2 + 0.5(I_B + I_C)$, so $0.5 I_B + 2.7 I_C = 8.8$.

Solving: I_B **= 56.3 µA and** I_C **= 3.25 mA. Check:** both are positive, and I_C = 3.25 mA < βI_B = 6.76 mA. **Saturated.** The forced β is 3.25/0.0563 = **57.7**.

I_B changed from attempt 1 to attempt 2 because R_E links the base loop to the collector current. The shortcut "I_B unchanged" only works when the emitter is grounded.

New Q2 [6 marks] (pnp).

(a) I_E flows from +10 V through R_E *into* the emitter. I_B flows *out* of the base, down through R_B to ground. I_C flows *out* of the collector, down through R_C to ground. KVL from +10 V through R_E, the emitter-base junction and R_B to ground:

$$10 - 2I_E - 0.7 - 220 I_B = 0, \quad I_E = 81 I_B \Rightarrow I_B = \frac{9.3}{220 + 81(2)} = \frac{9.3}{382} = \mathbf{24.3\ \mu A}$$

$$I_C = 80 I_B = \mathbf{1.948\ mA}, \qquad I_E = 1.972\ \text{mA}$$

(b) $V_E = 10 - 2(1.972) = 6.06$ V. $V_B = V_E - 0.7 = 5.36$ V, which equals 220×0.0243, a useful check. $V_C = 2.2 \times 1.948 = 4.28$ V. $V_{EC} = 6.06 - 4.28 = \mathbf{1.77\ V} > 0.2$ V and $I_B > 0$, so **active**.

(c) At the edge, $V_C = V_E - 0.2 = 5.86$ V with the same I_C, because the base loop does not involve R_C. So $R_{C,\max} = 5.86/1.948 = \mathbf{3.01\ k\Omega}$.

New Q3 [7 marks] (sensitivity).

(a) $I_B = 4.3/100$ = 43 μA for both values of β, because the emitter is grounded.

- β = 100: I_C = 4.30 mA and $V_{CE} = 12 - 4.3(2.2)$ = 2.54 V. **Active.**
- β = 200: in the active model I_C = 8.6 mA, which gives V_{CE} = −6.92 V, so the transistor is not active. In saturation, $I_C = 11.8/2.2$ = **5.36 mA** (< βI_B = 8.6 mA). **Saturated.**

(b) The worst case is $\beta_{\max}$ = 200, because it gives the most collector current. Active requires

$$V_{CE} = 12 - I_B\left[\beta R_C + (\beta+1)R_E\right] \geq 0.2, \qquad I_B = \frac{4.3}{100 + (\beta+1)R_E}$$

Multiplying out: $11.8[100 + 201R_E] \geq 4.3[440 + 201R_E]$, so $201R_E(7.5) \geq 1892 - 1180 = 712$. That gives $R_E \geq$ **0.472 kΩ**.

(c) The E12 value 470 Ω is just *below* the limit. At β = 200 it gives V_{CE} = 0.18 V, which is saturated. So choose **560 Ω**:

β	I_B	I_C	V_{CE}
100	27.5 μA	2.75 mA	4.40 V
200	20.2 μA	4.05 mA	0.82 V

I_C now changes by 47 % over the range instead of 100 %. The circuit is still not well stabilised, because $(\beta+1)R_E$ is only about the same size as R_B. That is why practical designs use a divider (New Q5).

New Q4 [8 marks] (two transistors, npn into pnp).

(a) $\beta_1 R_{E1}$ = 150 kΩ ≥ $10R_2$ = 120 kΩ, so the approximate method is valid. Exact: R_{Th} = 56 ∥ 12 = 9.88 kΩ and E_{Th} = 12(12/68) = 2.118 V.

$$I_{B1} = \frac{2.118 - 0.7}{9.88 + 151(1)} = 8.81\ \mu\text{A}, \qquad I_{C1} = \mathbf{1.322\ mA}, \qquad I_{E1} = 1.331\ \text{mA}$$

(b) $V_{C1} = 12 - 1.322(4.7)$ = 5.79 V. For the pnp, $V_{E2} = V_{B2} + 0.7$ = 6.49 V, and $I_{E2} = (12 - 6.49)/2.2$ = 2.51 mA.

(c) I_{B2} flows *out* of the pnp base into node C1. KCL at C1: (current down through R_{C1}) + I_{B2} = I_{C1}, so R_{C1} carries $I_{C1} - I_{B2}$. With $I_{B2} = I_{E2}/101$ and $I_{E2} = (12 - V_{C1} - 0.7)/2.2$:

$$V_{C1} = 12 - 4.7\left(1.322 - \frac{11.3 - V_{C1}}{222.2}\right) \Rightarrow V_{C1} = \mathbf{5.90\ V}$$

Then V_{E2} = 6.60 V, I_{E2} = 2.454 mA, I_{B2} = 24.3 µA and $I_{C2} = 100 I_{B2}$ = 2.429 mA.

$$V_{C2} = 2.429 \times 1.8 = \mathbf{4.37\ V}$$

Checks: Q_2: V_{EC2} = 6.60 − 4.37 = 2.23 V > 0.2 V, so active. Q_1: V_{E1} = 1.33 V and V_{CE1} = 5.90 − 1.33 = 4.57 V > 0.2 V, so active. Neglecting I_{B2} changed V_{C1} by only 0.11 V, because I_{B2} is 2 % of I_{C1}.

(d) At the edge, $V_{C2} = V_{E2} - 0.2$ = 6.40 V with the same I_{C2}, so $R_{C2,\max}$ = 6.40/2.429 = **2.64 kΩ**. With a 3.3 kΩ resistor, Q_2 would saturate.

New Q5 [8 marks] (design).

(a) V_E = 1.5 V. $R_E = V_E/I_E \approx 1.5/2$ = **0.75 kΩ**. $R_C = (15 - 6 - 1.5)/2$ = **3.75 kΩ**.

(b) $R_2 \le \beta_{\min} R_E/10 = 100(0.75)/10$ = **7.5 kΩ**. $V_B = 1.5 + 0.7$ = 2.2 V, so $R_1 = R_2(V_{CC} - V_B)/V_B = 7.5(12.8)/2.2$ = **43.6 kΩ**.

(c) Choose R_1 = 43 kΩ, R_2 = 7.5 kΩ, R_E = 750 Ω and R_C = 3.6 kΩ. That gives R_{Th} = 6.39 kΩ and E_{Th} = 15(7.5/50.5) = 2.228 V.

β	$I_B = \dfrac{2.228 - 0.7}{6.39 + (\beta+1)0.75}$	I_{CQ}	$V_{CEQ} = 15 - I_C(3.6) - I_E(0.75)$
100	18.6 µA	**1.86 mA**	**6.90 V**
300	6.58 µA	**1.97 mA**	**6.41 V**

Tripling β changes I_C by only 6 %. The 3.9 kΩ standard value would also work: 6.34 V and 5.81 V. Both sit close to the 6 V target.

New Q6 [6 marks] (collector feedback, β range).

(a) R_C carries I_E. Loop from 12 V through R_C, R_F, V_{BE} and R_E:

$$I_B = \frac{12 - 0.7}{180 + (\beta+1)(3.3 + 0.5)}, \qquad V_C = 12 - 3.3 I_E, \qquad V_{CE} = 12 - 3.8 I_E$$

β	I_B	I_C	V_C	V_{CE}
80	23.2 µA	1.85 mA	5.81 V	4.87 V
160	14.3 µA	2.28 mA	4.42 V	3.27 V
240	10.3 µA	2.47 mA	3.80 V	2.56 V

All three are active. Tripling β raises I_C by 34 %.

(b) $I_B = 2.283/160$ = 14.27 µA, so $R_B = 11.3/0.01427$ = **792 kΩ**. At β = 240, $I_C = 240(11.3/792)$ = 3.43 mA and $V_{CE} = 12 - 3.43(3.3)$ = **0.70 V**, which is almost saturated. At β = 80 it is 1.14 mA and 8.23 V. With fixed bias, I_C rises in proportion to β (×3). With collector feedback, a rising I_C lowers V_C, which reduces I_B. See the figure in Section 1.3.

6. Quick recall check

Question

1. Saturation. Use $V_{CE} = 0.2$ V (with $V_{BE} = 0.7$ V) and solve the base and collector loops together with $I_E = I_B + I_C$. Then check that $I_C < \beta I_B$.
2. $\beta_{\max}$ = 200. It gives the largest I_C and so the smallest V_{CE}, which is the case closest to saturation.
3. $I_E = I_C + I_B$, because R_F's current (= I_B) also leaves the collector node through R_C.
4. V_B = 2.3 V. I_B and I_C both flow *out* of the transistor; I_E flows in.
5. $V_E \approx 0.1 V_{CC}$ (so $R_E = V_E/I_C$), and $R_2 \leq \beta_{\min} R_E/10$.

EDC · PAPER

EDC practice test: marking guide

Answers and worked solutions · Day 08 · Sat 31 Oct · pack EPT

Question-by-question marking guide

Question 1: multiple choice [5 marks]

Paper wording. Circle the correct answer in the following multiple-choice questions (I to V).

I. When a PN junction is formed, why does a depletion region develop near the junction?
a) Electrons and holes recombine near the junction, leaving behind ionized dopants
b) The semiconductor becomes electrically insulating everywhere
c) All free electrons move from the N-region into the P-region permanently
d) The applied voltage removes all charge carriers from the semiconductor

II. The small-signal or dynamic resistance of a diode is related to:
a) The slope of the I–V characteristic at the operating point
b) The total length of the diode leads
c) The reverse breakdown voltage only
d) The DC supply resistance only

III. The main advantage of full-wave rectification over half-wave rectification is that full-wave rectification:
a) Uses only the negative half of the input waveform
b) Uses both half-cycles of the input waveform
c) Completely eliminates ripple without a capacitor
d) Requires no semiconductor devices

IV. Why is the base region of a BJT made relatively thin?
a) To allow most injected carriers to cross the base and reach the collector
b) To prevent carriers from reaching the collector
c) To increase the base resistance as much as possible
d) To eliminate the depletion region at the collector-base junction

V. Why are DC and AC analyses of a transistor amplifier often performed separately?
a) A transistor can operate with either DC or AC but never both
b) DC analysis establishes the operating point, while AC analysis examines small variations around that point
c) AC analysis determines the transistor’s doping concentration
d) DC analysis is only needed when the transistor is damaged

Part	Answer	One-line reason
I	**(a)**	Carriers diffuse across and recombine near the junction, leaving immobile ionised donors (+, n side) and acceptors (−, p side). That bound charge is the depletion region.
II	**(a)**	$r_d = \Delta V_D / \Delta I_D = nV_T / I_D$, the inverse slope of the i–v curve at the Q-point.
III	**(b)**	It uses both half-cycles. That gives a higher DC output and twice the ripple frequency. It does *not* remove ripple without a capacitor, so (c) is wrong.
IV	**(a)**	Injected carriers cross the thin base before they recombine, so most reach the collector. That is why $\alpha \to 1$ and β is large.
V	**(b)**	DC analysis sets the Q-point. AC analysis linearises the device around that point (small-signal model).

This agrees with the Test 1 Reference.

Mark breakdown: 1 mark per correct letter. There are no part marks.

Most likely ways to lose marks:

- Changing (I) to (c) because “electrons move to the p side” sounds right. Diffusion stops when the field balances it, and it is not “all” electrons, “permanently”.
- Picking (c) in III because full-wave ripple is *smaller*. Smaller is not the same as eliminated.

Question 2: exponential model [5 marks]

Paper wording. Consider a silicon diode with its exponential model and $n = 1.5$. Find the change in the voltage across it when the diode operates at 27°C and the current changes from 0.1 mA to 10 mA.

Step 1: thermal voltage. $T = 273 + 27 = 300$ K.

$$V_T = \frac{kT}{q} = \frac{1.38 \times 10^{-23} \times 300}{1.6 \times 10^{-19}} = 25.875 \text{ mV}$$

Step 2: ratio form (removes the unknown I_S). For forward bias, $I_D \approx I_S e^{V_D/nV_T}$, so

$$\frac{I_2}{I_1} = e^{(V_2 - V_1)/nV_T} \Rightarrow \Delta V_D = nV_T \ln \frac{I_2}{I_1}$$

Step 3: substitute. $I_2/I_1 = 10/0.1 = 100$, and $\ln 100 = 4.605$.

$$\Delta V_D = 1.5 \times 25.875 \text{ mV} \times 4.605 = \mathbf{0.179 \text{ V } (179 \text{ mV})}$$

Sanity check: one decade costs $2.303\, nV_T = 89.4$ mV, so two decades cost 179 mV.

Python: 0.17874 V using T = 300 K. Using T = 300.15 K gives 0.17883 V, which is the same to 3 s.f. Using the rough $V_T = 25$ mV gives 0.173 V. That is acceptable only if you say you rounded V_T, and it may cost the "correct value" mark. This agrees with the Test 1 Reference (0.179 V).

Mark breakdown (5):

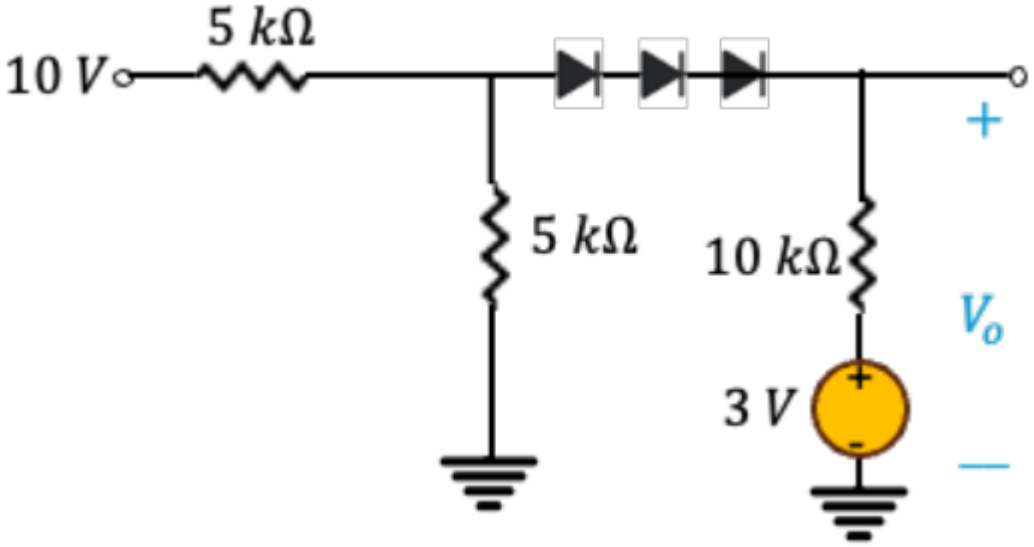
- 1: V_T computed at 300 K (25.9 mV), using the constants from the formula sheet.
- 2: the ratio derivation, showing that I_S cancels and arriving at $\Delta V = nV_T \ln(I_2/I_1)$.
- 1: correct substitution (n = 1.5, ratio 100).
- 1: final answer ≈ 0.179 V with units.

Most likely ways to lose marks:

- Dropping n (writing $V_T \ln 100 = 119$ mV), or using $\log_{10}$ instead of $\ln$ (which gives 77.6 mV).
- Using T = 27 instead of 300 K, or not showing the I_S cancellation, which costs the derivation marks.

Question 3: three series diodes, third PWL model [8 marks]

Paper wording. Consider the circuit below, which consists of the series combination of three identical diodes. Using the third piecewise linear model for the diodes with $V_D = 0.6$ V and $r_D = 0.20$ kΩ, answer the following questions.



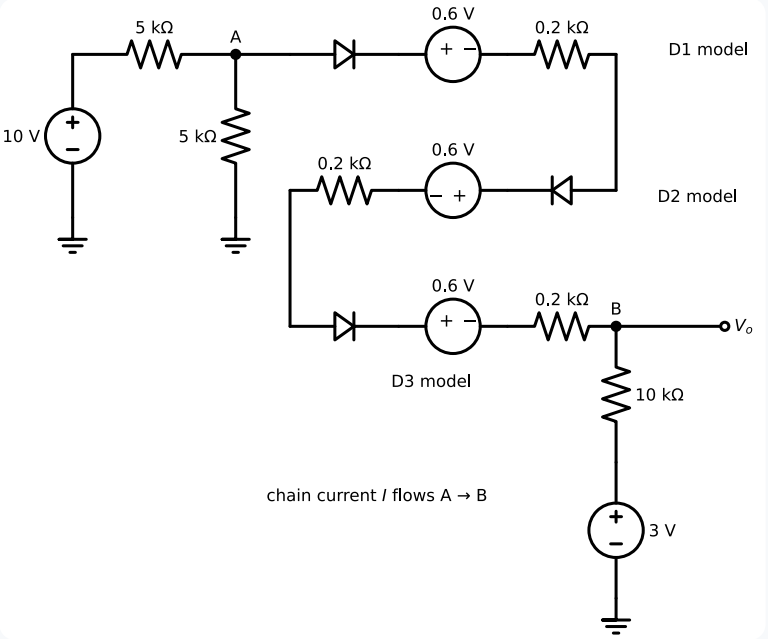
Q3 figure (from the practice-test paper)

Assume that all diodes are **conducting**.
(a) Redraw the entire circuit using the third piecewise linear equivalent circuit model of diodes. You do not need to simplify the circuit. [2 marks]
(b) Verify whether all three diodes are **conducting** under the given circuit conditions. [4 marks]
(c) Find the output voltage, V_o, as shown on the circuit. [2 marks]

READING THE FIGURE: 10 V source, then 5 kΩ to node A. Node A has 5 kΩ to ground. From A, three diodes in series (anodes towards A) run to node B = V_o. From B, 10 kΩ goes to the + terminal of a 3 V source whose − terminal is grounded.

(a) Redraw [2]. Replace *each* diode with an ideal diode in series with a 0.6 V source (+ towards A) and a 0.2 kΩ resistor. Draw all three separately, because the question says not to simplify:

10 V — 5 kΩ — **A** — [ideal D1, 0.6 V, 0.2 kΩ] — [ideal D2, 0.6 V, 0.2 kΩ] — [ideal D3, 0.6 V, 0.2 kΩ] — **B** (V_o) — 10 kΩ — (+) 3 V (−) — ground, with 5 kΩ from A to ground.



Q3(a) model answer: each diode drawn as an ideal diode, a 0.6 V source (+ towards A) and 0.2 kΩ.

(b) Verify conduction [4]. Assume all three conduct, with current I flowing A → B (work in mA, kΩ, V).

KCL at A:

$$\frac{10 - V_A}{5} = \frac{V_A}{5} + I \;\Rightarrow\; V_A = 5 - 2.5I$$

At B: $V_B = 3 + 10I$.

Diode chain: $V_A - V_B = 3(0.6) + 3(0.2)I = 1.8 + 0.6I$.

Substitute:

$$(5 - 2.5I) - (3 + 10I) = 1.8 + 0.6I \;\Rightarrow\; 0.2 = 13.1I \;\Rightarrow\; I = 0.01527\ \text{mA} = 15.3\ \mu\text{A}$$

$I > 0$ and the same current flows through all three ideal diodes, so **all three are conducting**.

Quick alternative (Thevenin at A: 5 V, 2.5 kΩ): with the diodes removed, the voltage across the string is $5 - 3 = 2$ V. That is more than the 1.8 V needed, so the diodes turn on. Then $I = (5 - 1.8 - 3)/(2.5 + 0.6 + 10) = 15.3$ µA, the same answer.

Check: $V_A = 4.962$ V, $V_B = 3.153$ V, so the chain drops 1.809 V $= 1.8 + 0.6(0.0153)$.

(c) Output [2].

$$V_o = V_B = 3 + 10\ \text{k}\Omega \times 15.3\ \mu\text{A} = \mathbf{3.15\ V}$$

Python: I = 15.267 µA, V_o = 3.1527 V. This agrees with the Test 1 Reference.

Mark breakdown (8):

- a. 1: each diode drawn as ideal diode + 0.6 V + 0.2 kΩ, with correct polarity. 1: the rest of the circuit is correct, including the 3 V source polarity (+ towards the 10 kΩ).
- b. 1: a correct KCL/KVL equation at A, or a correct Thevenin at A. 1: the diode-chain equation including $3r_D I$. 1: solving for I = 15.3 µA. 1: the stated conclusion that I > 0, so all three conduct.
- c. 1: $V_o = 3 + 10I$. 1: 3.15 V.

Most likely ways to lose marks:

- Forgetting the 5 kΩ shunt from A to ground and using 10 V straight into the string. This gives I = (10 − 1.8 − 3)/(15.6) = 0.333 mA, which is wrong.
- Using 0.6 V per diode but leaving out the r_D terms, or writing the 3 V source with the wrong sign. With the wrong sign you get I = 0.473 mA and $V_o = 1.73$ V. Check the sign: a 30-fold jump in current from a sign slip is the clue.

Question 4: Zener regulator design [5 marks]

Paper wording. Design the network below to maintain V_L at 12 V for a load variation (I_L) from 0 mA to 200 mA. That is, determine R_S and V_Z. (Word for word, this is WS3 Q13.)

Q4 figure (from the practice-test paper)

V_Z: the Zener sets the load voltage, so $\mathbf{V_Z = 12\ V}$.

R_S: design at the worst case, full load $I_L = 200$ mA, where the Zener is just in breakdown ($I_Z \to 0$):

$$R_S = \frac{V_i - V_Z}{I_{L,\max} + I_{Z,\min}} \approx \frac{16 - 12}{200\ \text{mA}} = \mathbf{20\ \Omega}$$

Workshop method (WS3 solution, Q13): at full load $R_L = 12/0.2 = 60\ \Omega$. With the Zener drawing nothing, it is a divider:

$$\frac{60 \times 16}{60 + R_S} = 12 \ \Rightarrow \ R_S = 20\ \Omega$$

For the last mark: at $I_L = 0$ the Zener carries all of $I_S = 4/20 = 200$ mA, so it needs $P_{Z,\max} \geq 12 \times 0.2 = 2.4$ W. (R_S dissipates $4 \times 0.2 = 0.8$ W.)

This agrees with the WS3 Q13 solution and the Test 1 Reference.

Mark breakdown (5):

- 1: $V_Z = 12$ V, with the reason given.
- 1: identifying full load (200 mA) as the case that sets R_S, with $I_{Z,\min} \approx 0$.
- 2: the equation (divider or $(V_i - V_Z)/I$) and $R_S = 20\ \Omega$.
- 1: checking the other extreme, $I_L = 0$: $I_Z = 200$ mA and $P_Z = 2.4$ W.

Most likely ways to lose marks:

- Sizing R_S at $I_L = 0$. You cannot get an R_S from that case, and some students divide by a made-up I_Z instead.
- Leaving out the no-load check. The workshop solution stops at 20 Ω, but “design” questions usually reward the Zener current and power rating.

Question 5: series clipper [5 marks]

Paper wording. Determine v_o of the network below for the input shown. (Word for word, this is WS3 Q12.)

Q5 figure (from the practice-test paper)

READING THE FIGURE: v_i is a square wave between +20 V and −5 V. The circuit is v_i(+), then a 4 V battery (− on the input side, + on the diode side), then an ideal diode (cathode

towards the battery, anode towards the output), then a node v_o with 1 kΩ to ground.

Orientation. The only direction current can flow through the diode is from the output node towards the battery. That means current flows *up* through the 1 kΩ, so v_o can only be zero or negative.

Transition voltage (ideal diode, so $v_d = 0$ and $i_d = 0$ at the boundary): $v_i + 4 = 0 \Rightarrow v_{iT} = -4$ V. The diode conducts only for $v_i < -4$ V.

$v_i = +20$ V: the diode is reverse biased (it sees 20 + 4 = 24 V reverse). It is an open circuit, so no current flows and

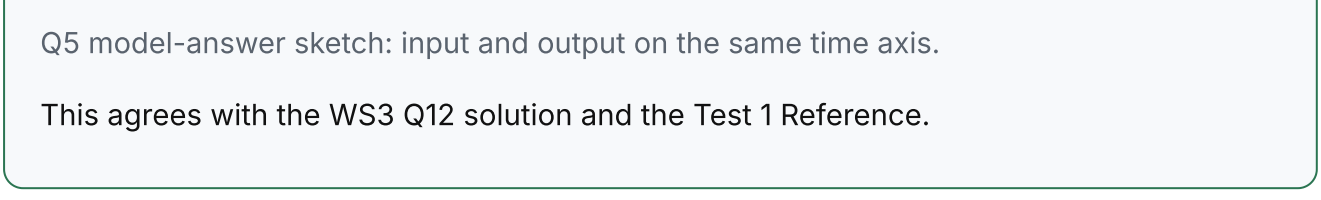
$$v_o = \mathbf{0\ V}$$

$v_i = -5$ V: the input overpowers the 4 V battery, so the diode is on (short). KVL round the loop:

$$+5 - 4 + v_o = 0 \ \Rightarrow \ v_o = \mathbf{-1\ V}$$

Equivalently, $v_o = v_i + 4 = -1$ V, and the current is 1 mA up through the 1 kΩ.

Sketch: a square wave in step with v_i: 0 V while $v_i = 20$ V, and −1 V while $v_i = -5$ V.



Q5 model-answer sketch: input and output on the same time axis.

This agrees with the WS3 Q12 solution and the Test 1 Reference.

Mark breakdown (5):

- 1: transition voltage $v_i = -4$ V, or a correct statement of when the diode conducts.
- 1: $v_i = +20$ V, diode off, $v_o = 0$ V.
- 2: $v_i = -5$ V, diode on, KVL giving $v_o = -1$ V.
- 1: a sketch of v_o against time, aligned with v_i and with levels labelled.

Most likely ways to lose marks:

- Misreading the diode direction and getting $v_o = 20 - 4 = 16$ V for the positive half. Trace the arrow before writing anything.
- Getting the battery sign wrong in the KVL and writing $v_o = -9$ V ($-5 - 4$). The + terminal faces the diode, so it *adds* 4 V to v_i.

Question 6: load line and Q-point [6 marks]

Paper wording. Draw the load line for the network below on the characteristics.
a. Find the Q-point and resulting values I_{C_Q} and V_{CE_Q}.
b. Find the value of β at the Q-point.

Q6 circuit (from the practice-test paper)

Q6 output characteristics (from the practice-test paper)

READING THE FIGURE: emitter-bias circuit with $V_{CC} = 20$ V, $R_B = 270$ kΩ, $R_C = 470\ \Omega$, $R_E = 2.2$ kΩ and $\beta = 125$ printed on the transistor.

Step 1: load line. $V_{CE} = V_{CC} - I_C(R_C + R_E)$ with $I_E \approx I_C$, and $R_C + R_E = 2.67$ kΩ.

- $I_C = 0 \Rightarrow V_{CE} = 20$ V.
- $V_{CE} = 0 \Rightarrow I_C = 20/2.67\text{k} = 7.49$ mA.

Join (20 V, 0) to (0, 7.49 mA) on the graph with a ruler.

Step 2: base current (base loop, which picks the curve):

$$I_{BQ} = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} = \frac{20 - 0.7}{270\text{k} + 126(2.2\text{k})} = \frac{19.3}{547.2\text{k}} = 35.3\ \mu\text{A}$$

Step 3 (a): Q-point. Read where the load line crosses an imagined $I_B \approx 35\ \mu$A curve, halfway between the 30 µA and 40 µA curves. On the printed graph, in the range $V_{CE} \approx 7$–8 V, the 30 µA curve sits at about 4.0 mA and the 40 µA curve at about 5.2 mA. Interpolating to 35.3 µA:

$$I_{CQ} \approx 4.6\ \text{mA (graph)}, \qquad V_{CEQ} \approx 20 - 4.66(2.67) \approx 7.6\ \text{V (graph)}$$

Analytical cross-check using the stated $\beta = 125$:

$$I_{CQ} = 125 \times 35.3\ \mu\text{A} = 4.41\ \text{mA}, \quad I_E = 4.44\ \text{mA}, \quad V_{CEQ} = 20 - 4.41\text{m}(470) - 4.44\text{m}(2200) = 8.15\ \text{V}$$

Either answer is accepted if you show the method: graph values of about 4.6 mA and 7.6 V, or analytical values of 4.41 mA and 8.15 V. The question says “on the characteristics”, so the graph reading is the intended answer. Quote the analytical one as a check. (The E4 pack reads the same graph as about 4.5 mA, 8.0 V, β ≈ 127. Treat **4.5–4.6 mA, 7.6–8.0 V, β ≈ 127–130** as the graph answer.)

Step 3 (b): β at the Q-point.

$$\beta_{dc} = \frac{I_{CQ}}{I_{BQ}} \approx \frac{4.6\ \text{mA}}{35.3\ \mu\text{A}} \approx \mathbf{130}$$

A reading anywhere in 4.5–4.7 mA gives about 128–133.

Where this differs from the Test 1 Reference. The Reference (p. 19) reads $I_{CQ} \approx 4.2$–4.4 mA off the graph and gets $\beta \approx 120$. That graph reading is too low. On the printed characteristics the 30 µA curve is already at about 4.0 mA (it sits on the “4” gridline). The 40 µA curve is at about 5.2 mA. So a 35 µA operating point cannot be at 4.2 mA. It is about 4.6–4.7 mA, which gives $\beta \approx 130$ and $V_{CEQ} \approx 7.6$ V. I checked this by measuring the curve positions on the rasterised page: 30 µA gives 4.03 mA and 40 µA gives 5.22 mA at $V_{CE} = 7.5$ V. The Reference’s analytical values (4.41 mA, 8.15 V) are correct for $\beta = 125$. The difference simply shows that the printed device has $\beta \approx 130$, a little above the stated 125.

Mark breakdown (6):

- 1: load-line equation and both intercepts (20 V and 7.49 mA).
- 1: load line drawn correctly on the characteristics.
- 1: $I_{BQ} = 35.3\ \mu$A, with the $(\beta + 1)R_E$ term.
- 2: the Q-point marked, and I_{CQ} and V_{CEQ} read off (about 4.6 mA and 7.6 V; analytical values of 4.41 mA and 8.15 V with working are also accepted).
- 1: $\beta = I_{CQ}/I_{BQ}$, using your *graph* values (about 130).

Most likely ways to lose marks:

- Writing $I_B = 19.3/270\text{k} = 71.5\ \mu$A, which forgets that R_E is reflected into the base loop as $(\beta + 1)R_E$.
- Using only R_C for the load line ($I_{C,sat} = 42.6$ mA) and forgetting R_E. Or, in (b), quoting $\beta = 125$ back instead of dividing your graph readings.

Question 7: Thevenin at the base [10 marks]

Paper wording. For the circuit shown on the right, $\beta = 50$, $V_{BE,on} = 0.7$ V, and $V_{CE,sat} = 0.2$ V. Answer the following questions assuming that the BJT operates in the **active state**.
a) Find the Thevenin equivalent circuit (i.e., V_{Th} and R_{Th}) seen from the base of the transistor. [4 marks]
b) Redraw the circuit by replacing the transistor with its **DC equivalent circuit** model and Thevenin equivalent model in the base. [4 marks]
c) Determine collector current (I_C) and voltage (V_C). [2 marks]

Q7 figure (from the practice-test paper)

(a) Thevenin [4].

Resistance: zero both sources (short them to their rails):

$$R_{Th} = R_1 \parallel R_2 = \frac{15 \times 30}{45} = \mathbf{10\ k\Omega}$$

Voltage: open circuit at the base, by superposition:

$$V_{Th} = 9 \cdot \frac{30}{45} + (-12) \cdot \frac{15}{45} = 6 - 4 = \mathbf{2\ V}$$

Check: the node sits 30/45 of the way from −12 V to +9 V, which is $-12 + 21(2/3) = 2$ V.

(b) Redraw [4]. The base side is $V_{Th} = 2$ V in series with $R_{Th} = 10$ kΩ, feeding the base. Between B and E (ground) there is a 0.7 V source (+ at B). Between C and E there is a dependent current source $\beta I_B = 50 I_B$, pointing C → E. Above it, $R_C = 1.5$ kΩ goes to $V_{CC} = 12$ V. Label I_B, I_C and V_C.

Q7(b) model answer: Thévenin base source with the active-mode DC model.

(c) I_C and V_C [2]. Base loop (emitter grounded, so there is no R_E term):

$$I_B = \frac{V_{Th} - V_{BE}}{R_{Th}} = \frac{2 - 0.7}{10\text{k}} = 130\ \mu\text{A}$$

$$I_C = \beta I_B = 50 \times 130\ \mu\text{A} = \mathbf{6.5\ mA}$$

$$V_C = V_{CC} - I_C R_C = 12 - 6.5\text{m} \times 1.5\text{k} = \mathbf{2.25\ V}$$

Check: $V_{CE} = V_C = 2.25\ \text{V} > 0.2$ V and $I_B > 0$, so the transistor is active as assumed.

This agrees with the Test 1 Reference.

Mark breakdown (10):

- a. 2: R_{Th} (method of zeroing the sources, then 10 kΩ). 2: V_{Th} (superposition or divider between the two rails, then 2 V).
- b. 1: base Thevenin source and resistor. 1: $V_{BE} = 0.7$ V source with correct polarity. 1: βI_B current source C → E, with correct direction. 1: R_C and V_{CC}, emitter to ground, and labels.
- c. 1: I_B and $I_C = 6.5$ mA. 1: $V_C = 2.25$ V, with the active-state check.

Most likely ways to lose marks:

- Treating −12 V as ground and getting $V_{Th} = 9 \times 30/45 = 6$ V. This also gives $I_C = 26.5$ mA and an impossible V_C. Always include both rails.

- In (b), drawing the current source in the wrong direction or as an independent source, or leaving out the 0.7 V source. These are easy marks that are often lost.

Question 8: two transistors [6 marks]

Paper wording. Consider the circuit below where Q_1 and Q_2 have different current gain, $\beta_1 = 100$ and $\beta_2 = 50$. You can assume $V_{BE,on}$ = 0.7 V and $V_{CE,sat} = 0.2$ V for both BJTs and both BJTs are in **active state**. Find the output voltage, V_{C2} by showing your work.

Q8 figure (from the practice-test paper)

READING THE FIGURE: Q_1 (NPN) has its base and collector tied together at 2.5 V. Its emitter drives the emitter of Q_2 (PNP). Q_2's base goes through 10 kΩ to a 1 V terminal. Its collector is V_{C2}, with 1 kΩ to ground.

Step 1: Q_1 sets a level. Q_1 is diode-connected (B tied to C), so it simply drops one V_{BE}:

$$V_{E1} = 2.5 - 0.7 = 1.8 \text{ V} = V_{E2}$$

Step 2: Q_2 base. Q_2 is a PNP in active mode, so its emitter is 0.7 V above its base:

$$V_{B2} = 1.8 - 0.7 = 1.1 \text{ V}$$

Step 3: base current. Current flows out of the base, through 10 kΩ, to the 1 V terminal. That is the correct direction for a PNP.

$$I_{B2} = \frac{1.1 - 1}{10 \text{ k}\Omega} = 10 \ \mu\text{A}$$

Step 4: collector.

$$I_{C2} = \beta_2 I_{B2} = 50 \times 10 \ \mu\text{A} = 0.5 \text{ mA} \ (\text{out of the collector, down through } 1 \text{ k}\Omega)$$

$$V_{C2} = 0.5 \text{ mA} \times 1 \text{ k}\Omega = \mathbf{0.5 \text{ V}}$$

> **CHECK:** $V_{EC2} = 1.8 - 0.5 = 1.3 \text{ V} > 0.2$ V and $I_{B2} > 0$, so Q_2 is active. $I_{E2} = 51 \times 10 \ \mu\text{A} = 0.51$ mA, which Q_1 supplies. β_1 does not affect V_{C2}, because Q_1 only fixes the 1.8 V level. Write this down, because it shows you understood the circuit.

This agrees with the Test 1 Reference.

Mark breakdown (6):

- 1: recognising Q_1 as diode-connected, so $V_{E1} = V_{E2} = 1.8$ V.
- 1: $V_{B2} = 1.1$ V, using the PNP polarity.
- 1: $I_{B2} = 10$ μA, with the correct direction.
- 1: $I_{C2} = 0.5$ mA.
- 1: $V_{C2} = 0.5$ V.
- 1: active-state check ($V_{EC2} = 1.3$ V > 0.2 V), or the comment that β_1 is irrelevant.

Most likely ways to lose marks:

- Using NPN polarity on Q_2 ($V_{B2} = 1.8 + 0.7 = 2.5$ V). This gives $I_B = 150$ μA into the base, and the whole answer goes wrong.
- Trying to push β_1 into the calculation, for example by computing I_{B1} and calling it I_{B2}. If Q_1's numbers appear anywhere in your V_{C2} line, check again.

After marking

Q	Marks	Your score	Topic	Revise in
1 (I)	1		pn junction, depletion region	E1
1 (II)	1		dynamic resistance r_d	E2
1 (III)	1		full-wave vs half-wave rectification	E3
1 (IV)	1		BJT physics (thin base, α, β)	E4
1 (V)	1		DC vs AC (small-signal) analysis	E4 / E6
2	5		exponential (Shockley) model, $\Delta V = nV_T \ln$ ratio	E2
3	8		third piecewise-linear diode model, verify conduction	E2 (Thevenin step: E1)
4	5		Zener regulator design	E3
5	5		series clipper with a battery	E3
6	6		emitter bias, load line, Q-point from characteristics	E4 / E5

Q	Marks	Your score	Topic	Revise in
7	10		Thevenin at the base, active-mode DC model	E5 (Thevenin: E1)
8	6		multi-transistor DC, PNP polarity	E4 / E5

How to read your score. If you scored 40 or more, you are on track. Redo only the questions where you lost marks, a day before Test 1. If you scored 30–39, do the "Revise in" pack for every question where you lost 2 or more marks. If you scored under 30, prioritise Q3, Q6 and Q7 (24 marks between them, all method marks) and their packs, E2 and E5.

Quick recall check

1. Two decades of diode current at $n = 1.5$, 27 °C: how much does V_D change?
2. In a Zener regulator, at which load current do you size R_S, and which case do you use to check the Zener power rating?
3. For an emitter-bias circuit, what resistance does R_E look like from the base loop?
4. In a series clipper with an ideal diode and a battery, how do you find the transition voltage?
5. A PNP in active mode has its emitter at 1.8 V. What is its base voltage, and which way does the base current flow?

1. $\Delta V = nV_T \ln 100 = 1.5 \times 25.9\ \text{mV} \times 4.605 \approx 179$ mV, which is about 89 mV per decade.
2. Size R_S at maximum load current ($I_Z \approx 0$): $R_S = (V_i - V_Z)/I_{L,\max}$. Check the power at no load, where the Zener takes all of I_S: $P_Z = V_Z I_S$.
3. $(\beta + 1)R_E$, so $I_B = (V_{CC} - V_{BE})/(R_B + (\beta + 1)R_E)$.
4. Set the diode at the boundary ($v_d = 0$, $i_d = 0$) and write KVL round the input loop. Here $v_i + 4 = 0$ gives $v_{iT} = -4$ V.
5. $V_B = 1.8 - 0.7 = 1.1$ V. Base current flows *out* of the base, and I_C flows *out* of the collector.

BJT small-signal I

Answers and worked solutions · Day 10 · Mon 2 Nov · pack E6

5. Practice

From your course files

WS6 Q1 (course question).

Take V_T = 25 mV, as the solution does.

$$g_m = \frac{I_C}{V_T} = \frac{1\ \text{mA}}{0.025\ \text{V}} = \mathbf{40\ mA/V}$$

$$\alpha = \frac{\beta}{1+\beta} = \frac{100}{101}, \qquad r_e = \frac{\alpha}{g_m} = \frac{100}{101} \times \frac{1}{40\ \text{mA/V}} = \mathbf{24.8\ \Omega}$$

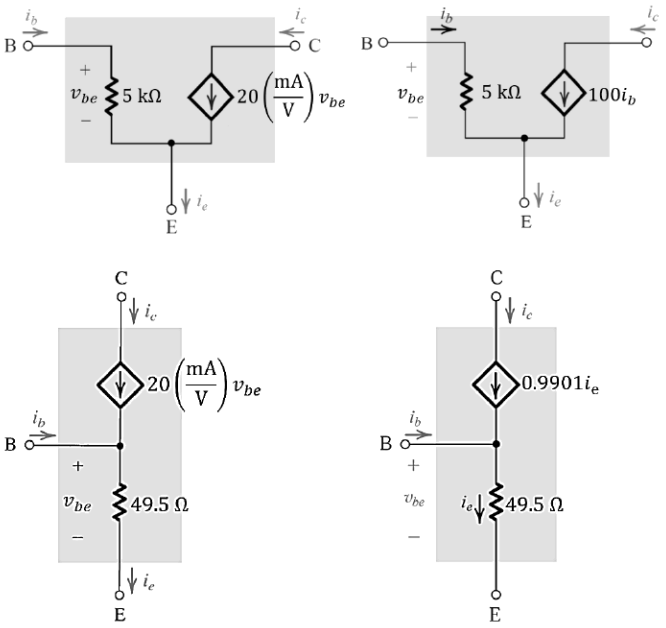
$$r_\pi = \frac{\beta}{g_m} = \frac{100}{40\ \text{mA/V}} = \mathbf{2.5\ k\Omega}$$

The solution's first line shows "0.5 mA" in the g_m fraction. That is a typo: I_C = 1 mA, and the 40 mA/V result is right.

WS6 Q2 (course question).

$$r_\pi = \frac{\beta}{g_m} = \frac{100}{20\ \text{mA/V}} = \mathbf{5\ k\Omega}, \qquad r_e = \frac{\alpha}{g_m} = \frac{100}{101 \times 20\ \text{mA/V}} = \mathbf{49.5\ \Omega}$$

The four models are hybrid-π with $g_m v_{be}$ or with βi_b, and T with $g_m v_{be}$ or with αi_e (α = 0.9901). The official drawings:



WS6 Q2: the four small-signal models from the official solution. Top: hybrid-π with $g_m v_{be}$ and with $100 i_b$. Bottom: T model with $g_m v_{be}$ and with $0.9901 i_e$.

WS6 Q4 (course question).

$\beta = \infty$, so $\alpha = 1$. In the small-signal circuit the +3 V rail is ground, so the 10 kΩ runs from the collector to ground. The base is grounded, the current source is an open circuit, and v_i drives the emitter through the (shorted) capacitor.

T model: r_e = 0.125 kΩ from the base (ground) to the emitter, and αi_e from the collector to the base, with i_e flowing from base to emitter.

- Collector: $v_c = -\alpha i_e (10\ \text{k}\Omega)$
- Emitter: $i_e = -\dfrac{v_i}{r_e}$

$$v_c = -\alpha \left(-\frac{v_i}{r_e} \right) (10\ \text{k}\Omega) \ \Rightarrow\ \frac{v_c}{v_i} = \frac{\alpha (10\ \text{k}\Omega)}{r_e} = \frac{1 \times 10\ \text{k}}{0.125\ \text{k}} = \mathbf{80}$$

Positive: a **non-inverting** amplifier. The signal enters at the emitter with the base at AC ground, so this is a common-base stage (more on CB in E7). The DC values (V_C = 1 V, I = 0.2 mA) only confirm that the transistor is active; r_e is given.

WS6 Q5 (course question).

1. The small-signal circuit. DC sources off and capacitors shorted:

- v_{sig} in series with 2 kΩ drives the base; the 10 kΩ runs from base to ground.
- Hybrid-π: r_π from base to emitter; $g_m v_{be}$ and r_o from collector to emitter.

- Z is grounded through its capacitor, so the emitter is at AC ground (the 1 mA source is open).
- At the collector: the 8 kΩ R_C (its top is AC ground) and the 8 kΩ load on Y, both to ground.

2. Gain. The output is the voltage across $r_o \parallel 8\text{k} \parallel 8\text{k}$, with $g_m v_{be}$ flowing into the collector from that network:

$$v_y = -g_m v_{be}\,(r_o \parallel 8\text{k} \parallel 8\text{k}) \qquad (1)$$

v_{be} is the voltage across $r_\pi \parallel 10\text{k}$, which forms a divider with the 2 kΩ source resistance:

$$v_{be} = v_{sig}\,\frac{r_\pi \parallel 10\text{k}}{(r_\pi \parallel 10\text{k}) + 2\text{k}} \qquad (2)$$

Numbers: $r_\pi \parallel 10\text{k}$ = 2.016 kΩ, so the divider is 0.502. $r_o \parallel 8\text{k} \parallel 8\text{k} = 101\text{k} \parallel 4\text{k}$ = 3.848 kΩ.

$$\frac{v_y}{v_{sig}} = -39.604 \times 0.502 \times 3.848 = \mathbf{-76.49\ V/V}$$

An inverting amplifier.

WS6 Q6 (course question).

Which DC method. $\beta R_E = 180(2.2\text{ k}) = 396$ kΩ is less than $10R_2 = 560$ kΩ, so the approximate method is not allowed. Use the exact (Thevenin) method.

DC (exact): $R_{Th} = 220\text{ k} \parallel 56\text{ k} = 44.64$ kΩ and $E_{Th} = \dfrac{56}{276}(20) = 4.06$ V.

$$I_B = \frac{4.06 - 0.7}{44.64\text{ k} + 181(2.2\text{ k})} = 7.58\ \mu\text{A}, \quad I_E = 181 I_B = 1.372\text{ mA}, \quad I_C = 180 I_B = 1.365\text{ mA}$$

1. $r_e = 26\text{ mV}/1.372\text{ mA} = \mathbf{18.94\ \Omega}$. The solution gives 18.92 Ω because it rounds I_B to 7.59 µA.

2. $V_B = V_{BE} + I_E R_E = 0.7 + 1.372(2.2) = \mathbf{3.72\ V}$ (the same as $E_{Th} - I_B R_{Th}$). The OCR'd solution did not show a V_C value; from the solution's own currents, $V_C = V_{CC} - I_C R_C = 20 - 1.365(4.7) = \mathbf{13.58\ V}$. Check: $V_E = 3.02$ V, so $V_{CE} = 10.6$ V. Active.

3. $\beta r_e = 180(18.94) = 3.41$ kΩ, so $Z_i = R_{Th} \parallel \beta r_e = 44.64\text{ k} \parallel 3.41\text{ k} = \mathbf{3.17\ k\Omega}$.

The figure gives the output conductance $g_{os} = 30\ \mu$S (the same quantity as h_{oe}), so $r_o = 1/g_{os} = 33.3$ kΩ. That is only 7× R_C, so keep it: $R_C \parallel r_o = 4.12$ kΩ, and

$$A_V = -\frac{R_C \parallel r_o}{r_e} = -\frac{4119}{18.94} = \mathbf{-217.4}$$

The solution, with r_e = 18.92 Ω and $R_C \parallel r_o$ rounded to 4.12 kΩ, gets −217.76. Ignoring r_o would give −248.

New questions

N1 (easy) [4 marks].

a. $g_m = 2/26 = \mathbf{76.9\ mA/V}$ and $r_\pi = 150/0.0769 = \mathbf{1.95\ k\Omega}$.

$I_E = 2(151/150) = 2.013$ mA, so $r_e = 26/2.013 = \mathbf{12.91\ \Omega}$. Also $r_o = 80/2\ \text{mA} = \mathbf{40\ k\Omega}$.

b. $(\beta+1)r_e = 151(12.91) = 1.95\ \text{k}\Omega = r_\pi$. ✓ The identity is exact because $I_E = (\beta+1)I_B$.

N2 (medium) [7 marks].

a. $I_B = (15 - 0.7)/680\ \text{k} = 21.03\ \mu$A, $I_C = 120 I_B = 2.524$ mA, and $I_E = 121 I_B = 2.545$ mA. $V_{CE} = 15 - 2.524(4.7) = \mathbf{3.14\ V}$ > 0.2 V, so the BJT is active.

b. $r_e = 26/2.545 = \mathbf{10.22\ \Omega}$ and $r_o = 100/2.524\ \text{mA} = \mathbf{39.6\ k\Omega}$.

c. $\beta r_e = 1.226$ kΩ, so $Z_i = 680\ \text{k} \parallel 1.226\ \text{k} = \mathbf{1.22\ k\Omega}$.

Without r_o: $Z_o = 4.7$ kΩ and $A_v = -4700/10.22 = \mathbf{-460}$.

With r_o: $Z_o = 4.7\ \text{k} \parallel 39.6\ \text{k} = \mathbf{4.20\ k\Omega}$ and $A_v = -4202/10.22 = \mathbf{-411}$. Here $r_o/R_C = 8.4 < 10$, so r_o should be kept.

d. $v_o = -411(5\ \text{mV})\sin\omega t = \mathbf{-2.06 \sin \omega t\ V}$. The downward headroom is $V_{CE} - 0.2 = 2.94$ V and the upward headroom is $V_{CC} - V_{CE} = 11.9$ V. A 2.06 V peak fits, so there is no clipping, though there is not much margin on the negative side. The output is inverted (180°).

N3 (medium) [8 marks].

a. $\beta R_E = 150$ kΩ ≥ $10R_2 = 100$ kΩ, so the approximate method is allowed.

$V_B = 10(12)/57 = 2.105$ V, $V_E = 1.405$ V and $I_E = 1.405$ mA. Then $V_C = 12 - 1.405(3.3) = \mathbf{7.36\ V}$ and $V_{CE} = 12 - 1.405(4.3) = \mathbf{5.96\ V}$. Active.

b. $r_e = 26/1.405 = \mathbf{18.50\ \Omega}$

c. $R' = 47\ \text{k} \parallel 10\ \text{k} = 8.25$ kΩ and $\beta r_e = 2.78$ kΩ, so $Z_i = \mathbf{2.08\ k\Omega}$.

$Z_o = 3.3\text{ k} \parallel 60\text{ k} = \mathbf{3.13\ k\Omega}$ and $A_v = -3128/18.50 = \mathbf{-169}$, with a 180° phase shift. If r_o is ignored, A_v = −178.

For comparison, the exact (Thevenin) method gives $I_E = 1.33$ mA, $r_e = 19.5\ \Omega$ and $A_v = -160$. The test only guarantees about 10 % accuracy, and here the error is about 5 %.

d. $h_{ie} = \beta r_e = 2.78$ kΩ, $h_{fe} = \beta = 150$, $h_{oe} = 1/r_o = 16.7\ \mu$S. ($h_{re} \approx 0$.)

N4 (multi-part, chains) [9 marks] — your Practical 3 amplifier.

a. $\beta R_E = 270$ kΩ ≥ 100 kΩ, so the approximate method is allowed. $V_B = 10(10)/56 = \mathbf{1.786\ V}$, $V_E = 1.086$ V and $I_E = 1.086/0.98\text{ k} = \mathbf{1.108\ mA}$. The measured values (1.735 V, 1.12 mA) are within 3 % and 1 %. The lower V_B is base current loading the divider.

b. $r_e = 26/1.12 = \mathbf{23.2\ \Omega}$ and $r_o = 100/1.12\text{ mA} \approx \mathbf{89\ k\Omega}$.

c. $R' = 46\text{ k} \parallel 10\text{ k} = 8.21$ kΩ and $\beta r_e = 275.5(23.2) = 6.40$ kΩ, so $Z_i = \mathbf{3.60\ k\Omega}$ (measured 3.51 kΩ with V_s = 50 mV).

With $r_o = \infty$: $Z_o = 3.81$ kΩ and $|A_v| = 3810/23.21 = \mathbf{164}$. With r_o: $Z_o = 3.65$ kΩ and $|A_v| = 3654/23.21 = \mathbf{157}$.

d. C_E is not a perfect short. At 1 kHz, $X_{C_E} = 1.59\ \Omega$, which adds about 7 % to r_e in the emitter. Your findings show this brings the prediction to 147. The rest is component and β tolerance. (Dividing by $r_e + X_{C_E}$ is a preview of the unbypassed analysis in E7.)

e. $R_C \parallel r_o \parallel R_L = 3.81\text{ k} \parallel 89\text{ k} \parallel 4.7\text{ k} = 2.06$ kΩ, so $|A_{vL}| = 2055/23.21 = \mathbf{88.6}$. The measured value was 77.1; with the X_{C_E} correction your findings predicted 82.9. Loading roughly halves the gain, which is why E7 treats R_L properly.

N5 (hard, design) [8 marks].

a. $|A_v| = g_m R_C = \dfrac{I_C}{V_T} R_C = \dfrac{I_C R_C}{V_T} = \dfrac{V_{R_C}}{V_T}$. The individual values of I_C and R_C do not matter, only their product.

b. $V_{R_C} = 12 - 6 = 6$ V, so $|A_v|_{max} = 6/0.026 = \mathbf{231}$. More gain needs a larger V_{R_C}, which leaves less V_{CE} and less swing. That is the basic gain-versus-swing trade-off.

c. $V_{R_C} = 150(0.026) = 3.9$ V, so $R_C = 3.9\text{ V}/1\text{ mA} = \mathbf{3.9\ k\Omega}$ (a standard value). $V_{CEQ} = 12 - 3.9 = \mathbf{8.1\ V}$. $I_B = 10\ \mu$A, so $R_B = (12 - 0.7)/10\ \mu\text{A} = \mathbf{1.13\ M\Omega}$.

d. $I_E = 1.01$ mA, so $r_e = 25.74\ \Omega$ and $-R_C/r_e = \mathbf{-151.5}$ (the α difference). With $r_o = 100$ kΩ: $R_C \parallel r_o = 3.75$ kΩ and $A_v = \mathbf{-145.8}$. That is 3 % short of the target. To compensate, raise R_C slightly (about 4.02 kΩ, nearest standard 3.9 kΩ or 4.3 kΩ).

> Also note that fixed bias makes I_C proportional to β, so a real part with a different β shifts both the Q-point and the gain. This is why the divider bias is preferred.

6. Quick recall check

Question

1. $g_m = I_C/V_T$; $r_\pi = \beta/g_m = V_T/I_B$; $r_e = V_T/I_E \approx 1/g_m$; $r_o = V_A/I_C = 1/h_{oe}$.
2. R_1 goes base to ground (in parallel with R_2), R_C goes collector to ground, and R_E is shorted, so it disappears.
3. $v_\pi = i_b r_\pi$ and $g_m r_\pi = \beta$, so $g_m v_\pi = \beta i_b$.
4. $Z_i = R_1 \parallel R_2 \parallel \beta r_e$; $Z_o = R_C \parallel r_o$; $A_v = -(R_C \parallel r_o)/r_e$, with a 180° shift.
5. $|A_v| \approx V_{R_C}/V_T = 5/0.026 \approx 192$, inverting.

BJT small-signal II

Answers and worked solutions · Day 12 · Wed 4 Nov · pack E7

5. Practice

From your course files

WS6 Q6 (course question).

Which DC method. $\beta R_E = 180(2.2\text{ k}) = 396$ kΩ < $10R_2 = 560$ kΩ, so use the exact method.

$R_{Th} = 220\text{ k} \parallel 56\text{ k} = 44.64$ kΩ, $E_{Th} = \dfrac{56}{276}(20) = 4.06$ V.

$$I_B = \frac{4.06 - 0.7}{44.64\text{ k} + 181(2.2\text{ k})} = 7.59\ \mu\text{A}, \qquad I_E = 181 I_B = 1.374\text{ mA}$$

1. $r_e = 26\text{ mV}/1.374\text{ mA} = \mathbf{18.92\ \Omega}$ (the official value; unrounded I_B = 7.58 µA gives 18.94 Ω).

2. $V_B = V_{BE} + I_E R_E = 0.7 + 1.374(2.2) = \mathbf{3.72\ V}$. V_C is not given in the solution: $V_C = 20 - I_C R_C = 20 - 1.365(4.7) = \mathbf{13.58\ V}$, so V_{CE} = 10.6 V (active).

3. $Z_i = R_{Th} \parallel \beta r_e = 44.64\text{ k} \parallel 180(18.92) = \mathbf{3.17\ k\Omega}$.

$r_o = 1/g_{os} = 1/30\ \mu\text{S} = 33.3$ kΩ. That is less than $10R_C$ = 47 kΩ, so keep it: $R_C \parallel r_o = 4.12$ kΩ.

$$A_V = -\frac{R_C \parallel r_o}{r_e} = -\frac{4.12\text{ k}}{18.92} = \mathbf{-217.8}$$

With unrounded r_e this is −217.4. Either is the same answer.

WS6 Q7 (course question).

This is a pnp common-base stage: the emitter goes through 6.8 kΩ to +6 V, the base is grounded and the collector goes through 4.7 kΩ to −10 V.

1. DC emitter loop:

$$I_E = \frac{V_{EE} - V_{BE}}{R_E} = \frac{6 - 0.7}{6.8\text{ k}\Omega} = 0.779\text{ mA}, \qquad r_e = \frac{26\text{ mV}}{0.779\text{ mA}} = \mathbf{33.38\ \Omega}$$

2. $Z_i = R_E \parallel r_e = 6.8\text{ k} \parallel 33.38 = \mathbf{33.22\ \Omega}$ and $Z_o = R_C = \mathbf{4.7\ k\Omega}$. ($r_o = 1/g_{os}$ = 100 kΩ ≥ $10R_C$, so it is ignored.)

3.

$$A_V \approx \frac{R_C}{r_e} = \frac{4.7\ \text{k}}{33.38} = \mathbf{140.8}$$

Positive: CB does not invert. Keeping α = 0.998 gives 140.5, so α hardly matters.

WS6 Q4 (course question).

$\beta = \infty$, so $\alpha = 1$. In the small-signal circuit the 10 kΩ goes from the collector to ground, the base is grounded, the current source is open, and v_i drives the emitter.

T model: r_e = 0.125 kΩ from base to emitter, αi_e from collector to base.

- Collector: $v_c = -\alpha i_e(10\ \text{k}\Omega)$
- Emitter: $i_e = -v_i/r_e$

$$\frac{v_c}{v_i} = \frac{\alpha(10\ \text{k})}{r_e} = \frac{1 \times 10\ \text{k}}{0.125\ \text{k}} = \mathbf{80}$$

A non-inverting (common-base) amplifier.

WS6 Q5 (course question).

1. DC sources off, capacitors shorted: v_{sig} and 2 kΩ drive the base; 10 kΩ from base to ground; r_π from base to emitter; the emitter is at AC ground through Z; $g_m v_{be}$ and r_o from collector to emitter; R_C = 8 kΩ and the 8 kΩ load both from collector to ground.

2.

$$v_y = -g_m v_{be}(r_o \parallel 8\text{k} \parallel 8\text{k}), \qquad v_{be} = v_{sig}\frac{r_\pi \parallel 10\text{k}}{(r_\pi \parallel 10\text{k}) + 2\text{k}}$$

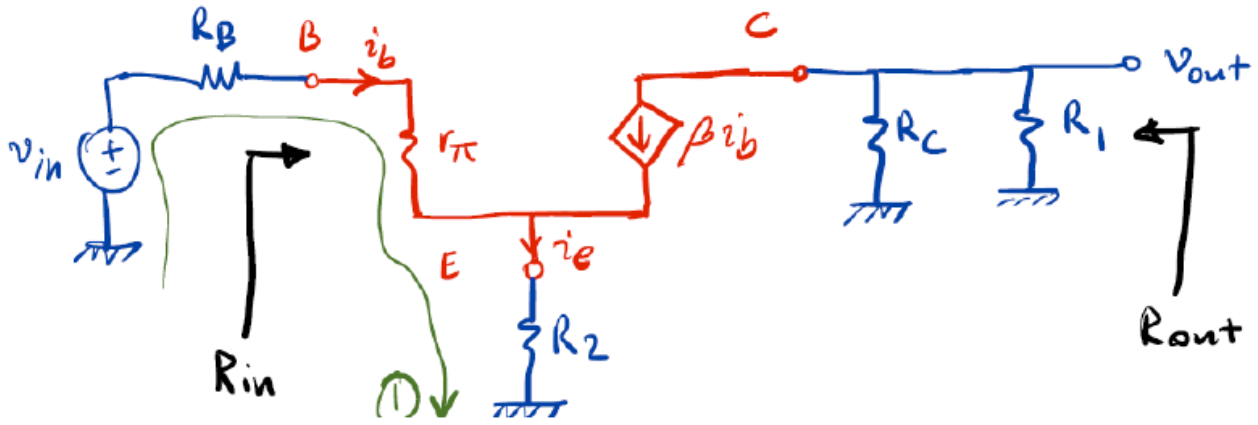
$r_\pi \parallel 10\text{k}$ = 2.016 kΩ (divider 0.502) and $r_o \parallel 8\text{k} \parallel 8\text{k}$ = 3.848 kΩ, so

$$\frac{v_y}{v_{sig}} = -39.604(0.502)(3.848) = \mathbf{-76.49\ V/V}$$

Inverting.

WS6 Q9 (course question).

Small-signal circuit. C_1 grounds the R_1–R_2 node, so R_1 ends up from the collector to ground (in parallel with R_C) and R_2 from the emitter to ground. The current source I_1 is open. The official drawing:



WS6 Q9: hybrid-π small-signal circuit from the official solution

Voltage gain.

$$v_{out} = -\beta i_b (R_C \parallel R_1) \qquad (1)$$

KVL around loop 1 with $i_e = (1+\beta) i_b$:

$$v_{in} = [R_B + r_\pi + (1+\beta) R_2] i_b \qquad (2)$$

$$\boxed{A_v = \frac{v_{out}}{v_{in}} = \frac{-\beta (R_C \parallel R_1)}{R_B + r_\pi + (1+\beta) R_2}}$$

Input impedance (looking into the base, which is how the solution defines it):

$$R_{in} = \frac{v_\pi + v_{R_2}}{i_b} = r_\pi + (1+\beta) R_2$$

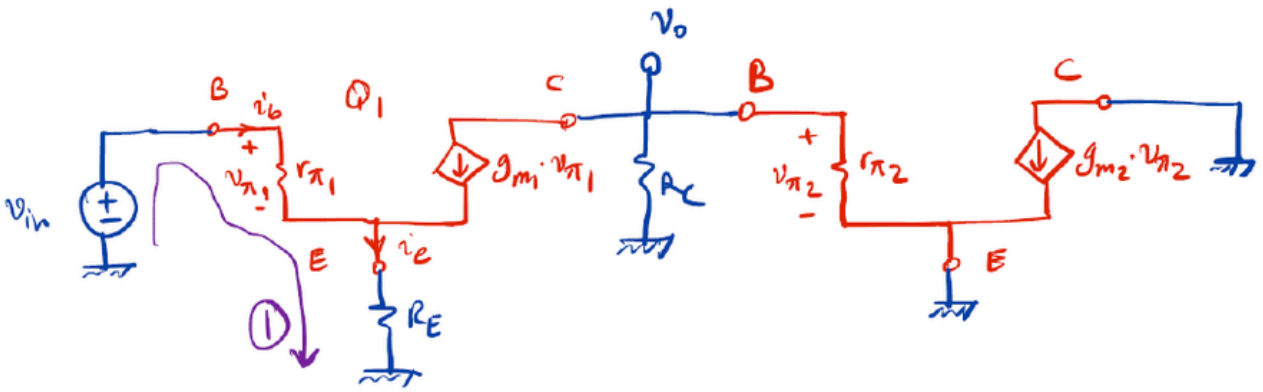
Seen from v_{in} it would be $R_B + r_\pi + (1+\beta) R_2$. Say which one you give.

Output impedance: with $v_{in} = 0$, $i_b = 0$, so the source is open and $R_{out} = R_C \parallel R_1$.

Check: with $R_2 = 0$ and $R_B = 0$ the gain collapses to $-g_m (R_C \parallel R_1)$, the bypassed CE result.

WS6 Q10 (course question).

Small-signal circuit (official drawing): Q_2's emitter is grounded, so its $r_{\pi 2}$ sits directly from Q_1's collector node to ground. Q_2's own collector goes to V_{CC} (AC ground) and does not affect v_{out}.



WS6 Q10: hybrid-π small-signal circuit from the official solution

The output node carries $R_C \parallel r_{\pi 2}$, and $g_{m1} v_{\pi 1}$ flows into it:

$$v_o = -g_{m1} v_{\pi 1} (R_C \parallel r_{\pi 2}) \qquad (1)$$

KVL in loop 1 with $i_e = (1+\beta) i_b$ and $i_b = v_{\pi 1} / r_{\pi 1}$:

$$v_{in} = [r_{\pi 1} + (1+\beta) R_E] \frac{v_{\pi 1}}{r_{\pi 1}} \qquad (2)$$

$$\boxed{\frac{v_o}{v_{in}} = \frac{-g_{m1} r_{\pi 1} (R_C \parallel r_{\pi 2})}{r_{\pi 1} + (1+\beta) R_E} = \frac{-(R_C \parallel r_{\pi 2})}{\dfrac{1}{g_{m1}} + \dfrac{1+\beta}{\beta} R_E}}$$

Here β is β_1; β_2 enters only through $r_{\pi 2}$.

WS6 Q11 (course question).

Why it fails. The speaker is connected directly from the collector of Q_1 to ground. At DC its solenoid is almost a short (under 1 Ω), so it pulls the collector to ground: $V_C \approx 0$. With $V_E = 0$, $V_{CE} \approx 0$, so Q_1 is driven into deep saturation and no longer works as an amplifier. The input coupling capacitor C_1 fixed the base bias, but the output is still DC-coupled.

Fix. Insert a coupling capacitor between the collector of Q_1 and the speaker. It blocks the DC component while passing the AC signal, so the speaker no longer disturbs the bias, and the microphone signal is still amplified.

New questions

N1 (easy) [5 marks].

a. Test: $\beta R_E = 150$ kΩ ≥ $10 R_2 = 120$ kΩ, so it is allowed. $V_B = \frac{12}{80}(15) = 2.25$ V, $V_E = 1.55$ V, $I_E = 1.55$ mA, $V_{CE} = 15 - 1.55(3.3 + 1) = 8.34$ V. Active. (The exact method gives 1.45 mA. The approximation is 7 % high, which is acceptable because the test passed.)

b. $r_e = 26/1.55 = 16.8\ \Omega$. $R' = 68\text{k} \parallel 12\text{k} = 10.2$ kΩ.
$Z_b = 150(16.8 + 1000) = 152.5$ kΩ. $Z_i = 10.2\text{k} \parallel 152.5\text{k} = \mathbf{9.56\ k\Omega}$.
$Z_o = R_C = \mathbf{3.3\ k\Omega}$. $A_v = -\dfrac{3300}{1016.8} = \mathbf{-3.25}$, against $-R_C/R_E = -3.30$ (1.7 % apart). Bypassed it would be $-3300/16.8 = -197$.

N2 (medium) [7 marks].

a. $I_E = (5 - 0.7)/2.2\text{k} = 1.955$ mA, $r_e = 26/1.955 = 13.30\ \Omega$,
$V_{CB} = 10 - 0.99(1.955)(3.9) = 2.45$ V > 0, so the CBJ is reverse biased (active).

b. $Z_i = R_E \parallel r_e = 2200 \parallel 13.30 = \mathbf{13.2\ \Omega}$. $Z_o = R_C = \mathbf{3.9\ k\Omega}$.
$A_{v_{NL}} = \alpha R_C/r_e = 0.99(3900)/13.30 = \mathbf{290}$.
$A_{v_L} = 0.99(3.9\text{k} \parallel 10\text{k})/13.30 = 0.99(2806)/13.30 = \mathbf{209}$. Both in phase.

c. $R_s = 50\ \Omega$: $A_{v_s} = \frac{13.2}{63.2}(209) = \mathbf{43.7}$. $R_s = 1$ kΩ: $A_{v_s} = \frac{13.2}{1013.2}(209) = \mathbf{2.72}$. The CB input is so low that a 1 kΩ source throws away 98.7 % of the signal. CB only makes sense when it is driven from a very low impedance (or from a current source, such as a CE collector in a cascode).

N3 (medium–hard, multi-part) [10 marks].

a. $A_{v_L} = -164 \times \frac{500}{500+3810} = \mathbf{-19.0}$. $A_{v_s} = \frac{3.60}{4.60}(-19.0) = \mathbf{-14.9}$. The 500 Ω load kills 88 % of the gain.

b. Emitter AC resistance: $R_E \parallel R_L = 3.3\text{k} \parallel 0.5\text{k} = 434.2\ \Omega$.
$Z_{b2} = 100(12.60 + 434.2) = 44.7$ kΩ, $Z_{i2} = 220\text{k} \parallel 44.7\text{k} = \mathbf{37.1\ k\Omega}$.

c. Stage 1 is loaded by Z_{i2}: $A_{v1} = -\frac{3810 \parallel 37.1\text{k}}{23.2} = -\frac{3455}{23.2} = \mathbf{-148.9}$. Follower: $A_{v2} = \frac{434.2}{434.2+12.60} = \mathbf{0.972}$. Overall: $A_{v_s} = \frac{3.60}{4.60}(-148.9)(0.972) = \mathbf{-113}$. That is 7.6 times better than (a).

d. The follower's source is the CE output, $Z_o = 3.81$ kΩ, in parallel with R_B:
$R'_s = 3.81\text{k} \parallel 220\text{k} = 3.745$ kΩ.
$Z_o = R_E \parallel \left(r_e + \frac{R'_s}{\beta}\right) = 3300 \parallel (12.60 + 37.45) = \mathbf{49.3\ \Omega}$. Using the slide's $R_E \parallel r_e = 12.6\ \Omega$ would be wrong by a factor of 4.

N4 (design) [6 marks].

a. Keep $R_{E1} + R_{E2} = 980\ \Omega$ so the DC loop, and therefore the Q-point, is unchanged. Put C_E across R_{E2} only. $\frac{R_C}{r_e + R_{E1}} = 10 \Rightarrow R_{E1} = 381 - 23.2 = 357.8\ \Omega$. Choose $R_{E1} = 360\ \Omega$ and $R_{E2} = 620\ \Omega$ (both E24, and they sum to exactly 980 Ω).

b. $A_v = -\frac{3810}{23.2+360} = \mathbf{-9.94}$.
$Z_i = 8.214\text{k} \parallel 275.5(383.2) = 8.214\text{k} \parallel 105.6\text{k} = \mathbf{7.62\ k\Omega}$. With β = 100: $8.214\text{k} \parallel 38.3\text{k} = \mathbf{6.76\ k\Omega}$.

c. r_e becomes $26/1.232 = 21.1\ \Omega$. Bypassed: $-164.1 \to -180.5$, a **10 %** rise. Your design: $-9.94 \to -10.00$, a **0.6 %** rise. The partial bypass makes the gain about 17 times less sensitive to bias drift.

N5 (hard) [8 marks].

a. At AC, R_C, R_F (at the output end) and R_L are in parallel:
$A_{v_L} = -\frac{3.9\text{k} \parallel 220\text{k} \parallel 3.9\text{k}}{13.08} = -\frac{1933}{13.08} = \mathbf{-147.7}$.

b. $\frac{R_F}{1-A_{v_L}} = \frac{220\text{k}}{148.7} = 1.479$ kΩ. $Z_i = \beta r_e \parallel 1.479\text{k} = 1570 \parallel 1479 = \mathbf{762\ \Omega}$. The load halves the gain, so the voltage across R_F is about half as large and R_F draws about half the current. It looks like a bigger resistor from the base, so Z_i rises.

c. $A_{v_s} = \frac{762}{762+600}(-147.7) = \mathbf{-82.6}$. A full nodal solution gives -82.63.

d. $\frac{507}{507+600}(-292.9)\frac{3900}{3900+3832} = -67.7$. That is 18 % low, because it uses the unloaded Z_i (507 Ω) when the real Z_i with R_L connected is 762 Ω. The two-port shortcut assumes Z_i does not depend on R_L. For a feedback stage it does.

6. Quick recall check

Question

1. $A_v \approx -R_C/(r_e + R_E) \approx -4.7/1.2 \approx -3.9$. β is missing (and nearly r_e too), so the gain is stable.
2. $Z_o = 2\text{k} \parallel (10 + 2000/100) = 2\text{k} \parallel 30\ \Omega \approx 29.6\ \Omega$.
3. The emitter follower (R_L is in the emitter and is reflected into Z_b) and collector feedback (Miller: Z_i depends on the gain).
4. $A_{v_s} = \dfrac{Z_i}{Z_i + R_s} A_{v_L}$.
5. DC: $R_{E1} + R_{E2}$. AC: R_{E1} only, $A_v \approx -R_C/(r_e + R_{E1})$.

Multistage amplifiers

Answers and worked solutions · Day 14 · Fri 6 Nov · pack E8

5. Practice

From your course files

1. Prac 4 calculation section

DC is unchanged from Worked Example 2 (capacitors are open at DC): V_B = 4.772 V, I_E = 4.072 mA, r_e = 26 mV/4.072 mA = 6.39 Ω in both stages, V_{CE} = 6.97 V.

Stage 2 is now bypassed, so it presents only βr_e to stage 1:

$$Z_{i_2} = 15\ \text{k} \parallel 4.7\ \text{k} \parallel 200(6.39) = 941\ \Omega$$

$$A_{v_1} = -\frac{R_C \parallel Z_{i_2}}{r_e} = -\frac{2.2\ \text{k} \parallel 941}{6.39} = -103.2, \qquad A_{v_{2(NL)}} = -\frac{R_C}{r_e} = -\frac{2200}{6.39} = -344.5$$

$$A_{v_{T(NL)}} = (-103.2)(-344.5) \approx \mathbf{35.6 \times 10^3}$$

This is the Week 5 lecture example (Worked Example 1), which gets 36.1×10^3 with the rounded r_e = 6.34 Ω.

Why step 7 sees clipping. The output can swing only about $V_{CE} - 0.2 \approx 6.8$ V before stage 2 saturates, so the input must stay below $6.8/35.6 \times 10^3 \approx 0.19$ mV peak. Even the handout's suggested 4 mV would ask for about 140 V at the output, so the output clips hard until V_i is well under 1 mV.

WS6 Q10 (course question).

Q_2's emitter is grounded, so its input resistance $r_{\pi 2}$ sits directly from Q_1's collector to ground, in parallel with R_C. Q_2's collector is at V_{CC} and does not affect v_{out}.

$$v_o = -g_{m1} v_{\pi 1}(R_C \parallel r_{\pi 2}), \qquad v_{in} = [r_{\pi 1} + (1+\beta)R_E]\frac{v_{\pi 1}}{r_{\pi 1}}$$

$$\frac{v_o}{v_{in}} = \frac{-g_{m1} r_{\pi 1}(R_C \parallel r_{\pi 2})}{r_{\pi 1} + (1+\beta)R_E} = \frac{-(R_C \parallel r_{\pi 2})}{\dfrac{1}{g_{m1}} + \dfrac{1+\beta}{\beta}R_E}$$

β here is β_1. The point for multistage work: stage 2's input impedance ($r_{\pi 2}$) loads stage 1's collector.

WS6 Q12 (course question).

I_X = 1 mA and I_Y = 0.5 mA are the collector currents, and both BJTs are active:

$$I_{B1} = \frac{I_X}{\beta} = 0.01\ \text{mA}, \qquad I_{B2} = \frac{I_Y}{\beta} = 0.005\ \text{mA}$$

KCL at node A (the joined bases): $I_1 = I_{B1} + I_{B2}$ = 0.015 mA. Both emitters are grounded, so $V_B = V_{BE} = 0.7$ V, and

$$V_{BB} = V_B + R_1 I_1 = 0.7 + 5(0.015) = \mathbf{0.775\ V}$$

WS6 Q13 (course question).

1. Start from Q_2, where the emitter current is known:

$$I_{C_2} = \alpha_2 I_{E_2} = 0.98(120) = \mathbf{117.6\ mA}, \qquad I_{B_2} = I_{E_1} = (1-\alpha_2) I_{E_2} = \mathbf{2.4\ mA}$$

Then Q_1:

$$I_{B_1} = (1-\alpha_1) I_{E_1} = 0.01(2.4) = \mathbf{0.024\ mA}, \qquad I_{C_1} = \alpha_1 I_{E_1} = \mathbf{2.376\ mA}$$

KCL at node A (the joined collectors): $I_C = I_{C_1} + I_{C_2}$ = **119.976 mA**.

2. KVL from 20 V through the 0.1 kΩ resistor:

$$v_o = 20 - 0.1 I_C = 20 - 0.1(119.976) = \mathbf{8.00\ V}$$

WS6 Q14 (course question).

READING THE FIGURE: $+20$ V and -10 V rails. Node A is Q_1's base, tied to ground through 10 kΩ. Q_1 is npn: collector C goes through 10 kΩ to +20 V, and emitter B goes through 9.3 kΩ to −10 V. Q_2 is pnp: its base is at node C, emitter D goes through 9.3 kΩ to +20 V, and collector E goes through 10 kΩ to −10 V.

1. $\beta = \infty$ means $I_{B1} = 0$. No current flows in the 10 kΩ resistor, so $V_A = \mathbf{0}$ V.
2. Q_1 is npn, so $V_B = V_A - 0.7 = \mathbf{-0.7}$ V.
3. $I_{E1} = I_{C1} = \dfrac{-0.7 - (-10)}{9.3\ \text{k}} = 1.00$ mA, so $V_C = 20 - (10\ \text{k})(1\ \text{mA}) = \mathbf{10.0}$ V.
   Q_1 check: $V_{CE1} = 10.7$ V > 0.2 V, so it is active ✓.
4. Q_2 is **pnp**, so its emitter sits 0.7 V *above* its base: $V_D = V_C + 0.7 = \mathbf{10.7}$ V.
5. $I_{E2} = I_{C2} = \dfrac{20 - 10.7}{9.3\ \text{k}} = 1.00$ mA, so $V_E = -10 + (10\ \text{k})(1\ \text{mA}) = \mathbf{0}$ V.
6. Q_2 check: $V_{EC2} = 10.7 - 0 = 10.7$ V > 0.2 V, so it is **active** ✓.

Where the official solution goes wrong: it writes $V_D = 0.7$ V, which treats the emitter as 0.7 V above *ground* instead of above the base (node C = 10 V). It then gets $I_{C2} = 2.075$ mA and $V_E = 10.75$ V, and concludes that Q_2 is saturated. That conclusion is wrong. $I_{B2} = 0$ does not disconnect the base from node C; it only means no current flows in that wire. The

> symmetric resistor values (10 k / 9.3 k on both sides) are a strong hint that the designer intended 1 mA in each transistor and both active.

WS6 Q16 (course question).

> **Stage 1** (exact Thevenin; C_{E_1} is open at DC, so the whole 2.2 kΩ counts):
>
> $$R_{Th} = 33\ \text{k} \parallel 10\ \text{k} = 7.67\ \text{k}\Omega, \qquad E_{Th} = \frac{10(14)}{43} = 3.26\ \text{V}$$
>
> $$I_{B_1} = \frac{3.26 - 0.7}{7.67 + 101(2.2)} = 11.1\ \mu\text{A}, \qquad I_{C_1} = 100 I_{B_1} = 1.11\ \text{mA}$$
>
> V_{E_1} = 2.47 V.
>
> **Shared node** ($V_{B_2} = V_{C_1}$, with I_{B_2} neglected, as the course does):
>
> $V_{B_2} = 14 - 1.11(6.8) = 6.44\ \text{V}, \qquad V_{E_2} = 6.44 - 0.7 = 5.74\ \text{V}, \qquad I_{E_2} = \frac{5.74}{1.2\ \text{k}} = 4.78\ \text{mA}$
>
> $V_{C_2} = V_{CC}$ = 14 V, so V_{CE_2} = 14 − 5.74 = **8.26 V**, and V_{CE_1} = 6.44 − 2.47 = 3.97 V. Both active.
>
> The official solution gives 11.17 µA, 1.12 mA, 6.38 V, 5.68 V, 4.73 mA and 8.32 V. Its denominator 229.2 kΩ should be 229.9 kΩ, and it rounds I_{C_1} up; the method is the same.
>
> > **NOTE ON I_{B_2}:** $\beta_2 = 50$, so $I_{B_2} = 4.78\ \text{mA}/51 = 94$ µA. That is about 8 % of I_{C_1}, and it flows through the 6.8 kΩ resistor. Including it with node KCL gives $V_{B_2} = 5.87$ V and $I_{E_2} = 4.30$ mA. The course answer neglects it. Give the course answer, then add one line saying it is an approximation.

WS6 Q17 (course question).

> Q_1 is a pnp whose collector drives the base of the npn Q_2; Q_1's emitter and Q_2's collector share the node under the 220 Ω resistor.
>
> **(a)** Loop from 12 V through 220 Ω (carrying $\approx I_{C_2} = \beta_1\beta_2 I_{B_1}$), Q_1's emitter-base junction and the 1.8 MΩ to ground:
>
> $$I_{B_1} = \frac{V_{CC} - V_{EB_1}}{R_B + \beta_1\beta_2 R_C} = \frac{12 - 0.7}{1.8\ \text{M} + 80(160)(220)} = \mathbf{2.45\ \mu A}$$
>
> $I_{C_1} = \beta_1 I_{B_1} = 80(2.45\ \mu\text{A}) = \mathbf{196\ \mu A} = I_{B_2}, \qquad I_{C_2} = \beta_2 I_{B_2} = 160(196\ \mu\text{A}) = \mathbf{31.36\ mA}$
>
> **(b)**
>
> - $V_{B_1} = I_{B_1} R_B = 2.45\ \mu\text{A} \times 1.8\ \text{M}\Omega$ = **4.41 V**

- $V_{E_1} = 12 - I_C(220\ \Omega) = 12 - 31.36(0.22)$ = **5.1 V** (check: pnp, so $V_{E_1} = V_{B_1} + 0.7$ = 5.11 V ✓)
- $V_{B_2} = V_{BE_2}$ = **0.7 V**, and $V_{C_1} = V_{B_2}$ = **0.7 V**
- $V_{C_2} = V_{E_1}$ = **5.1 V**, and V_{E_2} = **0 V**

The solution uses $I_C \approx I_{C_2}$ in the 220 Ω. Adding I_{E_1} (0.2 mA) would lower V_{E_1} to 5.06 V, which does not change the answer at this precision.

WS6 Q18 (course question).

1. The two BJTs are identical, have the same V_{BE} (bases joined, emitters joined), so they carry the same currents: $I_{B_1} = I_{B_2} = I_B$ and $I_{C_1} = I_{C_2} = I_o = \beta I_B$.

The base wire takes $I_x = I_{B_1} + I_{B_2} = 2I_B$ from Q_1's collector node. KCL at that node:

$$I_{REF} = I_{C_1} + I_x = I_o + 2I_B = I_o + \frac{2I_o}{\beta} = I_o\left(1 + \frac{2}{\beta}\right)$$

$$\boxed{\frac{I_o}{I_{REF}} = \frac{1}{1 + \dfrac{2}{\beta}}}$$

2. For very large β, $2/\beta \approx 0$, so $I_o/I_{REF} = 1$: the output current copies the reference. (With β = 100 the error is 2 %.)

WS6 Q19 (course question).

1. Assume active. The current source sets I_E = 1 mA, so

$$I_B = \frac{I_E}{1+\beta} = \frac{1}{101} = 0.0099\ \text{mA}, \qquad I_C = I_E - I_B = 0.9901\ \text{mA}$$

The base current flows out of ground through R_B into the base:

- $V_B = -R_B I_B = -100(0.0099)$ = **−0.990 V**
- $V_E = V_B - V_{BE} = -0.990 - 0.7$ = **−1.690 V**
- $V_C = V_{CC} - R_C I_C = 10 - 7.5(0.9901)$ = **2.574 V**

Check: $I_B > 0$ and $V_{CE} = 2.574 - (-1.690) = 4.26$ V > 0.2 V, so active.

2. With very large β, $I_{REF} = I$ = 1 mA. Q_1 is diode-connected, so $V_{C_1} = V_{B_1} = -V_{EE} + 0.7 = -10 + 0.7 = -9.3$ V. The resistor runs from V_{CC} to Q_1's collector:

$$R = \frac{V_{CC} - V_{C_1}}{I_{REF}} = \frac{10 - (-9.3)}{1\ \text{mA}} = \mathbf{19.3\ k\Omega}$$

New questions

N1 [5 marks]. Two-port cascade.

a. Stage 1 drives Z_{i_2} through Z_{o_1}:

$$A_{v_1} = A_{v_{NL1}} \frac{Z_{i_2}}{Z_{i_2} + Z_{o_1}} = -120 \cdot \frac{1.5}{4.5} = -40$$

Stage 2 drives R_L through Z_{o_2}:

$$A_{v_2} = -80 \cdot \frac{6}{10} = -48$$

$$A_{v_T} = (-40)(-48) = \mathbf{1920}$$

b. $A_{v_s} = \dfrac{2}{2+1}(1920) = \mathbf{1280}$

c. $A_{v_{NL1}} A_{v_{NL2}} = 9600$, so the loading costs a factor of $9600/1280 = \mathbf{7.5}$. Most of the loss is at the interstage divider (×1/3), because $Z_{i_2} < Z_{o_1}$.

N2 [10 marks]. Two-stage CE–CE cascade.

a. $\beta R_E = 150\ \text{k}\Omega \geq 10 R_2 = 100\ \text{k}\Omega$ ✓, so the approximate method holds (just).

$$V_B = \frac{10}{57}(12) = 2.105\ \text{V}, \quad V_E = 1.405\ \text{V}, \quad I_E = 1.405\ \text{mA}$$

$$V_{CE} = 12 - 1.405(3.3 + 1) = 5.96\ \text{V (active)}, \quad r_e = \frac{26}{1.405} = 18.5\ \Omega$$

The two stages are identical. (The exact Thévenin method gives $I_E = 1.33$ mA. That is a 5 % difference because the 10× rule is only just met.)

b. $Z_{i_2} = 47\ \text{k} \parallel 10\ \text{k} \parallel (150)(18.5) = 47\ \text{k} \parallel 10\ \text{k} \parallel 2.775\ \text{k} = \mathbf{2.08\ k\Omega}$

$$A_{v_1} = -\frac{3.3\ \text{k} \parallel 2.08\ \text{k}}{18.5} = -\frac{1274}{18.5} = \mathbf{-68.9}$$

c. $A_{v_2} = -\dfrac{3.3\ \text{k} \parallel 10\ \text{k}}{18.5} = -\dfrac{2481}{18.5} = \mathbf{-134.1}$

$$A_{v_T} = (-68.9)(-134.1) = \mathbf{+9.24 \times 10^3}$$

d. $Z_{i_1} = Z_{i_2} = \mathbf{2.08\ k\Omega}$ (identical stage).

$$A_{v_s} = \frac{2.08}{2.08 + 0.6}(9238) = \mathbf{7.17 \times 10^3}$$

$$v_o = (7167)(20\ \mu\text{V}) = \mathbf{143}\ \text{mV peak}$$

This is in phase with v_s, and well inside the swing allowed by $V_{CE} \approx 6$ V.

N3 [10 marks]. CE–EF cascade (buffering a low-resistance load).

a. Checks: $\beta R_{E1} = 144\text{ k} \geq 120\text{ k}$ ✓ and $\beta R_{E2} = 264\text{ k} \geq 220\text{ k}$ ✓. Stage 1: $V_{B1} = \frac{12}{68}(15) = 2.647$ V, so $I_{E1} = \frac{1.947}{1.2\text{ k}} = \mathbf{1.62}\text{ mA}$ and $r_{e1} = \mathbf{16.0}\ \boldsymbol{\Omega}$. Also $V_{C1} = 7.37$ V and $V_{CE1} = 5.43$ V, so Q1 is active. Stage 2: $V_{B2} = 7.5$ V and $V_{E2} = 6.8$ V, so $I_{E2} = \frac{6.8}{2.2\text{ k}} = \mathbf{3.09}\text{ mA}$ and $r_{e2} = \mathbf{8.41}\ \boldsymbol{\Omega}$. $V_{CE2} = 8.2$ V, so Q2 is active.

b. The follower's emitter load is $R_{E2} \parallel R_L = 2.2\text{ k} \parallel 500 = 407.4\ \Omega$.

$$Z_{b2} = \beta(r_{e2} + R_{E2} \parallel R_L) = 120(415.8) = 49.9\text{ k}\Omega$$

$$Z_{i_2} = 22\text{ k} \parallel 22\text{ k} \parallel 49.9\text{ k} = \mathbf{9.01}\ \mathbf{k\Omega}$$

c. $A_{v_1} = -\dfrac{4.7\text{ k} \parallel 9.01\text{ k}}{16.0} = -\dfrac{3089}{16.0} = \mathbf{-192.8}$

$$A_{v_2} = \frac{407.4}{8.41 + 407.4} = \mathbf{0.980}$$

$$A_{v_T} = \mathbf{-188.9}$$

d. $A_v = -\dfrac{4.7\text{ k} \parallel 500}{16.0} = \mathbf{-28.2}$

The follower raises the gain by a factor of about 6.7. A 500 Ω load directly on a 4.7 kΩ output impedance collapses the CE gain. The follower presents about 9 kΩ to Q1 instead. It also has a low output impedance, $Z_{o2} = R_{E2} \parallel \left(r_{e2} + \frac{R_C \| R_3 \| R_4}{\beta}\right) \approx 35\ \Omega$, so the 500 Ω load barely loads it.

N4 [7 marks]. Darlington emitter follower.

a. $\beta_D = (100)(80) = \mathbf{8000}$

$$I_B = \frac{18 - 1.6}{3.3\text{ M} + (8000)(390)} = \frac{16.4}{6.42\text{ M}} = \mathbf{2.55}\ \boldsymbol{\mu}\mathbf{A}$$

$I_E \approx \beta_D I_B = \mathbf{20.4}\text{ mA}$, $\quad V_E = (20.4\text{ mA})(390) = \mathbf{7.97}\text{ V}$, $\quad V_{CE} = 18 - 7.97 = \mathbf{10.0}\text{ V}$

b. $Z_{i_2} = \beta_D R_E = 3.12\ \text{M}\Omega$, so $Z_i = 3.3\text{ M} \parallel 3.12\text{ M} = \mathbf{1.60}\ \mathbf{M\Omega}$

$$A_i = \frac{\beta_D R_B}{R_B + \beta_D R_E} = \frac{8000(3.3)}{6.42} = \mathbf{4.11 \times 10^3}, \qquad A_v \approx \mathbf{1}\ (\text{in phase})$$

Note that R_B now limits Z_i. The Darlington itself presents 3.1 MΩ.

N5 [6 marks]. Current mirror design.

a. $V_{C1} = V_{B1} = 0.7$ V, so $R_{REF} = \dfrac{12 - 0.7}{2\ \text{mA}} = \mathbf{5.65\ k\Omega}$.

b. $I_{REF} = 2$ mA, so $I_o = \dfrac{I_{REF}}{1 + 2/\beta} = \dfrac{2}{1.025} = \mathbf{1.95\ mA}$ (2.4 % low).

c. $I_{REF} = I_o(1 + 2/\beta) = 2.05$ mA, so $R_{REF} = \dfrac{11.3}{2.05\ \text{mA}} = \mathbf{5.51\ k\Omega}$.

d. Q2 is active while $V_{CE2} \geq 0.2$ V: $R_L \leq \dfrac{12 - 0.2}{2\ \text{mA}} = \mathbf{5.9\ k\Omega}$.

6. Quick recall check

Question

1. Stage 1's gain changes, because its collector resistor is now in parallel with Z_{i_2}: $A_{v_1} = -(R_C \parallel Z_{i_2})/r_e$.
2. $Z_{i_2} = R_1 \parallel R_2 \parallel \beta(r_e + R_E)$.
3. $\beta_D = 6000$, $Z_i \approx \beta_D R_E = 6$ MΩ.
4. Q1 sees only r_{e2} as its load, so its gain is about −1. Its collector–base capacitance is therefore barely multiplied (little Miller effect), while Q2 (common base) supplies the full gain R_C/r_e.
5. $1/(1 + 2/50) = 0.962$.

FET physics

Answers and worked solutions · Day 15 · Sat 7 Nov · pack E9

5. Practice

From your course files

WS7 Q3 (course question).

This is Worked Example 2, where the full working is laid out.

a. −2 V lies between $V_P = -4.5$ V and 0, so Shockley applies:

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2 = 6\left(1 - \frac{-2}{-4.5}\right)^2 = 6(0.5556)^2 = \mathbf{1.852\ mA}$$

b. $$V_{GS} = V_P\left(1 - \sqrt{\frac{I_D}{I_{DSS}}}\right) = -4.5\left(1 - \sqrt{\tfrac{3}{6}}\right) = \mathbf{-1.318\ V}$$

Check: 3 mA is $I_{DSS}/2$, so this is the "$0.3V_P$" point ($0.3 \times -4.5 = -1.35$ V). The two values agree.

WS7 Q6 (course question).

Hint: $V_{GS} = V_P/2$, so $I_D = I_{DSS}/4$.

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2 \Rightarrow 3\ \text{mA} = I_{DSS}\left(1 - \frac{-3}{-6}\right)^2 = 0.25\, I_{DSS}$$

$$I_{DSS} = \frac{3\ \text{mA}}{0.25} = \mathbf{12\ mA}$$

WS7 Q5 (course question).

Use the four-point method (2.2):

V_{GS} (V)	0	$0.3V_P = -1.2$	$V_P/2 = -2$	−3 (extra point, Shockley)	$V_P = -4$
I_D (mA)	$I_{DSS} = 12$	$I_{DSS}/2 = 6$	$I_{DSS}/4 = 3$	0.75	0

Plot the points on I_D (mA) against V_{GS} (V) and draw a smooth concave-up curve that meets the V_{GS} axis tangentially at −4 V. Label I_{DSS} and V_P.

Notes on the official solution:

- It writes "Given $I_{DSS} = 6$ mA", which is a typo. All of its numbers use 12 mA, which is correct.
- The exact V_{GS} for $I_{DSS}/2$ is $-4(1 - \sqrt{0.5}) = -1.172$ V. The −1.2 V is the shorthand $0.3V_P$ value, which is fine for a sketch.

WS7 Q1 (course question).

Assume an n-channel JFET, so $V_P = -6$ V. Use method 2.3: compute the saturation current for each 1 V step of V_{GS} with Shockley, $I_D = 12(1 - V_{GS}/(-6))^2$ mA.

V_{GS} (V)	0	−1	−2	−3	−4	−5	−6 (V_P)
I_D (mA)	12 (I_{DSS})	8.33	5.33	3	1.33	0.333	0

Draw each as a horizontal line on the I_D–V_{DS} axes, starting at its knee $V_{DS} = V_{GS} - V_P$ (6, 5, 4, 3, 2, 1 V), with a curve back to the origin for the ohmic region. Join the knees with a dashed pinch-off locus. The $V_{GS} = -6$ V curve lies on the V_{DS} axis. Note how the curves crowd together as V_{GS} approaches V_P.

The official solution writes "$3V_{GS} = V_P$", which is a typo for "$V_{GS} = V_P$".

WS7 Q7 (course question).

"$V_{GS} = -V_P$ to 1 V" means from $V_{GS} = V_P = -8$ V up to +1 V (the sheet's sign is a slip: $-V_P$ would be +8 V).

Shockley, $I_D = 12(1 - V_{GS}/(-8))^2$ mA:

V_{GS} (V)	−8 (V_P)	−6	−4 ($V_P/2$)	−2.4 ($0.3V_P$)	0	+1
I_D (mA)	0	0.75	3	6 ($I_{DSS}/2$)	12 (I_{DSS})	**15.19**

- **Transfer curve:** plot these points and draw a smooth concave-up curve. It continues to the right of the I_D axis and rises above I_{DSS} for $V_{GS} > 0$ (enhancement mode).
- **Drain curves:** one flat saturation line per V_{GS} at the current in the table, with the knee at $V_{DS} = V_{GS} - V_P$, and the +1 V curve above the I_{DSS} curve.

Notes on the official solution:

- Its point list stops at 0 V. Add $V_{GS} = +1$ V: $I_D = 12(1 + 1/8)^2 = 15.19$ mA, so the curve continues above I_{DSS}.
- The exact V_{GS} for $I_{DSS}/2$ is −2.343 V. The −2.4 V is the shorthand $0.3V_P$ value.

WS7 Q2 (course question).

Read the flat (saturation) part of each curve. "$V_{DS} > V_P$" means V_{DS} beyond the pinch-off value, $|V_P| = 4$ V.

- $V_{GS} = 0$ V: $I_D = 8$ mA (I_{DSS}).
- $V_{GS} = -1$ V: $I_D = 4.5$ mA.

Difference: $8 - 4.5 = \mathbf{3.5\ mA}$.

Check: Fig. 1 is the slide-10 device. Shockley with $I_{DSS} = 8$ mA, $V_P = -4$ V gives $8(1 - 0.25)^2 = 4.5$ mA at −1 V. It matches.

WS7 Q4 (course question).

Read the scales first: 1 mA per vertical division, 1 V per horizontal division. So $V_{DS} = 10$ V is the right-hand edge of the grid. $I_{DSS} \approx 9$ mA and $V_P \approx -3$ V.

$V_{GS} = -0.7$ V lies between the −0.5 V curve (≈ 6.5 mA) and the −1 V curve (≈ 4 mA). The curves are not evenly spaced, so interpolate a little below the midpoint: $I_D \approx 5.5$ **mA** (the official answer).

Shockley cross-check with the figure's own values: $9(1 - 0.7/3)^2 = \mathbf{5.29\ mA}$. Both are acceptable, because a graph reading depends on how you interpolate between the −0.5 V and −1 V curves.

The official solution text writes "$V_{GS} = 0.7$ V", which is a typo for −0.7 V.

New questions

Q1 [3 marks]

a. $V_P > 0$ for p-channel, and +2 V lies between 0 and +6 V, so Shockley applies:

$$I_D = 6\left(1 - \frac{2}{6}\right)^2 = 6(0.6667)^2 = \mathbf{2.67\ mA}$$

b. 1.5 mA is $I_{DSS}/4$, so expect $V_P/2$:

$$V_{GS} = 6\left(1 - \sqrt{1.5/6}\right) = 6(0.5) = \mathbf{+3\ V}$$

c. Conventional current flows from **source to drain**, so it leaves by the drain terminal. V_{DS} is **negative**. The equation is the same as for n-channel, but with V_P and V_{GS} positive.

Q2 [4 marks]

a. $k = \dfrac{9\ \text{mA}}{(8-2)^2} = \dfrac{9}{36} = \mathbf{0.25\ mA/V^2}$

b. $I_D(5\ \text{V}) = 0.25(5-2)^2 = \mathbf{2.25\ mA}$. At 1.5 V the gate is below $V_T = 2$ V, so there is no channel and $I_D = \mathbf{0}$.

c. $V_{GS} = V_T + \sqrt{I_D/k} = 2 + \sqrt{4/0.25} = 2 + 4 = \mathbf{6\ V}$. The other root, $2 - 4 = -2$ V, is below V_T, so it is rejected.

d. $V_{DS(\text{sat})} = V_{GS} - V_T = 6 - 2 = \mathbf{4\ V}$. Saturation needs $V_{DS} \geq 4$ V.

Q3 [5 marks]

a.

- $I_D(-1) = 8\left(1 - \frac{-1}{-3}\right)^2 = 8(0.6667)^2 = \mathbf{3.56\ mA}$
- $I_D(-1.5) = 8(0.5)^2 = \mathbf{2.00\ mA}$ (this is the $V_P/2$ point, $I_{DSS}/4$)
- $I_D(+1) = 8\left(1 - \frac{1}{-3}\right)^2 = 8(1.3333)^2 = \mathbf{14.22\ mA}$

b. $V_{GS} < 0$ (−1 V and −1.5 V) is **depletion** mode, with $I_D < I_{DSS}$. $V_{GS} > 0$ (+1 V) is **enhancement** mode, with $I_D > I_{DSS}$.

c. $V_{GS} = -3(1 - \sqrt{4/8}) = \mathbf{-0.879\ V}$. The shorthand gives $0.3V_P = -0.9$ V, about 0.02 V away.

Q4 [8 marks] (chained)

a. Take the ratio of the two Shockley equations, so that I_{DSS} cancels:

$$\sqrt{\frac{4.5}{0.5}} = 3 = \frac{1 - (-1)/V_P}{1 - (-3)/V_P} = \frac{1+u}{1+3u}, \quad u = \frac{1}{V_P}$$

$$1 + u = 3 + 9u \ \Rightarrow\ u = -0.25 \ \Rightarrow\ \mathbf{V_P = -4\ V}$$

b. $I_{DSS} = \dfrac{4.5}{(1 - (-1)/(-4))^2} = \dfrac{4.5}{0.5625} = \mathbf{8\ mA}$

Check with the second point: $8(1 - 0.75)^2 = 0.5$ mA. It matches.

c. $I_D = 8(1 - 0.5)^2 = \mathbf{2\ mA}$. This is the $V_P/2 \to I_{DSS}/4$ point. (This is the slide-10 device.)

d. $V_{DS(\text{sat})} = V_{GS} - V_P = -2 - (-4) = \mathbf{2\ V}$

e. $r_d = \dfrac{r_o}{(1 - V_{GS}/V_P)^2} = \dfrac{250}{(0.5)^2} = \mathbf{1\ k\Omega}$

The resistance rises as V_{GS} goes more negative. That is the voltage-controlled-resistor behaviour.

Q5 [6 marks]

a. $V_{GS,n} = V_i$ and $V_{GS,p} = V_i - 5$.

V_i	$V_{GS,n}$	nMOS	$V_{GS,p}$	pMOS	V_o
0	0 V	off (< 1 V)	−5 V	on (< −1 V)	5 V
2.5 V	2.5 V	on	−2.5 V	on	transition region
5 V	5 V	on	0 V	off	0 V

At 2.5 V both transistors conduct. The output is in the transition (amplifier) region, and a current flows from V_{DD} to ground. In either logic state one device is off, so there is no static current.

b. $NM_L = 2.1 - 0.1 = \mathbf{2.0\ V}$. $NM_H = 4.9 - 2.9 = \mathbf{2.0\ V}$. The ideal inverter has $V_{IL} = V_{IH} = 2.5$ V, $V_{OL} = 0$ and $V_{OH} = 5$ V, giving 2.5 V each. The real margins are smaller because the transition region has finite width.

c. $t_P = \frac{1}{2}(12 + 18) = \mathbf{15\ ns}$

Q6 [6 marks] (hard, chained)

a. $k = \dfrac{5}{(-6.5 - (-1.5))^2} = \dfrac{5}{25} = \mathbf{0.2\ mA/V^2}$

b. $|I_D| = 0.2(-4 + 1.5)^2 = 0.2(6.25) = \mathbf{1.25\ mA}$

c. $(V_{GS} + 1.5)^2 = 1.8/0.2 = 9$, so $V_{GS} + 1.5 = \pm 3$. The roots are $V_{GS} = -4.5$ V and $V_{GS} = +1.5$ V. A pMOS is on only for $V_{GS} < V_T = -1.5$ V, so $\mathbf{V_{GS} = -4.5\ V}$. The +1.5 V root is not physical for this device. At that point $V_{DS(\text{sat})} = V_{GS} - V_T = -3$ V, so saturation needs $V_{DS} \le -3$ V.

d. $V_{GS,p} = V_i - V_{DD} = 0.5 - 5 = \mathbf{-4.5\ V}$, which is the part (c) point, so the saturation current is $\mathbf{1.8\ mA}$. The nMOS has $V_{GS} = 0.5\ \text{V} < 1.5$ V, so it is **off**. No

current path exists, the pMOS pulls the output to V_{DD}, and $V_o \approx 5$ V (logic 1). Because no current flows, the pMOS actually sits in its ohmic region with $V_{DS} \approx 0$. The 1.8 mA is only what it *could* supply, e.g. to charge a load capacitance.

6. Quick recall check

Question

1. $V_{GS} = 0$ V and $V_{DS} \geq |V_P|$ (saturation). It is the maximum JFET current.
2. $I_{DSS}/4$ (the $V_P/2$ point). $V_{GS} \approx 0.3V_P = -1.5$ V (exactly −1.46 V).
3. The enhancement MOSFET: $I_D = k(V_{GS} - V_T)^2$ for $V_{GS} > V_T$, and 0 otherwise.
4. $V_{GS} = 0 - 5 = -5$ V. It is **on**, and the output is high.
5. $NM_L = V_{IL} - V_{OL}$ and $NM_H = V_{OH} - V_{IH}$.

FET DC bias

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5. Practice

From your course files

WS8 Q2 (course question).

> **GIVEN:** $V_{DD} = 18$ V, $R_D = 2$ kΩ, $R_G = 2$ MΩ, $V_{GG} = 4$ V with its + terminal at ground, $I_{DSS} = 8$ mA, $V_P = -4$ V.
>
> $V_{GS} = -V_{GG} = -4\text{ V} = V_P$. So $I_D = 8(1-1)^2 = 0$ mA. The JFET is pinched off (cut off).
>
> $V_D = 18 - 0(2) = 18$ V.
>
> **ANSWER:** $V_{GS} = -4$ V, $V_D = 18$ V. The point of this question is to recognise $V_{GS} = V_P$ and avoid computing anything further.

WS8 Q1 (course question).

> **Given (from figure):** $V_{DD} = 14$ V, $R_D = 1.8$ kΩ, $R_G = 1$ MΩ, $V_{GG} = 1.5$ V with its + terminal at ground, $I_{DSS} = 12$ mA, $V_P = -4$ V.
>
> **(a) Transfer curve points:**
>
> | V_{GS} (V) | 0 | -1.2 ($0.3V_P$) | -2 ($0.5V_P$) | -4 |
> |---|---|---|---|---|
> | I_D (mA) | 12 | 6 | 3 | 0 |
>
> Draw a smooth parabola through these points.
>
> **(b) Graphical:** $V_{GS} = -V_{GG} = -1.5$ V. This is a vertical line. It meets the curve at about $I_{DQ} \approx 4.7$ mA. Then $V_{DSQ} = 14 - 4.7(1.8) \approx 5.5$ V.
>
> **(c) Shockley:**
>
> $$I_{DQ} = 12\left(1 - \frac{-1.5}{-4}\right)^2 = 12(0.625)^2 = 4.69\text{ mA}$$
>
> $$V_{DSQ} = 14 - (4.69)(1.8) = 5.56\text{ V}$$
>
> The two methods agree to within graph-reading accuracy.
>
> **Saturation check:** $V_{GS} - V_P = 2.5$ V, and $5.56 \geq 2.5$. Saturated.

ANSWER: $I_{DQ} = 4.69$ mA, $V_{DSQ} = 5.56$ V.

WS8 Q3 (course question).

GIVEN: $V_{DD} = 18$ V, $R_D = 1.5$ kΩ, $R_S = 750$ Ω, $R_G = 1$ MΩ to ground, $I_{DSS} = 10$ mA, $V_P = -4$ V.

Bias line: $V_G = 0$, so $V_{GS} = -0.75 I_D$ (I_D in mA).

Substitute:

$$I_D = 10\left(1 - \frac{-0.75 I_D}{-4}\right)^2 = 10\,(1 - 0.1875 I_D)^2$$

$$0.3516\, I_D^2 - 4.75\, I_D + 10 = 0$$

The roots are:

- $I_D = 10.90$ mA, which gives $V_{GS} = -8.18$ V. This is below $V_P = -4$ V, so reject it.
- $I_D = 2.61$ mA, which gives $V_{GS} = -1.96$ V. This is valid.

ANSWER: $I_{DQ} = 2.61$ mA (with $V_{GSQ} = -1.96$ V).

As an extra check, $V_{DS} = 18 - 2.61(2.25) = 12.1$ V, which is at least $V_{GS} - V_P = 2.04$ V, so it is saturated. The figure in Section 3 shows this crossing.

WS8 Q4 (course question).

GIVEN: $V_{DD} = 18$ V, $R_D = 2$ kΩ, $R_S = 0.51$ kΩ, $R_G = 1$ MΩ to ground, $V_P = -4$ V, measured $V_S = 1.7$ V.

a. $I_{DQ} = V_S / R_S = 1.7 / 0.51 = 3.33$ mA.

b. $V_G = 0$, so $V_{GSQ} = V_G - V_S = -1.7$ V.

c. Rearrange Shockley:

$$I_{DSS} = \frac{I_D}{(1 - V_{GS}/V_P)^2} = \frac{3.333}{(1 - 0.425)^2} = \frac{3.333}{0.3306} = 10.08\ \text{mA}$$

d. $V_D = 18 - 3.333(2) = 11.33$ V.

e. $V_{DS} = V_D - V_S = 11.33 - 1.7 = 9.63$ V.

Saturation: $V_{GS} - V_P = 2.3$ V, and $9.63 \geq 2.3$. Saturated.

ANSWERS: 3.33 mA, -1.7 V, 10.1 mA, 11.33 V, 9.63 V.

WS9 Q1 (course question).

> **READING THE FIGURE:** the gate (the terminal with the arrow) is grounded. The left channel terminal goes through 2.2 kΩ to +14 V, so that is the drain. The right terminal goes through 0.39 kΩ to ground, and that is the source, where V_S is measured. $I_{DSS} = 6$ mA, $V_P = -6$ V.
>
> Bias line: $V_G = 0$, so $V_{GS} = -0.39 I_D$.
>
> Substitute: $I_D = 6(1 - 0.065 I_D)^2$. The roots are:
>
> - 66.7 mA, which gives $V_{GS} = -26$ V. Reject it.
> - 3.55 mA, which gives $V_{GS} = -1.38$ V. This is valid.
>
> $V_S = I_D R_S = 3.55(0.39) = 1.38$ V.
>
> Check: $V_D = 14 - 3.55(2.2) = 6.19$ V and $V_{DS} = 4.80$ V. Also $V_{GS} - V_P = 4.62$ V, so it is saturated, but only by a small margin.
>
> **ANSWER:** $V_S = 1.38$ V.

WS9 Q3 (course question).

> **READING THE FIGURE:** the gate is grounded. $R_D = 3$ kΩ goes to +20 V. $R_S = 1.2$ kΩ goes to **+2 V**, which is positive. $V_{DS} = 4$ V is given. No device parameters are needed.
>
> Output KVL: $V_{DS} = 20 - 2 - I_D(3 + 1.2)$, so $4 = 18 - 4.2 I_D$.
>
> $I_D = 14/4.2 = 3.33$ mA.
>
> $V_D = 20 - 3.33(3) = 10$ V. $V_S = 2 + 3.33(1.2) = 6$ V. Check: $V_{DS} = 10 - 6 = 4$ V. This matches.
>
> $V_{GS} = V_G - V_S = 0 - 6 = -6$ V.
>
> **ANSWERS:** $I_D = 3.33$ mA, $V_D = 10$ V, $V_S = 6$ V, $V_{GS} = -6$ V. Because the source supply is positive, it adds to the source voltage instead of subtracting.

WS9 Q2 (course question).

> **READING THE FIGURE:** the gate is grounded. The drain goes through 2.2 kΩ to +16 V. The source goes through 2.2 kΩ to −4 V. $I_{DSS} = 6$ mA, $V_P = -6$ V. The figure labels the gate–source voltage "V_{DSQ}". This is a typo for V_{GSQ}, because the + sign is at the gate.
>
> Source voltage: $V_S = -4 + 2.2 I_D$. So $V_{GS} = 0 - V_S = 4 - 2.2 I_D$.

Substitute: $I_D = 6\left(1 + \dfrac{4 - 2.2I_D}{6}\right)^2 = 6\,(1.667 - 0.367I_D)^2$. The roots are:

- 7.62 mA, which gives $V_{GS} = -12.8$ V. Reject it.
- 2.71 mA, which gives $V_{GS} = -1.97$ V. This is valid.

a. $I_{DQ} = 2.71$ **mA,** $V_{GSQ} = -1.97$ **V.**

b. $V_S = -4 + 2.71(2.2) = +1.97$ V. $V_D = 16 - 2.71(2.2) = 10.03$ V.
$V_{DS} = 10.03 - 1.97 = 8.07$ V.

Using the formula from the slides:
$V_{DS} = V_{DD} + V_{SS} - I_D(R_D + R_S) = 20 - 2.71(4.4) = 8.07$ V. This agrees.

Saturation: $V_{GS} - V_P = 4.03$ V, and $8.07 \geq 4.03$. Saturated.

ANSWERS: 2.71 mA, -1.97 V, $V_{DS} = 8.07$ V, $V_S = 1.97$ V.

WS8 Q5 (course question).

GIVEN: $V_{DD} = 16$ V, $R_1 = 36$ kΩ, $R_2 = 12$ kΩ, $R_D = 2$ kΩ, $I_{DSS} = 12$ mA, $V_P = -8$ V. The target is $V_D = 10$ V.

1. $I_D = (16 - 10)/2 = 3$ mA.
2. Invert Shockley: $V_{GS} = V_P(1 - \sqrt{I_D/I_{DSS}}) = -8(1 - \sqrt{0.25}) = -4$ V.
3. $V_G = 12(16)/(36 + 12) = 4$ V.
4. $V_S = V_G - V_{GS} = 4 - (-4) = 8$ V.
5. $R_S = V_S/I_D = 8/3 = 2.67$ kΩ.

Answer (standard method): $R_S = 2.67$ kΩ.

Saturation check, which the question does not ask for: $V_{DS} = 10 - 8 = 2$ V, but $V_{GS} - V_P = -4 + 8 = 4$ V. Since $2 < 4$, the JFET would actually be in the **ohmic region** at this Q-point, so Shockley's equation does not strictly apply. The expected workshop answer is still 2.67 kΩ. I checked with an ohmic-region (triode) model, which is not part of this course: with $R_S = 2.67$ kΩ, the real circuit gives $I_D \approx 2.89$ mA and $V_D \approx 10.2$ V, so the design is still close. In an exam, give 2.67 kΩ and add one line noting the saturation check fails. That shows the examiner you know the limits of the model.

WS8 Q6 (course question).

GIVEN: $V_{DD} = 24$ V, $R_1 = 10$ MΩ, $R_2 = 6.8$ MΩ, $R_D = 2.2$ kΩ, $R_S = 0.75$ kΩ, $V_{GS(\text{Th})} = 3$ V, $I_{D(\text{on})} = 5$ mA at $V_{GS(\text{on})} = 6$ V.

1. $k = 5/(6 - 3)^2 = 0.556$ mA/V².
2. $V_G = 6.8(24)/16.8 = 9.71$ V. Bias line: $V_{GS} = 9.71 - 0.75I_D$.

3. Substitute: $I_D = 0.556\,(9.71 - 0.75I_D - 3)^2 = 0.556\,(6.71 - 0.75I_D)^2$.
4. The roots are:
   - $I_D = 16.1$ mA, which gives $V_{GS} = -2.39$ V. This is below V_T, so reject it.
   - $I_D = 4.97$ mA, which gives $V_{GS} = 5.99$ V. This is valid.

a. $I_{DQ} = 4.97$ **mA,** $V_{GSQ} = 5.99$ **V.** The device is running almost exactly at its $I_{D(\text{on})}$ point.

b. $V_D = 24 - 4.97(2.2) = 13.07$ V. $V_S = 4.97(0.75) = 3.72$ V.

Saturation: $V_{DS} = 9.35$ V, and $V_{GS} - V_T = 2.99$ V, so it is saturated.

ANSWERS: 4.97 mA, 5.99 V, $V_D = 13.1$ V, $V_S = 3.72$ V.

WS9 Q4 (course question).

READING THE FIGURE: this is an n-channel D-MOSFET (continuous channel bar). The gate is grounded. $R_D = 2.2$ kΩ goes to +18 V. $R_S = 0.39$ kΩ goes to −4 V. $I_{DSS} = 8$ mA, $V_P = -8$ V.

Standard method:

$V_S = -4 + 0.39I_D$, so $V_{GS} = 4 - 0.39I_D$.

Substitute: $I_D = 8\left(1 + \dfrac{4 - 0.39I_D}{8}\right)^2 = 8(1.5 - 0.04875I_D)^2$. The roots are:

- 105 mA, which gives $V_{GS} = -37$ V. Reject it.
- 9.01 mA, which gives $V_{GS} = +0.49$ V. This is allowed for a D-MOSFET.

a. $I_{DQ} = 9.01$ mA, $V_{GSQ} = +0.49$ V.

b. $V_S = -4 + 9.01(0.39) = -0.49$ V. $V_D = 18 - 9.01(2.2) = -1.81$ V. So $V_{DS} = V_D - V_S = -1.32$ V.

This result is impossible. V_{DS} cannot be negative for an n-channel device biased this way. The largest current the resistors can pass is $(18 + 4)/(2.2 + 0.39) = 8.49$ mA, which happens when $V_{DS} = 0$. That is less than the 9.01 mA that Shockley's equation predicts. So the saturation check fails, and the MOSFET is in the ohmic region.

An ohmic-region (triode) model, which is not part of this course, gives about $I_D \approx 7.0$ mA, $V_{GS} \approx +1.26$ V, $V_S \approx -1.26$ V and $V_{DS} \approx 3.8$ V. You cannot do this calculation with this course's tools.

What to write in an exam: give the Shockley values for (a). For (b), state $V_{DS} = -1.3$ V, then add that this violates $V_{DS} \geq V_{GS} - V_P$, so the device is not in saturation and the Shockley Q-point is not valid. I believe this is the textbook problem the question is taken

from. Its answer is usually quoted without this check, so the intended answer is probably $I_D \approx 9$ mA, $V_{GS} \approx 0.5$ V, $V_{DS} \approx -1.3$ V, $V_S \approx -0.49$ V.

New questions

N1. [4 marks]

$I_D = 8(1 - 2/6)^2 = 8(0.6667)^2 = 3.56$ mA.

$V_{DS} = 15 - 3.56(1.5) = 9.67$ V.

Saturation: $V_{GS} - V_P = 4$ V, and $9.67 \geq 4$. Saturated.

ANSWER: $I_{DQ} = 3.56$ mA, $V_{DSQ} = 9.67$ V.

N2. [6 marks]

a. The bias line is $V_{GS} = -I_D(1\text{ k}) = -I_D$. Substitute: $I_D = 12(1 - 0.2I_D)^2$, which gives $0.48I_D^2 - 5.8I_D + 12 = 0$.

The roots are:

- 9.43 mA, which gives $V_{GS} = -9.43$ V. This is below $V_P = -5$ V, so reject it.
- 2.65 mA, which gives $V_{GS} = -2.65$ V. This is in $[-5, 0]$, so keep it.

$I_{DQ} = 2.65$ **mA,** $V_{GSQ} = -2.65$ **V.**

b. $V_{DS} = 20 - 2.65(2.2 + 1) = 11.5$ V. Also $V_{GS} - V_P = 2.35$ V, and $11.5 \geq 2.35$, so it is saturated.

N3. [6 marks] Design.

1. $V_G = 100(20)/1000 = 2$ V.
2. $V_{GS} = -4(1 - \sqrt{4/10}) = -4(1 - 0.6325) = -1.47$ V.
3. $V_S = V_G - V_{GS} = 2 + 1.47 = 3.47$ V, so $R_S = 3.47/4 = 0.868$ kΩ.
4. $R_D = (20 - 12)/4 = 2$ kΩ.
5. Check: $V_{DS} = 12 - 3.47 = 8.53$ V, and $V_{GS} - V_P = 2.53$ V. Since $8.53 \geq 2.53$, it is saturated.

ANSWER: $R_S \approx 868$ Ω (use the nearest standard value, 820 Ω or 910 Ω), $R_D = 2$ kΩ.

N4. [6 marks]

$V_G = 1(18)/9 = 2$ V. The bias line is $V_{GS} = 2 - 0.2I_D$.

Substitute: $I_D = 8\left(1 + \dfrac{2 - 0.2I_D}{4}\right)^2 = 8(1.5 - 0.05I_D)^2$.

The roots are:

- 101 mA, which gives $V_{GS} = -18.2$ V. Reject it.
- 8.90 mA, which gives $V_{GS} = +0.22$ V. This is valid.

$I_{DQ} = 8.90$ **mA,** $V_{GSQ} = +0.22$ **V.** V_{GSQ} is positive, so the D-MOSFET is in enhancement mode and $I_{DQ} > I_{DSS}$.

$V_D = 18 - 8.90(1.2) = 7.32$ V. $V_S = 8.90(0.2) = 1.78$ V. $V_{DS} = 5.54$ V.

Saturation: $V_{GS} - V_P = 4.22$ V, and $5.54 \geq 4.22$. Saturated.

N5. [8 marks] Chained.

a. $k = 4/(6-2)^2 = 0.25$ mA/V².

b. The bias line is $V_{GS} = 15 - 1.5I_D$. Substitute: $I_D = 0.25\,(13 - 1.5I_D)^2$. The roots are:

- 13.6 mA, which gives $V_{GS} = -5.37$ V. This is below V_T, so reject it.
- 5.53 mA, which gives $V_{GS} = 6.70$ V. This is valid.

$I_{DQ} = 5.53$ **mA,** $V_{GSQ} = 6.70$ **V.**

c. $V_{DSQ} = V_{GSQ} = 6.70$ V. No check is needed, because saturation needs $V_{DS} \geq V_{GS} - V_T$, and with $V_{DS} = V_{GS}$ this always holds.

d. For $V_{DS} = V_{GS} = 8$ V, $I_D = 0.25(8-2)^2 = 9$ mA. Then $R_D = (15-8)/9 = 0.778$ kΩ.

ANSWER: $R_D = 778\ \Omega$.

N6. [8 marks] Common-gate design, hard.

a. $V_{GS} = -5(1 - \sqrt{0.5}) = -1.46$ V. With the gate grounded, $V_S = -V_{GS} = +1.46$ V. Also $V_S = -10 + 5R_S$, so $R_S = (10 + 1.46)/5 = 2.29$ kΩ.

b. $R_D = (15-6)/5 = 1.8$ kΩ. $V_S = 1.46$ V, so $V_{DS} = 6 - 1.46 = 4.54$ V. Also $V_{GS} - V_P = 3.54$ V, and $4.54 \geq 3.54$, so it is saturated.

c. With $R_D = 2.2$ kΩ, $V_D = 15 - 11 = 4$ V and $V_{DS} = 2.54$ V, which is less than 3.54 V. **It is not saturated.**

The limit is $V_D \geq V_S + (V_{GS} - V_P) = 1.46 + 3.54 = 5$ V. So $R_D \leq (15-5)/5 = 2.0$ kΩ.

> Note that I_D does not depend on R_D while the JFET stays saturated, because R_D is not in the gate–source loop.

6. Quick recall check

Question

1. $V_{GS} = -I_D R_S$. It passes through the origin. Pick any I_D to get a second point.
2. Keep the root with $V_P \leq V_{GS} \leq 0$. Reject the one with $V_{GS} < V_P$.
3. Yes, for a D-MOSFET when $V_{GSQ} > 0$ (enhancement mode). It never happens for a JFET.
4. JFET: $V_{DS} \geq V_{GS} - V_P$. E-MOSFET: $V_{DS} \geq V_{GS} - V_{GS(\mathrm{Th})}$.
5. $V_{DS} = V_{GS}$, because $I_G = 0$ means no voltage drops across R_G, so $V_G = V_D$.

FET small-signal I

Answers and worked solutions · Day 18 · Tue 10 Nov · pack E11

5. Practice

From your course files

WS8 Q7 (course question).

$$g_{m0} = \frac{2I_{DSS}}{|V_P|} = \frac{2(12\text{ mA})}{4\text{ V}} = \mathbf{6}\text{ mS}$$

WS8 Q8 (course question).

At $V_{GS} = 0$, Shockley gives $I_D = I_{DSS}$. Rearrange $g_{m0} = 2I_{DSS}/|V_P|$:

$$I_{DSS} = \frac{g_{m0}|V_P|}{2} = \frac{(5\text{ mS})(4\text{ V})}{2} = \mathbf{10}\text{ mA}$$

WS8 Q9 (course question).

The device alone (no R_D):

$$Z_o(\text{FET}) = r_d = \frac{1}{g_{os}} = \frac{1}{25\ \mu\text{S}} = \mathbf{40}\text{ k}\mathbf{\Omega}$$

$$A_v(\text{FET}) = -g_m r_d = -(4.5\text{ mS})(40\text{ k}\Omega) = \mathbf{-180}$$

This is the largest gain the device could ever give (the limit $R_D \to \infty$). Any real R_D gives less.

WS8 Q10 (course question).

READ THE FIGURE. The curve meets the I_D axis at 10 mA, so $I_{DSS} = 10$ mA. It meets the V_{GS} axis at −5 V, so $V_P = -5$ V.

(a) $g_{m0} = \dfrac{2(10\text{ mA})}{5\text{ V}} = \mathbf{4}\text{ mS}$.

(b) Graphically. Draw the tangent at $V_{GS} = -0.5$ V. A clean way to do it from the plotted dots is to take the chord from $V_{GS} = -1$ V ($I_D \approx 6.4$ mA, the dot just below 7) to $V_{GS} = 0$ ($I_D = 10$ mA). That chord is symmetric about −0.5 V:

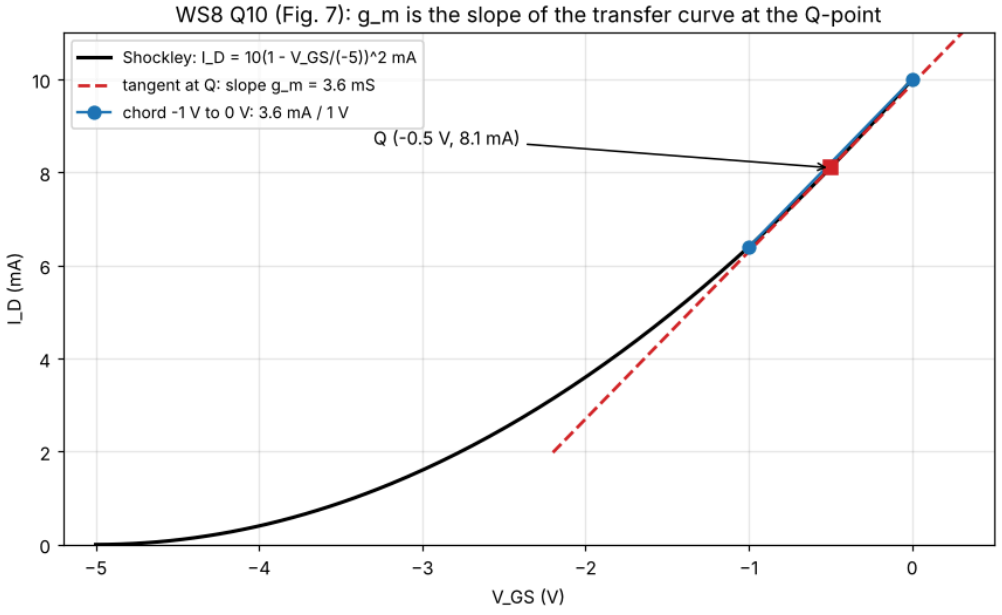
$$g_m \approx \frac{\Delta I_D}{\Delta V_{GS}} = \frac{10 - 6.4\ \text{mA}}{1\ \text{V}} \approx \mathbf{3.6}\ \text{mS}$$

(A hand-drawn tangent will give roughly 3.4–3.8 mS.)

(c) Calculated.

$$g_m = g_{m0}\left(1 - \frac{V_{GS}}{V_P}\right) = 4\ \text{mS}\left(1 - \frac{-0.5}{-5}\right) = 4(0.9) = \mathbf{3.6}\ \text{mS}$$

Comparison: the two agree. For a parabola, a chord symmetric about a point has exactly the tangent slope there. A drawn tangent may differ by about ±5 %.



WS8 Q10: tangent and symmetric chord at the Q-point

WS8 Q11 (course question).

Circuit (Fig. 8): $V_{DD} = +18$ V, $R_D = 1.8$ kΩ, $R_G = 1$ MΩ returned to a 1.5 V battery with its − terminal at the gate side. The source is grounded.

DC: $I_G = 0$, so $V_{GS_Q} = -1.5$ V.

$$I_{D_Q} = 10\left(1 - \frac{-1.5}{-6}\right)^2 = 5.625\ \text{mA}, \qquad V_{DS} = 18 - (5.625)(1.8) = 7.88\ \text{V}\ (\text{saturation, since } > 4.5\ \text{V})$$

g_m: $g_{m0} = 2(10)/6 = 3.33$ mS, so $g_m = 3.33(1 - 0.25) = 2.5$ mS.

ac:

- $Z_i = R_G = \mathbf{1}\ \mathbf{M\Omega}$
- $Z_o = R_D \parallel r_d = 1.8 \parallel 40 = \mathbf{1.72}\ \mathbf{k\Omega}$

- $A_v = -g_m(R_D \parallel r_d) = -(2.5\ \text{mS})(1.722\ \text{k}\Omega) = \mathbf{-4.31}$ (−4.5 ignoring r_d)

WS8 Q12 (course question).

Circuit (Fig. 9): $V_{DD} = 12$ V, $R_D = 3.3$ kΩ, $R_G = 10$ MΩ, $R_S = 1.1$ kΩ bypassed by $C_S = 20$ µF.

You do not need a DC analysis here, because $g_m = g_{fs}$ is given directly. R_S is bypassed, so the source is at ac ground.

- $r_d = 1/50\ \mu\text{S} = 20$ kΩ. Note that r_d is only 6× R_D, so you must keep it.
- $Z_i = R_G = \mathbf{10\ \text{M}\Omega}$
- $Z_o = R_D \parallel r_d = 3.3 \parallel 20 = \mathbf{2.83\ \text{k}\Omega}$

(Not asked, but for completeness: $A_v = -(3\ \text{mS})(2.83\ \text{k}\Omega) = -8.50$.)

WS8 Q13 (course question).

Circuit (Fig. 10): $V_{DD} = 20$ V, $R_1 = 82$ MΩ, $R_2 = 11$ MΩ, $R_D = 2$ kΩ, $R_S = 610\ \Omega$ bypassed by C_S. $I_{DSS} = 12$ mA, $V_P = -3$ V, $r_d = 50$ kΩ.

DC:

$$V_G = 20 \cdot \frac{11}{82 + 11} = 2.366\ \text{V}, \qquad V_{GS} = 2.366 - 610\, I_D$$

Substitute into $I_D = 12\ \text{mA}(1 + V_{GS}/3)^2$ and solve. The valid root is $V_{GS_Q} = -0.974$ V, $I_{D_Q} = 5.47$ mA. (A graphical solution gives about −0.95 V and 5.5 mA.)

Check: $V_S = 3.34$ V, $V_D = 20 - 5.47(2) = 9.05$ V, $V_{DS} = 5.71$ V $> V_{GS} - V_P = 2.03$ V, so the device is in saturation.

g_m: $g_{m0} = 2(12)/3 = 8$ mS, so $g_m = 8\left(1 - \frac{-0.974}{-3}\right) = 5.40$ mS.

ac (R_S bypassed):

- $Z_i = R_1 \parallel R_2 = 82 \parallel 11 = \mathbf{9.70\ \text{M}\Omega}$
- $Z_o = R_D \parallel r_d = 2 \parallel 50 = \mathbf{1.92\ \text{k}\Omega}$
- $A_v = -(5.403\ \text{mS})(1.923\ \text{k}\Omega) = -10.39$
- $V_o = A_v V_i = -10.39 \times 20\ \text{mV} = \mathbf{-208}\ \text{mV}$ (about 0.21 V peak, inverted)

(Ignoring r_d gives −216 mV.)

WS8 Q14 (course question).

Circuit (Fig. 11): $V_{DD} = 16$ V, $R_D = 1.1$ kΩ, $R_G = 10$ MΩ from gate to ground, and the source grounded. n-channel D-MOSFET with $I_{DSS} = 8$ mA, $V_P = -3$ V.

DC: the gate is at 0 V (no gate current through R_G) and the source is at 0 V, so $V_{GS_Q} = 0$ V and $I_{D_Q} = I_{DSS} = 8$ mA. $V_{DS} = 16 - 8(1.1) = 7.2$ V > 3 V, so the device is in saturation.

g_m: $V_{GS_Q} = 0$, so $g_m = g_{m0} = 2(8)/3 = 5.33$ mS.

ac: $r_d = 1/20\ \mu\text{S} = 50$ kΩ, $R_D \parallel r_d = 1.076$ kΩ.

$$A_v = -(5.333\ \text{mS})(1.076\ \text{k}\Omega) = -5.74, \qquad V_o = -5.74 \times 2\ \text{mV} = \mathbf{-11.5}\ \text{mV}$$

(−11.7 mV ignoring r_d.) Also, $Z_i = 10$ MΩ and $Z_o = 1.08$ kΩ.

WS8 Q15 (course question).

Circuit (Fig. 12): $V_{DD} = 30$ V, $R_1 = 40$ MΩ, $R_2 = 10$ MΩ, $R_D = 3.3$ kΩ, $R_S = 1.2$ kΩ bypassed. n-channel E-MOSFET with $V_{GS(Th)} = 3$ V, $k = 0.4 \times 10^{-3}$ A/V².

DC:

$$V_G = 30 \cdot \frac{10}{50} = 6\ \text{V}, \qquad V_{GS} = 6 - 1200\, I_D, \qquad I_D = 0.4 \times 10^{-3}(V_{GS} - 3)^2$$

Substitute the second equation into the first: $0.48(V_{GS} - 3)^2 + V_{GS} - 6 = 0$. The roots are $V_{GS} = 4.667$ V and −0.75 V. Reject −0.75 V, because it is below $V_{GS(Th)}$ and the device would be off.

So $V_{GS_Q} = 4.67$ V and $I_{D_Q} = 0.4(1.667)^2 = 1.11$ mA. $V_{DS} = 30 - 1.111(4.5) = 25.0$ V $> V_{GS} - V_{GS(Th)} = 1.67$ V, so the device is in saturation.

g_m **(E-MOSFET formula):** $g_m = 2k(V_{GS_Q} - V_{GS(Th)}) = 2(0.4 \times 10^{-3})(1.667) = 1.33$ mS.

ac:

- $Z_i = 40 \parallel 10 = 8$ MΩ
- $Z_o = 3.3 \parallel 40 = 3.05$ kΩ
- $A_v = -(1.333\ \text{mS})(3.048\ \text{k}\Omega) = -4.06$
- $V_o = -4.06 \times 0.8\ \text{mV} = \mathbf{-3.25}\ \text{mV}$

New questions

N1 [3 marks] (easy).

a. $g_{m0} = 2(8)/4 = \mathbf{4}$ mS.

b. $g_m = 4\left(1 - \frac{-1.5}{-4}\right) = 4(0.625) = \mathbf{2.5}$ mS.

c. $g_m = g_{m0}\sqrt{I_D/I_{DSS}} = 4\sqrt{2/8} = \mathbf{2}$ mS. Check: $I_D = 2$ mA corresponds to $V_{GS} = -4(1 - 0.5) = -2$ V, and $4(1 - 0.5) = 2$ mS.

N2 [4 marks] (design).

DC: $I_{D_Q} = 12(1 - 2/6)^2 = 5.33$ mA. $g_{m0} = 2(12)/6 = 4$ mS, so $g_m = 4(1 - 1/3) = 2.67$ mS.

a. $R_D \parallel r_d = 6/2.667\ \text{mS} = 2.25$ kΩ. Then $R_D = \dfrac{1}{1/2.25\text{k} - 1/60\text{k}} = \mathbf{2.34\ k\Omega}$. The nearest standard value is 2.4 kΩ, which gives $A_v = -6.15$. (2.2 kΩ would give −5.66.)

b. $V_{DS} = 24 - (5.33\ \text{mA})(2.34\ \text{k}\Omega) = \mathbf{11.5}$ V (11.2 V with 2.4 kΩ). This is greater than $V_{GS} - V_P = 4$ V, so the device is in saturation.

N3 [8 marks] (multi-part, chained).

a. $V_{GS} = -680\, I_D$ with $I_D = 6(1 + V_{GS}/3)^2$ mA. The valid root is $V_{GS_Q} = \mathbf{-1.304}$ V, $I_{D_Q} = \mathbf{1.92}$ mA. Then $V_{DS} = 18 - 1.918(3.3 + 0.68) = \mathbf{10.37}$ V, which is above $V_{GS} - V_P = 1.7$ V, so the device is in saturation.

b. $g_{m0} = 2(6)/3 = 4$ mS, so $g_m = 4\left(1 - \frac{-1.304}{-3}\right) = \mathbf{2.26}$ mS. $r_d = 1/20\ \mu\text{S} = \mathbf{50\ k\Omega}$.

c. $Z_i = \mathbf{1\ M\Omega}$, $Z_o = 3.3 \parallel 50 = \mathbf{3.10\ k\Omega}$, $A_v = -(2.261)(3.096) = \mathbf{-7.00}$.

d. $A_v = -\dfrac{(2.261)(3.3)}{1 + (2.261)(0.68)} = \mathbf{-2.94}$ (−2.85 with r_d included). The gain falls by a factor of 2.4.

e. $A_v = -g_m(r_d \parallel R_D \parallel R_L) = -2.261\,(50 \parallel 3.3 \parallel 4.7\ \text{k}) = -4.22$. Then

$$A_{vs} = \frac{1\ \text{M}}{1\ \text{M} + 600}(-4.22) = \mathbf{-4.22}$$

The FET's huge Z_i makes R_{sig} negligible, while the load costs about 40 % of the gain.

N4 [5 marks].

a. $V_G = 20(20/120) = 3.33$ V and $V_{GS} = 3.33 - 330 I_D$. With Shockley, the valid root is $V_{GS_Q} = \mathbf{+0.293}$ V, $I_{D_Q} = \mathbf{9.21}$ mA (more than I_{DSS}, as expected for positive

V_{GS}). $V_{DS} = 20 - 9.214(1.33) = \mathbf{7.75}$ V, which is above $V_{GS} - V_P = 4.29$ V, so the device is in saturation.

b. $g_{m0} = 2(8)/4 = 4$ mS, so $g_m = 4\left(1 - \frac{0.293}{-4}\right) = \mathbf{4.29}$ mS. That is 7 % above g_{m0}, because $V_{GS_Q} > 0$.

c. $Z_i = 100 \parallel 20 = \mathbf{16.7}$ MΩ, and $A_v = -g_m R_D = \mathbf{-4.29}$.

(Design note: with $R_D = 1.5$ kΩ the drain would drop to $V_{DS} = 3.1$ V, below 4.29 V. The device would then be ohmic and the small-signal model would not apply. Always do the check.)

N5 [6 marks] (hard).

a. $V_G = 24(6/24) = 6$ V, $V_{GS} = 6 - 1000 I_D$, $I_D = 0.5\,\text{mA}(V_{GS} - 2)^2$. Substituting gives $0.5(V_{GS} - 2)^2 + V_{GS} - 6 = 0$, so $V_{GS} = 4$ V or −2 V. Reject −2 V. So $V_{GS_Q} = \mathbf{4}$ V, $I_{D_Q} = \mathbf{2}$ mA, and $V_{DS} = 24 - 2(5.7) = \mathbf{12.6}$ V (above 2 V, so saturation).

b. $g_m = 2k(V_{GS_Q} - V_{GS(Th)}) = 2(0.5)(2) = \mathbf{2}$ mS. Check: $2\sqrt{kI_D} = 2\sqrt{0.5 \times 2} = 2$ mS.

c. Unbypassed: $A_v = -\dfrac{(2)(4.7)}{1 + (2)(1)} = \mathbf{-3.13}$. Bypassed: $A_v = -(2)(4.7) = \mathbf{-9.4}$.

d. We need $R_D \parallel R_L = 5/2$ mS $= 2.5$ kΩ, so $R_L = \dfrac{1}{1/2.5 - 1/4.7} = \mathbf{5.34}$ kΩ.

6. Quick recall check

Question

1. $g_{m0} = 2I_{DSS}/|V_P|$ and $g_m = g_{m0}(1 - V_{GS_Q}/V_P)$. Only V_P in g_{m0} is taken as a magnitude.
2. $\sqrt{1/4} = 0.5$, so $g_m = 0.5\, g_{m0}$.
3. $Z_i = R_1 \parallel R_2$, $Z_o = R_D \parallel r_d$, $A_v = -g_m(R_D \parallel r_d)$.
4. $A_v = -g_m R_D/(1 + g_m R_S)$.
5. An n-channel D-MOSFET can be biased at $V_{GS_Q} > 0$, which makes $(1 - V_{GS_Q}/V_P) > 1$. A JFET's gate junction must not be forward biased, so $V_{GS} \le 0$.

FET small-signal II

Answers and worked solutions · Day 20 · Thu 12 Nov · pack E12

5. Practice

From your course files

WS9 Q5 (course question).

Circuit (from Fig. 5): $V_{DD} = 22$ V. Gate on the divider 91 MΩ (top) / 11 MΩ (bottom) with a bypass capacitor to ground, so the gate is at ac ground. V_i is applied to the source, which has $R_S = 1$ kΩ to ground. $R_D = 2.2$ kΩ, output at the drain. $I_{DSS} = 7.5$ mA, $V_P = -4$ V, $r_d = 30$ kΩ. This is a common-gate amplifier.

DC.

$$V_G = 22 \cdot \frac{11}{91+11} = 2.373 \text{ V}, \qquad V_{GS} = V_G - I_D R_S = 2.373 - 1000 I_D$$

Substituting into $I_D = 7.5\text{ mA}\,(1 - V_{GS}/(-4))^2$ and solving the quadratic gives (the valid root, $V_P < V_{GS} < 0$):

$$I_{DQ} = 3.60 \text{ mA}, \qquad V_{GSQ} = -1.23 \text{ V}$$

Check: $V_{DS} = 22 - 3.60\text{ mA} \times 3.2\text{ k}\Omega = 10.48$ V, which is above $V_{GS} - V_P = 2.77$ V. Saturation holds.

g_m.

$$g_{m0} = \frac{2(7.5)}{4} = 3.75 \text{ mS}, \qquad g_m = 3.75\left(1 - \frac{-1.23}{-4}\right) = 2.60 \text{ mS}$$

> r_d **CHECK:** $r_d = 30$ kΩ and $10R_D = 22$ kΩ, so $r_d \geq 10R_D$ just holds. Use the full formulas anyway since r_d is given.

$$Z_i = R_S \parallel \frac{r_d + R_D}{1 + g_m r_d} = 1\text{ k}\Omega \parallel \frac{32.2\text{ k}\Omega}{1 + (2.60\text{ mS})(30\text{ k}\Omega)} = 1\text{ k}\Omega \parallel 408\ \Omega = \mathbf{290\ \Omega}$$

$$Z_o = R_D \parallel r_d = 2.2\text{k} \parallel 30\text{k} = \mathbf{2.05\ k\Omega}$$

$$A_v = \frac{g_m R_D + R_D/r_d}{1 + R_D/r_d} = \frac{(2.60\text{ mS})(2.2\text{ k}\Omega) + 0.0733}{1.0733} = \mathbf{5.39}$$

Positive, so V_o is in phase with V_i. Approximations: $Z_i \approx R_S \parallel 1/g_m = 278\ \Omega$ and $A_v \approx g_m R_D = 5.72$, within about 6%.

WS9 Q6 (course question).

"Match" means the CG input resistance equals the source resistance: $R_{in} = 1/g_m = R_{sig} = 100\ \Omega$, so

$$g_m = \frac{1}{100\ \Omega} = 10\ \text{mA/V}$$

With $V_{ov} = 0.20$ V and $g_m = 2I_D/V_{ov}$:

$$I_D = \frac{g_m V_{ov}}{2} = \frac{(10\ \text{mA/V})(0.20\ \text{V})}{2} = \mathbf{1\ mA}$$

Overall gain with $R_D = 2$ kΩ (total drain resistance):

$$G_v = \frac{R_{in}}{R_{in}+R_{sig}}\, g_m R_D = \frac{100}{200} \times (10\ \text{mS})(2\ \text{k}\Omega) = 0.5 \times 20 = \mathbf{10\ V/V}$$

Matching costs half the signal at the input. In course notation: $Z_i \approx 1/g_m$, $A_v \approx g_m R_D$, $A_{v_s} = \frac{Z_i}{Z_i+R_{sig}} A_v$.

WS9 Q7 (course question).

Fig. 6: $v_{sig} = 1$ V, $R_{sig} = 1$ MΩ, R_{in} very large, $R_o = 100\ \Omega$, $R_L = 1$ kΩ, $v_o \approx 0.9$ V.

R_{in} is very large (gate draws no current), so $v_i \approx v_{sig} = 1$ V.

The follower's output resistance is $R_o = 1/g_m$ (no R_S shown, r_o ignored), so $g_m = 1/100\ \Omega = 10$ mA/V. With $V_{ov} = 0.25$ V:

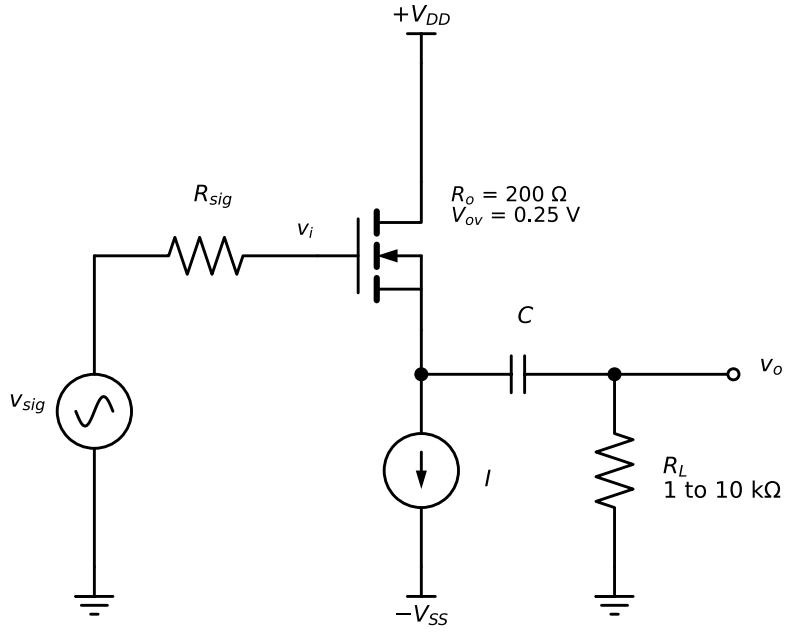
$$I_D = \frac{g_m V_{ov}}{2} = \frac{(10\ \text{mA/V})(0.25)}{2} = \mathbf{1.25\ mA}$$

Output: $v_o = v_i \dfrac{R_L}{R_L + 1/g_m} = 1 \times \dfrac{1000}{1100} = \mathbf{0.909\ V}$ (the figure's "≈ 0.9 V").

Gate–source signal: $v_{gs} = v_i - v_o = v_i \dfrac{1/g_m}{1/g_m + R_L} = \dfrac{100}{1100} = \mathbf{90.9\ mV}$.

Check: v_{gs} = 91 mV is well below $2V_{ov} = 0.5$ V, so the small-signal model is reasonable.

WS9 Q8 (course question).



WS9 Q8: source follower biased by a current source I

Here $k = 0.4$ mA/V² is the process parameter $\mu_n C_{ox}$, because W/L is the unknown. (If it were the course k, $g_m = 2kV_{ov} = 0.2$ mS and R_o would be 5 kΩ, not 200 Ω.)

$$R_o = \frac{1}{g_m} = 200\ \Omega \ \Rightarrow\ g_m = 5\ \text{mA/V}$$

$$\frac{W}{L} = \frac{g_m}{\mu_n C_{ox} V_{ov}} = \frac{5\ \text{mA/V}}{(0.4\ \text{mA/V}^2)(0.25\ \text{V})} = \mathbf{50}$$

$$I = I_D = \frac{g_m V_{ov}}{2} = \frac{(5)(0.25)}{2} = \mathbf{0.625\ mA}$$

Check: $I_D = \frac{1}{2}(0.4)(50)(0.25)^2 = 0.625$ mA. In course notation $k = \frac{1}{2}(0.4)(50) = 10$ mA/V² and $g_m = 2kV_{ov} = 5$ mS. Same answer.

Gain range ($G_v = R_L/(R_L + R_o)$, R_{sig} negligible against the gate):

$$R_L = 1\ \text{k}\Omega:\ G_v = \frac{1000}{1200} = 0.833, \qquad R_L = 10\ \text{k}\Omega:\ G_v = \frac{10\,000}{10\,200} = 0.980$$

So $\mathbf{0.83 \le G_v \le 0.98}$ V/V.

WS9 Q9 (course question).

Fig. 7: two identical self-bias stages. $V_{DD} = 18$ V, $R_D = 2.2$ kΩ, $R_S = 390\ \Omega$ bypassed (50 µF), $R_G = 10$ MΩ each, $V_i = 20$ mV. $I_{DSS} = 8$ mA, $V_P = -4.5$ V. r_d not given, so $r_d = \infty$.

DC (both stages). $V_{GS} = -390 I_D$. Solving with $I_D = 8\ \text{mA}(1 - V_{GS}/(-4.5))^2$ (valid root):

$$I_{DQ} = 3.70\ \text{mA}, \qquad V_{GSQ} = -1.44\ \text{V}$$

Check: $V_{DS} = 18 - 3.70\ \text{mA} \times 2.59\ \text{k}\Omega = 8.43$ V, above $V_{GS} - V_P = 3.06$ V.

g_m.

$$g_{m0} = \frac{2(8)}{4.5} = 3.556\ \text{mS}, \qquad g_m = 3.556\left(1 - \frac{-1.441}{-4.5}\right) = 2.42\ \text{mS}$$

Stage 2 (unloaded): $A_{v2} = -g_m R_D = -(2.417\ \text{mS})(2.2\ \text{k}\Omega) = \mathbf{-5.32}$.

Stage 1 (loaded by $R_{G2} = 10$ MΩ): $A_{v1} = -g_m(2.2\text{k} \parallel 10\text{M}) = \mathbf{-5.32}$ (the load changes it only in the 4th figure).

Overall: $A_v = A_{v1}A_{v2} = (-5.32)(-5.32) = \mathbf{28.3}$, in phase.

Output: $V_o = A_v V_i = 28.3 \times 20\ \text{mV} = \mathbf{565\ mV}$ (0.57 V).

Also $Z_i = 10$ MΩ, $Z_o = 2.2$ kΩ.

New questions

N1.

$I_{DQ} = k(V_{GSQ} - V_{GS(\text{Th})})^2 = 0.3\ \text{mA/V}^2 \times (3\ \text{V})^2 = \mathbf{2.7\ mA}$.

$g_m = 2k(V_{GSQ} - V_{GS(\text{Th})}) = 2(0.3)(3) = \mathbf{1.8\ mS}$.

Check: $V_{ov} = 3$ V, $g_m = 2I_D/V_{ov} = 2(2.7)/3 = 1.8$ mS. Same.

N2.

a. Self-bias: $V_{GS} = -1500 I_D$. Solve $I_D = 12\ \text{mA}(1 - 1500 I_D/4)^2$. Roots 1.67 mA and 4.25 mA; the second gives $V_{GS} = -6.38\ \text{V} < V_P$, reject.

$I_{DQ} = \mathbf{1.67\ mA}$, $V_{GSQ} = \mathbf{-2.51\ V}$.

$g_{m0} = 2(12)/4 = 6$ mS, $g_m = 6(1 - 2.507/4) = \mathbf{2.24\ mS}$.

b. $Z_i = R_G = \mathbf{1\ M\Omega}$. $Z_o = R_S \parallel 1/g_m = 1.5\text{k} \parallel 446.6 = \mathbf{344\ \Omega}$.

$$A_v = \frac{g_m R_S}{1 + g_m R_S} = \frac{3.359}{4.359} = \mathbf{0.771}.$$

c. $R_S \parallel R_L = 600\ \Omega$, $g_m(600) = 1.344$:

$$A_{v_L} = \frac{1.344}{2.344} = \mathbf{0.573}.$$

$A_{v_s} = \dfrac{1\text{ M}}{1\text{ M} + 10\text{ k}} \times 0.573 = \mathbf{0.568}$. R_{sig} barely matters; R_L matters a lot because it is comparable with Z_o.

N3.

a. $Z_i = R_S \parallel 1/g_m = 1.2\text{k} \parallel 250 = \mathbf{207\ \Omega}$. $Z_o = R_D = \mathbf{3.3\ k\Omega}$.

b. $A_{v_{NL}} = g_m R_D = 4 \times 3.3 = 13.2$.

$A_{v_L} = g_m(R_D \parallel R_L) = 4\text{ mS} \times 2.481\text{ k}\Omega = \mathbf{9.92}$.

$A_{v_s} = \dfrac{207}{207 + 200} \times 9.92 = \mathbf{5.05}$.

c. The CG input resistance (207 Ω) is about equal to R_{sig} (200 Ω), so the input divider passes only about half of V_s.

N4.

a. $V_{GS} = V_P(1 - \sqrt{I_D/I_{DSS}}) = -5(1 - \sqrt{0.4}) = \mathbf{-1.84\ V}$.
$R_S = 1.838/4\text{ mA} = \mathbf{459\ \Omega}$ (use 470 Ω standard).

b. $g_{m0} = 2(10)/5 = 4$ mS, $g_m = 4(1 - 1.838/5) = \mathbf{2.53\ mS}$.

c. $r_d = 1/25\ \mu\text{S} = 40$ kΩ. $R_D \parallel r_d = 10/2.53\text{ mS} = 3.953$ kΩ.

$R_D = \dfrac{1}{1/3.953\text{k} - 1/40\text{k}} = \mathbf{4.39\ k\Omega}$.

With 4.3 kΩ: $A_v = -2.53\text{ mS} \times (4.3\text{k} \parallel 40\text{k}) = -9.82$. With 4.7 kΩ: $A_v = -10.6$. Pick 4.3 kΩ if the gain must not exceed 10, or 4.7 kΩ if it must reach at least 10.

d. $V_{DS} = 30 - 4\text{ mA}(4.386 + 0.459)\text{ k}\Omega = 10.6$ V. Needs $V_{DS} > V_{GS} - V_P = 3.16$ V. OK.

(Note: with $V_{DD} = 20$ V, V_{DS} would be only 0.62 V and the design would fail. Always recheck.)

e. $A_v = \dfrac{-g_m R_D}{1 + g_m R_S + (R_D + R_S)/r_d} = \dfrac{-(2.53)(4.386)}{1 + 1.162 + 0.121} = \mathbf{-4.86}$.

N5.

a. Drain feedback: $V_{GS} = V_{DS} = 15 - 2200 I_D$. Solve $I_D = 0.5\text{ mA}(13 - 2200 I_D)^2$: roots 4.54 mA and 7.69 mA. The second gives $V_{GS} = -1.92$ V, below threshold, reject.

$I_{DQ} = \mathbf{4.54\ mA}$, $V_{GSQ} = \mathbf{5.01\ V}$, $g_m = 2(0.5)(5.013 - 2) = \mathbf{3.01\ mS}$.

b. Stage 1 load: $R_F \parallel R_D \parallel Z_{i2} = 12\text{M} \parallel 2.2\text{k} \parallel 1\text{M} \approx 2.195$ kΩ.

$A_{v1} = -g_m(2.195\ \text{k}\Omega) = \mathbf{-6.61}$.

$Z_{i1} = \dfrac{R_F}{1 + g_m(R_D \parallel R_{L1})} = \dfrac{12\ \text{M}}{1 + 6.61} = \mathbf{1.58\ M\Omega}$ (1.57 MΩ using R_D alone).

c. $A_{v2} = \dfrac{g_m(R_S \parallel R_L)}{1 + g_m(R_S \parallel R_L)} = \mathbf{0.573}$ (from N2c).

$A_v = A_{v1}A_{v2} = (-6.61)(0.573) = \mathbf{-3.79}$.

$V_i = \dfrac{1.576}{1.576 + 0.5} \times 50\ \text{mV} = 37.96$ mV, so $V_o = -3.79 \times 37.96 = \mathbf{-144\ mV}$ (144 mV, 180° out of phase). Overall $A_{v_s} = -2.88$.

R_{sig} matters here because the drain-feedback Z_i (1.6 MΩ) is not much bigger than 500 kΩ.

d. $Z_o = Z_{o2} = R_S \parallel 1/g_m = \mathbf{344\ \Omega}$.

N6.

a. $V_{ov} = \sqrt{\dfrac{2I_D}{\mu_n C_{ox}(W/L)}} = \sqrt{\dfrac{1}{8}} = \mathbf{0.354\ V}$. $V_{GS} = 0.6 + 0.354 = \mathbf{0.954\ V}$.

$g_m = 2I_D/V_{ov} = 1/0.354 = \mathbf{2.83\ mA/V}$.

b. $k = \frac{1}{2}(0.2)(40) = \mathbf{4\ mA/V^2}$. $2kV_{ov} = 2(4)(0.354) = 2.83$ mA/V. Same.

c. $A_v = -g_m(R_D \parallel R_L) = -2.83 \times 3.33 = \mathbf{-9.43}$.

6. Quick recall check

Question

1. $Z_o = R_S \parallel 1/g_m$.
2. $Z_i \approx R_F/(1 + g_m R_D)$.
3. $Z_i \approx 1/g_m = 200\ \Omega$.
4. $g_m = 2I_D/V_{ov} = 2(2)/0.4 = 10$ mA/V.
5. $A_v = (-4)(-4) = +16$, in phase.

Op-amps, differential amps, power amps

Answers and worked solutions · Day 21 · Fri 13 Nov · pack E13

5. Practice

From your course files

WS10 Q1 (course question).

> See Worked Example 1. $A_d = 120$, $A_c = 20$, CMRR $= 6$, **CMRR = 15.56 dB**.

WS10 Q2 (course question).

> See Worked Example 2. $V_d = 60$ mV, $V_c = 170$ mV.
>
> a. CMRR $= 200$: $V_o = 360\ \text{V}(1 + 0.01417) =$ **365.1 V**.
>
> b. CMRR $= 10^5$: $V_o = 360\ \text{V}(1 + 2.83 \times 10^{-5}) =$ **360.01 V**.
>
> If the inputs are read as µV (Boylestad's original), the answers are 365.1 mV and 360.01 mV. Mention that a real op-amp would saturate at its rails for the mV version.

WS10 Q3 (course question).

> READ THE FIGURE: V_1 goes straight into the **+** input (pin 5). A 36 kΩ resistor runs from the − input (pin 4) to ground, and 750 kΩ runs from the output (pin 10) back to −. Supplies are ±9 V (741). So this is a **non-inverting amplifier** with $R_1 = 36$ kΩ and $R_f = 750$ kΩ.
>
> $$A_v = 1 + \frac{R_f}{R_1} = 1 + \frac{750}{36} = 21.83$$
>
> $$V_o = 21.83 \times 150\ \text{mV} = \mathbf{3.275\ V\ rms}\ (\text{in phase with } V_1)$$
>
> Rail check: the peak is $3.275\sqrt{2} = 4.63$ V, which is inside ±9 V, so there is no clipping.
>
> Trap: $R_f/R_1 = 20.8$ alone would give 3.125 V. The "1 +" is worth the mark.

WS10 Q4 (course question).

> READ THE FIGURE: V_1 through 47 kΩ and V_2 through 12 kΩ both go to the − input (pin 4). There is 470 kΩ feedback, and + (pin 5) is grounded. Supplies are ±15 V. This is an **inverting summing amplifier**.

$$V_o = -\left(\frac{470}{47}V_1 + \frac{470}{12}V_2\right) = -\left(10(40\text{ mV}) + 39.17(20\text{ mV})\right)$$

$$V_o = -(400 + 783.3)\text{ mV} = \mathbf{-1.183\text{ V rms}}$$

So the output is 1.183 V rms, 180° out of phase with the (in-phase) inputs. The peak is 1.67 V, well inside ±15 V.

WS10 Q5 (course question).

READ THE FIGURE:

- **Stage 1** is an inverting amplifier: $V_1 = 12$ mV through 33 kΩ to −, with 330 kΩ feedback and + grounded.
- **Stage 2** is an inverting summer. Stage 1's output goes through 47 kΩ, and $V_2 = 18$ mV goes through another 47 kΩ. Both go to the − input, with 470 kΩ feedback and + grounded.

Stage 1:

$$V_{o1} = -\frac{330}{33}(12\text{ mV}) = -10(12\text{ mV}) = -120\text{ mV}$$

Stage 2:

$$V_o = -\left(\frac{470}{47}V_{o1} + \frac{470}{47}V_2\right) = -10\left(-120 + 18\right)\text{ mV}$$

$$V_o = -(-1200 + 180)\text{ mV} = \mathbf{+1.02\text{ V}}$$

The two inversions cancel on the V_1 path (+100 overall). V_2 is inverted once (−10). This assumes V_1 and V_2 are in phase (or are dc values), as the figure implies.

WS10 Q6 (course question).

From Fig. 1 (page 3): $V_{CC} = 18$ V, $R_B = 1.2$ kΩ, $R_C = 16$ Ω, $\beta = 40$, fixed bias.

DC bias:

$$I_B = \frac{18 - 0.7}{1.2\text{ k}\Omega} = 14.42\text{ mA}, \quad I_{CQ} = 40(14.42\text{ mA}) = 576.7\text{ mA},$$

$$V_{CEQ} = 18 - (0.5767)(16) = 8.77\text{ V (active region, close to centred)}.$$

Input power:

$$P_i(\text{dc}) = V_{CC}I_{CQ} = 18(0.5767) = \mathbf{10.38\text{ W}}.$$

Output power, with $I_b = 5$ mA rms:

$I_c(\text{rms}) = \beta I_b = 40(5\text{ mA}) = 200\text{ mA}, \qquad P_o(\text{ac}) = I_c^2(\text{rms})R_C = (0.2)^2(16) = \mathbf{0.64\text{ W}}.$

Check for clipping: $I_c(p) = 0.2\sqrt{2} = 283$ mA. This is below both $I_{CQ} = 577$ mA and $V_{CC}/R_C - I_{CQ} = 548$ mA, so the output is undistorted. Efficiency $= 0.64/10.38 = 6.2\%$, well under the 25% limit.

WS10 Q7 (course question).

Same circuit (values as in Q6) with $R_B = 1.5$ kΩ:

$$I_B = \frac{17.3}{1.5\text{ k}\Omega} = 11.53\text{ mA}, \quad I_{CQ} = 40(11.53) = 461.3\text{ mA}, \quad V_{CEQ} = 18 - 7.38 = 10.62\text{ V}.$$

$$P_i(\text{dc}) = 18(0.4613) = \mathbf{8.30\text{ W}}.$$

WS10 Q8 (course question).

With $R_B = 1.5$ kΩ: $I_{CQ} = 461.3$ mA. The saturation current is $V_{CC}/R_C = 18/16 = 1.125$ A.

The largest undistorted collector-current peak is the smaller of:

- the room down to cut-off, $I_{CQ} = 461.3$ mA, and
- the room up to saturation, $1.125 - 0.4613 = 663.7$ mA.

So $I_c(p) = 461.3$ mA (it clips at cut-off first, because the Q-point sits above the centre of the load line in V_{CE}).

$$P_o(\text{max}) = \frac{I_c^2(p)}{2}R_C = \frac{(0.4613)^2}{2}(16) = \mathbf{1.70\text{ W}}.$$

Efficiency at this output: $1.70/8.30 = 20.5\%$. It is below 25% because the Q-point is not centred. For comparison, a centred Q-point would give $V_{CC}^2/(8R_C) = 2.53$ W.

WS10 Q9 (course question).

Full working: Worked Example 4.

$R_L = 16 \parallel 16 \parallel 16 \parallel 16 = 4\ \Omega$. $a = \sqrt{8000/4} = \sqrt{2000} = \mathbf{44.72 : 1}$ (step-down).

WS10 Q10 (course question).

See Worked Example 3. (a) $P_i(\text{max}) = \mathbf{49.74\text{ W}}$; (b) $P_o(\text{max}) = \mathbf{39.06\text{ W}}$; (c) $\eta_{\text{max}} = \mathbf{78.54\%}$.

New questions

N1. [3 marks]

a. CMRR $= 10^{80/20} = 10^4$, so $A_c = 5000/10^4 =$ **0.5**.

b. $V_d = 0.5$ mV, $V_c = 1.0$ mV.
$V_o = A_d V_d + A_c V_c = 5000(0.5\ \text{mV}) + 0.5(1.0\ \text{mV}) = 2.5\ \text{V} + 0.5\ \text{mV} =$ **2.5005 V**.

Check with the bracket form: $2.5(1 + 1.0/(10^4 \times 0.5)) = 2.5005$ V.

N2. [5 marks]

a. Stage 1: $V_{o1} = -(150/10)(0.2) =$ **−3.0 V**. Stage 2:
$V_o = (1 + 22/10)(-3.0) = 3.2(-3.0) =$ **−9.6 V** (fine with ±15 V rails).

b. $V_o = -\left(\frac{100}{20}(0.3) + \frac{100}{50}(-0.5) + \frac{100}{10}(0.1)\right) = -(1.5 - 1.0 + 1.0) =$ **−1.5 V**.

N3. [6 marks]

a. $I_E = (12 - 0.7)/56\ \text{k}\Omega =$ **201.8 µA**. $I_C \approx I_E/2 =$ **100.9 µA**.
$V_C = 12 - (100.9\ \mu\text{A})(47\ \text{k}\Omega) =$ **7.26 V**.

b. $r_e = 26\ \text{mV}/100.9\ \mu\text{A} =$ **257.7 Ω** (per transistor current).

c. Single-ended: $A_v = R_C/(2r_e) = 47000/515.4 =$ **91.2**. Double-ended:
$A_d = R_C/r_e =$ **182.4**.

N4. [6 marks]

a. $P_o = (22 - 2)(0.290 - 0.010)/8 = (20)(0.28)/8 =$ **0.70 W**.

b. $P_i = 12(0.15) =$ **1.80 W**; $\eta = 0.70/1.80 =$ **38.9%** (below 50%).

c. $P_Q = 1.80 - 0.70 =$ **1.10 W**.

d. $R'_L = \Delta V_{CE}/\Delta I_C = 20/0.28 =$ **71.4 Ω**; $a = \sqrt{71.4/8} =$ **2.99 ≈ 3 : 1**.

N5. [8 marks, chained]

a. $I_p = 16/16 = 1$ A. $P_i = 2(20)(1)/\pi =$ **12.73 W**. $P_o = 16^2/(2 \cdot 16) =$ **8.0 W**.
$\eta = 8/12.73 =$ **62.8%** (check: $\frac{\pi}{4} \cdot \frac{16}{20} = 0.628$). $P_{2Q} = 4.73$ W, so **2.37 W per transistor**.

b. $P_o(\max) = 20^2/(2 \cdot 16) =$ **12.5 W**.

c. $P_{2Q}(\max) = 2(20)^2/(\pi^2 \cdot 16) =$ **5.07 W** total (2.53 W each), at $V_L(p) = 2V_{CC}/\pi =$ **12.73 V**.

d. Transistor dissipation peaks at about 64% of full swing, where it is higher than at full output, so the heatsinks must handle that worst case.

6. Quick recall check

Question

1. $V_d = 2$ mV, $V_c = 2$ mV.
2. 10^5.
3. $1 + 9 = 10$.
4. 25%, 50%, 78.5%.
5. $a = \sqrt{1000/10} = 10:1$.

EDC · MOCK

EDC mock A

Answers and worked solutions · Day 27 · Thu 19 Nov · pack EX1

Marking guide

Before you open any solution: write your raw mark per question in the table at the end. Award a mark only if your script shows the line that earns it. If your formula and substitution are right but an arithmetic slip gives a wrong number, take the method marks and lose the answer mark. Accept answers within about 2% of the model answer when the difference comes from rounding.

Where the questions come from. Three parts are workshop-style variants (same method, changed numbers), because the course reuses workshop questions: **Q2 is WS3 Q3**, **Q8(a) is WS10 Q2** and **Q10(b) is WS10 Q10**. Everything else is new. Every number below was computed in Python.

Suggested timing (82 marks, 120 min):

Q	Topic	Marks	Time	Clock
1	MCQ	10	8 min	0:08
2	Diode string design	8	11 min	0:19
3	BJT VDB, DC and AC	13	18 min	0:37
4	Emitter follower	9	13 min	0:50
5	JFET self-bias	8	12 min	1:02
6	E-MOSFET bias	7	10 min	1:12
7	Two-stage cascade	8	12 min	1:24
8	CMRR and diff amp	7	11 min	1:35
9	Op-amps	5	6 min	1:41
10	Power amps	7	10 min	1:51
	Check		9 min	2:00

Question 1: MCQ [10 marks]

Part	Answer	Reason
I	**(a)**	More reverse bias uncovers more fixed ion charge, so W grows. The junction behaves like a capacitor with a wider gap, so C_T falls (this is how a varactor works).
II	**(b)**	$V_r = V_P/(fCR)$, so ripple is inversely proportional to C. Doubling C roughly halves it. It never reaches zero.
III	**(c)**	The smallest $I_S = (V_{i,min} - V_Z)/R_S$, with most of it taken by the largest I_L, leaves the least for I_Z. If I_Z falls below I_{ZK}, regulation is lost. Case (a) is the worst case for Zener *power*, not for dropout.
IV	**(c)**	$V_{BC} = +0.5$ V means the collector-base junction is forward biased as well. Both junctions are forward biased, so the BJT is saturated. Check: $V_{CE} = V_{BE} - V_{BC} = 0.2$ V $= V_{CE,sat}$.
V	**(b)**	A capacitor is open at DC, so the bias is unchanged. At signal frequency it shorts R_E, so $\lvert A_v \rvert$ goes from $\approx R_C/(r_e + R_E)$ to $\approx R_C/r_e$. Z_i *falls* (from

Part	Answer	Reason
		$\beta(r_e + R_E)$ to βr_e), so (c) is wrong.
VI	**(a)**	At $V_{GS} = V_P$, Shockley gives $I_D = I_{DSS}(1-1)^2 = 0$. The channel is pinched off.
VII	**(c)**	A D-MOSFET has a built-in channel. It works in depletion mode ($V_{GS} < 0$) and in enhancement mode ($V_{GS} > 0$). A JFET must keep $V_{GS} \leq 0$, and an E-MOSFET needs $V_{GS} > V_{GS(Th)}$.
VIII	**(b)**	$g_m = g_{m0}(1 - V_{GS}/V_P)$ is a straight line falling from g_{m0} at $V_{GS} = 0$ to zero at V_P. At $I_D = I_{DSS}/4$, $g_m = 0.5 g_{m0}$.
IX	**(a)**	Virtual ground: high open-loop gain forces $V_- \approx V_+ = 0$, and the ideal input draws no current.
X	**(d)**	Each transistor needs about 0.7 V of V_{BE} before it conducts. Near zero crossing neither one conducts. The fix is to bias both slightly on with diodes (class AB).

MARKS: 1 per correct letter, no part marks.

COMMON MISTAKES: III: picking (a) because it is the worst case for P_Z. V: thinking the bypass raises Z_i. VII: choosing the JFET.

REREAD: E1 (I), E3 (II, III), E4–E5 (IV), E7 (V), E9 (VI, VII), E11 (VIII), E13 (IX, X).

Question 2: Diode string design [8 marks] (WS3 Q3 variant)

(a) Four equal diodes share 2.6 V, so each one drops

$$V_D = \frac{2.6}{4} = 0.65 \text{ V}$$

That is 0.05 V below 0.7 V, which is half a decade below 1 mA:

$$I_D = 1 \text{ mA} \times 10^{(0.65-0.7)/0.1} = 10^{-0.5} \text{ mA} = \mathbf{0.316 \text{ mA}}$$

$$R = \frac{V_{DD} - V_o}{I_D} = \frac{5 - 2.6}{0.316 \text{ mA}} = \mathbf{7.59 \text{ k}\Omega}$$

(b) Use the same diode law, $V_o = 4\,[0.7 + 0.1 \log_{10}(I_D/1 \text{ mA})]$, and KCL at the output: $I_D = (5 - V_o)/R - V_o/R_L$.

The load takes about 0.25 mA of the 0.32 mA, so the diode current falls to roughly a third. That is about half a decade, so each diode drops about 0.05 V and V_o falls by about 0.2 V. Start the iteration at $V_o = 2.4$ V:

Iteration	$I_R = (5 - V_o)/R$	$I_L = V_o/R_L$	$I_D = I_R - I_L$	New V_o
1 (V_o = 2.400 V)	0.3426 mA	0.2400 mA	0.1026 mA	2.4044 V
2 (V_o = 2.4044 V)	0.3420 mA	0.2404 mA	0.1016 mA	2.4027 V

$$\mathbf{V_o \approx 2.40 \text{ V}}, \qquad \mathbf{I_D \approx 0.102 \text{ mA}}$$

(The converged solution is $V_o = 2.403$ V and $I_D = 0.1018$ mA.) A starting guess of 2.6 V also converges, but it oscillates for several steps first.

(c) "0.1 V per decade" means $2.3\,nV_T = 0.1$ V, so $nV_T = 0.1/\ln 10 = 43.4$ mV.

$$r_d = \frac{nV_T}{I_D} = \frac{43.4 \text{ mV}}{0.316 \text{ mA}} = \mathbf{137\ \Omega} \text{ per diode}, \qquad 4r_d = 549\ \Omega$$

Small-signal divider:

$$\Delta V_o = \Delta V_{DD} \frac{4r_d}{R + 4r_d} = \pm 0.5 \times \frac{549}{7589 + 549} = \mathbf{\pm 33.7 \text{ mV}}$$

(Solving the exponential model exactly at 5.5 V gives $V_o = 2.631$ V, a change of 31 mV. The estimate is good.)

Marks (8):

- a. 1: 0.65 V per diode. 1: current of 0.316 mA from the decade rule. 1: $R = 7.59$ kΩ.
- b. 1: KCL with the load current written correctly. 1: a valid iteration (at least two steps). 1: $V_o \approx 2.40$ V.
- c. 1: $r_d = 137\ \Omega$, with nV_T found from the 0.1 V/decade figure. 1: ± 34 mV.

COMMON MISTAKES:

- Using 0.7 V per diode and then trying to "fit" 2.6 V. The decade rule is the whole point of the question.
- Using the small-signal estimate in (b). The load current is not small: the diode current changes by a factor of 3, so linearising gives 2.47 V instead of 2.40 V.
- Using $r_d = 26\ \mathrm{mV}/I_D$. Here $n \neq 1$, and the 0.1 V/decade data gives $nV_T = 43.4$ mV.

REREAD: E2 (multi-diode circuits, the decade rule), E1 (r_d).

Question 3: BJT voltage-divider bias, DC then AC [13 marks]

(a) Approximate method check: $\beta R_E = 120 \times 1\ \text{k}\Omega = 120\ \text{k}\Omega \geq 10R_2 = 82\ \text{k}\Omega$. The approximate method *may* be used, but the question asks for the exact method.

$$V_{Th} = \frac{R_2}{R_1 + R_2} V_{CC} = \frac{8.2}{47.2} \times 18 = \mathbf{3.127\ V}, \qquad R_{Th} = 39\ \text{k} \parallel 8.2\ \text{k} = \mathbf{6.775\ k\Omega}$$

$$I_B = \frac{V_{Th} - V_{BE}}{R_{Th} + (\beta + 1)R_E} = \frac{3.127 - 0.7}{6.775\ \text{k} + 121\ \text{k}} = \mathbf{19.0\ \mu A}$$

$$I_C = \beta I_B = \mathbf{2.279\ mA}, \qquad I_E = (\beta + 1) I_B = \mathbf{2.298\ mA}$$

$$V_{CE} = V_{CC} - I_C R_C - I_E R_E = 18 - 7.522 - 2.298 = \mathbf{8.18\ V}$$

Active check: $V_{CE} = 8.18\ \text{V} > V_{CE,sat} = 0.2\ \text{V}$, so the BJT is in the active region. (Also $V_B = 3.00\ \text{V}$ and $V_C = 10.48\ \text{V}$, so the CB junction is reverse biased.)

(The approximate method gives $I_E = 2.43$ mA, about 6% high.)

(b)

$$r_e = \frac{26\ \text{mV}}{I_E} = \frac{26}{2.298} = \mathbf{11.31\ \Omega}, \qquad \beta r_e = 1.357\ \text{k}\Omega$$

$$Z_i = R_1 \parallel R_2 \parallel \beta r_e = 39\ \text{k} \parallel 8.2\ \text{k} \parallel 1.357\ \text{k} = \mathbf{1.131\ k\Omega}$$

$$Z_o = R_C \parallel r_o = 3.3\ \text{k} \parallel 50\ \text{k} = \mathbf{3.096\ k\Omega}$$

$$A_v = -\frac{R_C \parallel r_o}{r_e} = -\frac{3096}{11.31} = \mathbf{-273.7} \quad (-291.7 \text{ if } r_o \text{ is ignored})$$

(c)

$$A_{v_L} = -\frac{R_C \parallel r_o \parallel R_L}{r_e} = -\frac{1866}{11.31} = \mathbf{-165.0}$$

Check with the Thevenin output form: $A_v \dfrac{R_L}{R_L + Z_o} = -273.7 \times \dfrac{4.7}{7.796} = -165.0$.

$$A_{v_s} = \frac{Z_i}{Z_i + R_s} A_{v_L} = \frac{1131}{1731}(-165.0) = \mathbf{-107.8}$$

$$v_o = 107.8 \times 10\ \text{mV} = \mathbf{1.08\ V\ peak}\ (180^\circ \text{ out of phase with } v_s)$$

Which costs more gain? R_L does. It cuts the gain by the factor $165/274 = 0.60$, because R_L (4.7 kΩ) is not much bigger than Z_o (3.1 kΩ). R_s cuts it by

$1131/1731 = 0.65$, because Z_i (1.13 kΩ) is only about twice R_s. The two losses are similar, and R_L is slightly worse.

Marks (13):

- a. 1: the $\beta R_E \geq 10R_2$ check, stated. 1: V_{Th} and R_{Th}. 1: the I_B formula with $(\beta + 1)R_E$. 1: I_C and I_E. 1: V_{CE} with the active-region statement.
- b. 1: r_e from I_E (not I_C). 1: Z_i. 1: Z_o including r_o. 1: A_v with its negative sign.
- c. 1: A_{v_L}. 1: A_{v_s}. 1: v_o. 1: the R_s versus R_L comparison with reasons.

COMMON MISTAKES:

- Using R_{Th} with β instead of $(\beta + 1)$ in the I_B denominator. This is a small error, but it is a method error.
- Including R_E in Z_i or A_v when it is bypassed.
- Using $Z_i = \beta r_e$ alone and forgetting $R_1 \parallel R_2$.
- Getting A_{v_s} by dividing by R_s alone instead of by the divider $Z_i/(Z_i + R_s)$.
- Leaving out the minus sign. CE inverts.

REREAD: E5 (exact versus approximate VDB), E6 (CE bypassed, r_e model), E7 (loading with R_L and R_s).

Question 4: Emitter follower [9 marks]

(a) Base loop: $V_{CC} = I_B R_B + V_{BE} + (\beta+1) I_B R_E$.

$$I_B = \frac{12 - 0.7}{220\ \text{k} + 101(2.2\ \text{k})} = \frac{11.3}{442.2\ \text{k}} = \mathbf{25.55\ \mu A}$$

$I_E = (\beta+1) I_B = \mathbf{2.581\ mA}$, $V_E = I_E R_E = \mathbf{5.68\ V}$, $V_{CE} = 12 - 5.68 = \mathbf{6.32\ V}$

(b) $r_e = 26/2.581 = \mathbf{10.07\ \Omega}$.

No load: $Z_b = \beta r_e + (\beta+1) R_E = 1.007\ \text{k} + 222.2\ \text{k} = 223.2\ \text{k}\Omega$, so

$$Z_i = R_B \parallel Z_b = 220\ \text{k} \parallel 223.2\ \text{k} = \mathbf{110.8\ k\Omega}$$

With R_L: the emitter sees $R_E \parallel R_L = 2.2\ \text{k} \parallel 1\ \text{k} = 687.5\ \Omega$.

$$Z_b = 1.007\ \text{k} + 101(687.5) = 70.44\ \text{k}\Omega, \qquad Z_i = 220\ \text{k} \parallel 70.44\ \text{k} = \mathbf{53.4\ k\Omega}$$

(The course approximation $Z_b \cong \beta(r_e + R_E \parallel R_L)$ gives 52.96 kΩ. Accept it.)

(c) Looking back into the emitter, the base-side resistance is $R_s' = R_s \parallel R_B = 1\ \text{k} \parallel 220\ \text{k} = 995.5\ \Omega$. It is reflected down by β:

$$Z_o = R_E \parallel \left(r_e + \frac{R_s'}{\beta} \right) = 2.2\ \text{k} \parallel (10.07 + 9.955) = \mathbf{19.8\ \Omega}$$

(If you ignore R_s you get $R_E \parallel r_e = 10.0\ \Omega$. Give 1 of the 2 marks.)

(d)

$$A_{v_L} = \frac{R_E \parallel R_L}{R_E \parallel R_L + r_e} = \frac{687.5}{697.6} = \mathbf{0.986}$$

$$A_{v_s} = \frac{Z_i}{Z_i + R_s} A_{v_L} = \frac{53.36}{54.36}(0.986) = \mathbf{0.967} \quad \text{(in phase)}$$

(e) It has a high input impedance (53 kΩ) and a low output impedance (about 20 Ω). It can therefore drive a low-resistance load from a high-resistance source with almost no loss of signal: it is a buffer, and its current gain is large.

Marks (9): (a) 1: the I_B loop with $(\beta+1)R_E$. 1: I_E, V_E and V_{CE}. (b) 1: r_e and the no-load Z_i. 1: the loaded Z_i using $R_E \parallel R_L$. (c) 1: the R_s'/β reflection. 1: 19.8 Ω. (d) 1: A_{v_L}. 1: A_{v_s}. (e) 1: buffer, with high Z_i and low Z_o both stated.

> **COMMON MISTAKES:**

- Forgetting that R_L changes Z_i for a follower (Trap 2 in E7).
- Quoting $Z_o = R_E \parallel r_e$ when the source has resistance (Trap 1 in E7).
- Getting the direction of the reflection wrong. Emitter resistance looks $(\beta + 1)$ times bigger from the base. Base resistance looks β times smaller from the emitter.

REREAD: E5 (emitter-bias DC loop), E7 (section 1.4, emitter follower).

Question 5: JFET self-bias [8 marks]

(a) Bias line: $V_{GS} = -I_D R_S$. Substitute it into Shockley's equation, working in mA and kΩ:

$$I_D = 10\left(1 - \frac{-0.68 I_D}{-4}\right)^2 = 10(1 - 0.17 I_D)^2$$

Expanding gives $10 - 3.4 I_D + 0.289 I_D^2 = I_D$, so

$$0.289 I_D^2 - 4.4 I_D + 10 = 0 \quad \Rightarrow \quad I_D = \frac{4.4 \pm \sqrt{4.4^2 - 4(0.289)(10)}}{2(0.289)} = 12.44 \text{ mA or } 2.781 \text{ mA}$$

- Root 1: $I_D = 12.44$ mA gives $V_{GS} = -8.46$ V. This is beyond $V_P = -4$ V, where the JFET is off, and it is also more than I_{DSS}. **Reject it.**
- Root 2: $I_D = 2.781$ mA gives $V_{GS} = -1.89$ V, which lies between V_P and 0. **Accept it.**

$$\mathbf{I_{DQ} = 2.78\ mA}, \qquad \mathbf{V_{GSQ} = -1.89\ V}$$

(b)

$$V_S = I_D R_S = \mathbf{1.89\ V}, \qquad V_D = 20 - 2.781(2.2) = \mathbf{13.88\ V}, \qquad V_{DS} = \mathbf{11.99\ V}$$

Saturation check: $V_{DS} = 12.0\ \text{V} \geq V_{GS} - V_P = -1.89 + 4 = 2.11$ V. The JFET is pinched off (saturation), so Shockley's equation was valid.

(c)

$$g_{m0} = \frac{2 I_{DSS}}{|V_P|} = \frac{20}{4} = 5\ \text{mS}, \qquad g_m = 5\left(1 - \frac{-1.89}{-4}\right) = \mathbf{2.64\ mS}$$

$$r_d = \frac{1}{y_{os}} = \mathbf{50\ k\Omega}, \qquad Z_i = R_G = \mathbf{1\ M\Omega}, \qquad Z_o = r_d \parallel R_D = \mathbf{2.11\ k\Omega}$$

$$A_v = -g_m(r_d \parallel R_D) = -2.637\ \text{mS} \times 2.107\ \text{k}\Omega = \mathbf{-5.56}$$

(d) With R_S unbypassed, the signal is $v_i = v_{gs} + g_m v_{gs} R_S$ (because $i_d = g_m v_{gs}$ flows through R_S), and $v_o = -g_m v_{gs} R_D$. So

$$A_v = -\frac{g_m R_D}{1 + g_m R_S} = -\frac{2.637(2.2)}{1 + 2.637(0.68)} = \mathbf{-2.08}$$

(Keeping r_d, the full formula $-g_m R_D / [1 + g_m R_S + (R_D + R_S)/r_d]$ gives -2.03.)

Marks (8): (a) 1: the correct quadratic. 1: both roots. 1: the correct root, with the reason. (b) 1: V_S, V_D and V_{DS}. 1: the saturation check. (c) 1: g_m. 1: Z_i, Z_o and A_v. (d) 1: the derivation and -2.08.

COMMON MISTAKES:

- Getting a sign wrong inside the bracket. V_{GS}/V_P is a ratio of two negatives, so it is positive: $(1 - 0.17I_D)$, not $(1 + 0.17I_D)$.
- Taking the larger root because it "looks like a bigger current".
- Using g_{m0} instead of g_m at the Q-point.
- In (d), forgetting the $(1 + g_m R_S)$ factor. It is the FET version of $R_C/(r_e + R_E)$.

REREAD: E10 (self-bias, choosing the root), E11 (g_m, CS gain), E12 (unbypassed R_S).

Question 6: E-MOSFET voltage-divider bias [7 marks]

(a)

$$k = \frac{I_{D(on)}}{\left(V_{GS(on)} - V_{GS(Th)}\right)^2} = \frac{5\ \text{mA}}{(8-3)^2} = \mathbf{0.2\ mA/V^2}$$

(b)

$$V_G = \frac{6.8}{16.8} \times 24 = \mathbf{9.714\ V}, \qquad V_{GS} = V_G - I_D R_S = 9.714 - I_D\ (\text{mA, k}\Omega)$$

Substitute into $I_D = k(V_{GS} - 3)^2$:

$$I_D = 0.2(6.714 - I_D)^2 \quad \Rightarrow \quad 0.2I_D^2 - 3.686 I_D + 9.016 = 0$$

$I_D = 15.52\ \text{mA}$ (gives $V_{GS} = -5.81\ \text{V}$) or $I_D = 2.904\ \text{mA}$ (gives $V_{GS} = 6.81\ \text{V}$)

The first root has $V_{GS} < V_{GS(Th)}$, so the device would be off and no current could flow. **Reject it.**

$$\mathbf{I_{DQ} = 2.90\ mA}, \qquad \mathbf{V_{GSQ} = 6.81\ V}$$

(c)

$$V_{DS} = V_{DD} - I_D(R_D + R_S) = 24 - 2.904(3) = \mathbf{15.29\ V}$$

$$V_{DS(sat)} = V_{GS} - V_{GS(Th)} = 3.81\ \text{V} < 15.29\ \text{V}$$

The MOSFET is in saturation, so the square law was valid.

(d)

$$g_m = 2k(V_{GSQ} - V_{GS(Th)}) = 2(0.2)(3.81) = \mathbf{1.52\ mS}, \qquad A_v \approx -g_m R_D = \mathbf{-3.05}$$

Marks (7): (a) 1. (b) 1: V_G and the bias line. 1: the quadratic and both roots. 1: the right root, with the reason. (c) 1: V_{DS}. 1: the saturation comparison. (d) 1: g_m and A_v.

COMMON MISTAKES:

- Using Shockley's equation (with I_{DSS} and V_P) for an E-MOSFET.
- Taking $k = I_{D(on)}/V_{GS(on)}^2$, which forgets to subtract the threshold.
- Picking the root without checking it against $V_{GS(Th)}$.
- Leaving out the saturation check. "Confirm" parts carry a mark for the stated comparison.

REREAD: E9 (E-MOSFET characteristic), E10 (E-MOSFET voltage-divider bias), E11 ($g_m = 2k(V_{GS} - V_T)$).

Question 7: Two-stage cascade, JFET into BJT [8 marks]

(a)

$$g_{m0} = \frac{2(8)}{4} = 4\ \text{mS}, \qquad g_{m1} = 4\left(1 - \frac{-1.5}{-4}\right) = \mathbf{2.5\ mS} \qquad (I_{DQ} = 3.125\ \text{mA})$$

$$r_{e2} = \frac{26\ \text{mV}}{2.5\ \text{mA}} = \mathbf{10.4\ \Omega}, \qquad \beta r_{e2} = 1.56\ \text{k}\Omega$$

$$Z_{i2} = R_1 \parallel R_2 \parallel \beta r_{e2} = 33\ \text{k} \parallel 10\ \text{k} \parallel 1.56\ \text{k} = \mathbf{1.296\ k\Omega}$$

(b) Each stage is loaded by whatever follows it.

$$A_{v1} = -g_{m1}(R_{D1} \parallel Z_{i2}) = -2.5\ \text{mS} \times (3.3\ \text{k} \parallel 1.296\ \text{k}) = -2.5 \times 0.9306 = \mathbf{-2.33}$$

$$A_{v2} = -\frac{R_{C2} \parallel R_L}{r_{e2}} = -\frac{2.2\ \text{k} \parallel 4.7\ \text{k}}{10.4} = -\frac{1498.6}{10.4} = \mathbf{-144.1}$$

$$A_{vT} = A_{v1}A_{v2} = (-2.327)(-144.1) = \mathbf{+335}$$

(c) $Z_i = R_G = \mathbf{1\ M\Omega}$ and $Z_o = R_{C2} = \mathbf{2.2\ k\Omega}$.

$$A_{v_s} = \frac{R_G}{R_G + R_{sig}} A_{vT} = \frac{1000}{1010}(335.3) = \mathbf{332}$$

$$v_o = 332 \times 2\ \text{mV} = \mathbf{0.664\ V\ peak},\ \text{in phase with } v_s\ (\text{two inversions})$$

(DC sanity check, not required: stage 1 has $V_{DS} = 15 - 3.125(3.3 + 0.48) = 3.19$ V, which is at least $V_{GS} - V_P = 2.5$ V, so it is pinched off.)

(d) The BJT stage's input impedance (1.3 kΩ) is much lower than R_{D1} (3.3 kΩ), so it shunts the JFET's drain load and the gain collapses from -8.25 to -2.33. Fix: put a buffer (an emitter follower or source follower) between the stages, or redesign stage 2 for a higher Z_i, for example with a partly unbypassed R_E.

Marks (8): (a) 1: g_{m1}. 1: r_{e2} and Z_{i2}. (b) 1: A_{v1} loaded by Z_{i2}. 1: A_{v2} loaded by R_L. 1: A_{vT} positive. (c) 1: Z_i and Z_o. 1: A_{v_s}, v_o and the phase. (d) 1.

COMMON MISTAKES:

- Multiplying the **no-load** gains: $(-8.25)(-211.5) = 1745$, which is five times too big. This is the most common cascade error.
- Loading stage 1 with R_L, or loading stage 2 with Z_{i1}.
- Forgetting that with $R_G = 1$ MΩ the source resistance hardly matters. The factor is 0.99.
- Giving a negative overall gain. Two inverting stages make it non-inverting.

REREAD: E8 (cascades and loaded-gain method), E11 (CS gain), E12 (multistage FET).

Question 8: CMRR and the differential amplifier [7 marks]

(a) (WS10 Q2 method.)

$$V_d = V_{i1} - V_{i2} = 0.1\ \text{mV}, \qquad V_c = \tfrac{1}{2}(V_{i1} + V_{i2}) = 1.0\ \text{mV}$$

$$V_o = A_d V_d \left(1 + \frac{1}{\text{CMRR}}\frac{V_c}{V_d}\right) = 4000(0.1\ \text{mV})\left(1 + \frac{10}{\text{CMRR}}\right) = 0.4\ \text{V}\left(1 + \frac{10}{\text{CMRR}}\right)$$

- i. 80 dB means CMRR $= 10^{80/20} = 10^4$: $V_o = 0.4(1.001) = \mathbf{400.4\ mV}$.
- ii. 40 dB means CMRR $= 100$: $V_o = 0.4(1.1) = \mathbf{440\ mV}$.

With a poor CMRR, the common-mode error is 10% of the output.

(b) Base at 0 V, so the emitters are at -0.7 V:

$$I_{tail} = \frac{-0.7 - (-12)}{15\ \text{k}} = 0.7533\ \text{mA}, \qquad I_E = \frac{I_{tail}}{2} = \mathbf{0.377\ mA}\ \text{each}$$

$$r_e = \frac{26}{0.3767} = \mathbf{69.0\ \Omega}, \qquad V_C = 12 - \alpha I_E R_C = 12 - 0.3729(10) = \mathbf{8.27\ V}$$

$$A_d = \frac{R_C}{2r_e} = \frac{10\ \text{k}}{138.1} = \mathbf{72.4}$$

$$A_c = \frac{\beta R_C}{r_i + 2(\beta+1)R_E} = \frac{100(10\ \text{k})}{6.90\ \text{k} + 202(15\ \text{k})} = \mathbf{0.329} \quad (\approx R_C/2R_E = 0.333)$$

$$\text{CMRR} = \frac{72.4}{0.329} = \mathbf{220} = 20\log_{10} 220 = \mathbf{46.8\ dB}$$

Marks (7): (a) 1: V_d and V_c. 1: the formula, or the $A_c = A_d/\text{CMRR}$ route. 1: both outputs. (b) 1: I_E, r_e and V_C. 1: A_d. 1: A_c. 1: CMRR as a ratio and in dB.

COMMON MISTAKES:

- Using the dB figure directly as the ratio (CMRR = 80). Convert it first: CMRR $= 10^{\mathrm{dB}/20}$.
- Using $V_c = V_{i1} + V_{i2}$ without the factor of one half.
- Giving each transistor the full tail current. It splits in two.
- Using $10\log_{10}$ for a voltage-gain ratio.

REREAD: E13 (differential amplifiers and CMRR; WS10 Q1–Q2).

Question 9: Op-amps [5 marks]

(a)

$$V_{o1} = \left(1 + \frac{R_f}{R_1}\right) V_1 = \left(1 + \frac{47}{4.7}\right)(20\ \mathrm{mV}) = 11 \times 20 = \mathbf{220\ mV\ rms}\ (\text{in phase})$$

(b)

$$V_o = -\left(\frac{100}{33} V_{o1} + \frac{100}{10} V_2\right) = -(3.030 \times 220 + 10 \times 50)\ \mathrm{mV} = -(666.7 + 500) = \mathbf{-1.167\ V\ rms}$$

That is 1.17 V rms, 180° out of phase with the inputs. Its peak is 1.65 V, well inside the ±15 V rails.

(c) $10\ \mathrm{k\Omega}$. Op-amp 2's inverting input is a virtual ground, so V_2 sees only its own series resistor.

Marks (5): (a) 1: the formula. 1: 220 mV. (b) 1: the summing formula with each signal's own gain. 1: -1.17 V. (c) 1.

COMMON MISTAKES:

- Using the inverting gain $R_f/R_1 = 10$ for op-amp 1. It is non-inverting, so the gain is 11.
- Dropping the minus sign, or adding the gains before multiplying.

REREAD: E13 (op-amp configurations; WS10 Q3–Q5).

Question 10: Power amplifiers [7 marks]

(a)

$$I_{BQ} = \frac{18 - 0.7}{1.2\ \text{k}} = 14.42\ \text{mA}, \qquad I_{CQ} = \beta I_{BQ} = \mathbf{576.7\ mA}$$

$$V_{CEQ} = 18 - 0.5767(20) = \mathbf{6.47\ V}$$

AC: $I_{c(p)} = \beta I_{b(p)} = 40(8\ \text{mA}) = 320$ mA peak, so $V_{c(p)} = 6.4$ V. This is just inside $V_{CEQ} = 6.47$ V, but it takes V_{CE} down to 0.07 V, slightly below $V_{CE,sat} = 0.2$ V, so the negative collector-voltage peaks just touch saturation (the undistorted limit is $(V_{CC} - 0.2)/R_C - I_{CQ} = 313$ mA). The clipping is about 2%, so use the ideal sinusoid formula and say it is marginal.

$$P_{o(ac)} = \frac{I_{c(p)}^2 R_C}{2} = \frac{0.32^2(20)}{2} = \mathbf{1.02\ W}$$

$$P_{i(dc)} = V_{CC} I_{CQ} = 18(0.5767) = \mathbf{10.38\ W}, \qquad \eta = \frac{1.024}{10.38} = \mathbf{9.9\%}$$

(This is well below the 25% ceiling for a series-fed class A stage. The Q-point is not centred, so the swing is limited by V_{CEQ}.)

(b) (WS10 Q10 method.)

$$P_{i(max)} = \frac{2V_{CC}^2}{\pi R_L} = \frac{2(900)}{16\pi} = \mathbf{35.8\ W}$$

$$P_{o(max)} = \frac{V_{CC}^2}{2R_L} = \frac{900}{32} = \mathbf{28.1\ W}$$

$$\eta_{max} = \frac{28.125}{35.81} = \frac{\pi}{4} = \mathbf{78.5\%}$$

(Not asked: the two transistors together dissipate at most $2V_{CC}^2/(\pi^2 R_L) = 11.4$ W. That maximum occurs at $V_{L(p)} = 0.636 V_{CC}$, not at full output.)

Marks (7): (a) 1: I_{CQ} and V_{CEQ}. 1: P_o, using peak$^2/2$ or rms^2 consistently. 1: P_i. 1: η. (b) 1 each for (i), (ii) and (iii).

COMMON MISTAKES:

- Using the peak current as if it were rms in $P_o = I^2 R$. That doubles the power.
- Taking P_i as P_o plus something guessed. Class A input power is simply $V_{CC} I_{CQ}$.

- In class B, using V_{CC}^2/R_L for P_o. The factor of 2 for a sine wave is missing.

REREAD: E13 (power amplifiers; WS10 Q6–Q10).

Score interpretation

Write your marks in this table first:

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Total
/10	/8	/13	/9	/8	/7	/8	/7	/5	/7	/82

Also add up your score in two blocks: **diodes and BJT** (Q1 I–V, Q2, Q3, Q4: 35 marks) and **FET, multistage, op-amps and power** (Q1 VI–X, Q5–Q10: 47 marks). The weaker block is where your last two days go.

Score	What it means	What to do in the two days left
66–82 (80%+)	Exam-ready. The methods are automatic.	Do EX2 (mock B) timed tomorrow. Spend the rest of the time on any question where you lost 2 or more marks. Read the traps lists in E8 and E12 once more. Finish your reference sheet (REFE).
49–65 (60–79%)	Pass with a margin. Some method gaps or slips.	For each question below 60%, reread its listed pack's method section (section 2) and redo the question with the guide closed. Then sit EX2. Check that your reference sheet has every formula you looked up today.
33–48 (40–59%)	Borderline. You have the ideas but not the full playbooks.	Focus on the canonical, high-mark types: Q3 (E5–E7), Q5 (E10–E11) and Q7 (E8). Rework the worked examples in those packs. Then do only Q3, Q5 and Q7 of EX2, timed. Make sure the formula table above is on your printed reference.
Below 33 (under 40%)	Not yet safe.	Do not try to cover everything. Learn four playbooks well: VDB DC to AC (E5, E7), JFET self-bias with the root choice (E10), the cascade loaded-gain rule (E8), and class B power (E13). Each one is worth 7–13 marks. Redo this paper's Q3, Q5, Q7 and Q10 the next day without notes on the method.

Patterns worth checking whatever your total:

- **Lost marks on Q1, VI–X:** the conceptual FET, op-amp and power material is under-revised. Reread E9 and E13 section 1.
- **Right method, wrong numbers on Q3–Q7:** usually units. Work in mA and kΩ throughout. Then $V = mA \times k\Omega$, g_m is in mS, and $g_m \times k\Omega$ is dimensionless.
- **Ran out of time:** you spent too long on Q3. It is the canonical question, so practise it until it takes 15 minutes.
- **Picked the wrong root in Q5 or Q6:** write the check line every time: "reject: $|V_{GS}| > |V_P|$" or "reject: $V_{GS} < V_{GS(Th)}$".

EDC mock B

Answers and worked solutions · Day 28 · Fri 20 Nov · pack EX2

Marking guide

Mark strictly. Give a mark only when your script shows the line that earns it. Fill in the table at the end, then use the “reread” column to plan the morning.

Question 1 (8 marks)

Part	Answer	Why
I	**b**	$p = n_i^2/N_D = (1.5\times10^{10})^2/10^{16} = 2.25\times10^4$ cm^{-3}
II	**c**	Bridge: PIV $= V_{s,\text{peak}} - V_D \approx V_m$. Centre tap: PIV $= 2V_s - V_D \approx 2V_m$
III	**b**	Saturation: both junctions forward biased, the collector cannot take βI_B, so the forced β is smaller
IV	**c**	$I_D = 6(1-1/(-3))^2 = 6(4/3)^2 = 10.7$ mA > I_{DSS}. The gate is insulated, so no gate current flows
V	**b**	$NM_L = V_{IL} - V_{OL} = 2.1 - 0.2 = 1.9$ V; $NM_H = V_{OH} - V_{IH} = 4.8 - 2.9 = 1.9$ V
VI	**c**	$20\log_{10}(4000/0.4) = 20\log_{10}10^4 = 80$ dB
VII	**b**	Each half needs $V_{BE} \approx 0.7$ V before it conducts, so the output is flat near zero crossing
VIII	**b**	$1/(1+2/50) = 0.962$

> **COMMON MISTAKES:** in V, swapping V_{IL} and V_{IH} (option c is $V_{IH} - V_{IL}$, the width of the transition region). In II, quoting the centre-tap PIV for the bridge.

REREAD: I E1; II E3; III E5; IV E9/E10; V E9; VI and VII E13; VIII E8.

Question 2 (7 marks)

(a)(i) p-type with $N_A \gg n_i$: $p \approx N_A = 5 \times 10^{15}$ cm^{-3}, and

$$n = \frac{n_i^2}{p} = \frac{(1.5 \times 10^{10})^2}{5 \times 10^{15}} = \mathbf{4.5 \times 10^{4}\ cm^{-3}} \quad [1]$$

(a)(ii) The electron term is negligible:

$\sigma = q(n\mu_n + p\mu_p) \approx 1.6 \times 10^{-19} \times 5 \times 10^{15} \times 500 = \mathbf{0.400\ (\Omega\,cm)^{-1}}$, $\quad \rho = 2.50\ \Omega\,\text{cm} \quad [1]$

Work in cm: $L = 0.2$ cm, $A = 0.01 \times 0.01 = 10^{-4}$ cm².

$$R = \frac{\rho L}{A} = \frac{2.5 \times 0.2}{10^{-4}} = \mathbf{5.00\ k\Omega} \quad [1]$$

(a)(iii) $T = 200 + 273.15 = 473.15$ K. Use k in eV/K in the exponent:

$$n_i = 7.3 \times 10^{15}(473.15)^{3/2} e^{-1.12/(2\times 8.62\times 10^{-5}\times 473.15)} = \mathbf{8.18 \times 10^{13}\ cm^{-3}}$$

This is 1.6 % of N_A, so the bar is **still extrinsic**: $p \approx 5.00 \times 10^{15}$ cm^{-3} and $n = n_i^2/p \approx 1.34 \times 10^{12}$ cm^{-3}. [1]

(Check: the same formula at 300 K gives 1.51×10^{10} cm^{-3}, which matches the given room-temperature value.)

(b)(i) At 27 °C, $V_T = \dfrac{1.38 \times 10^{-23} \times 300}{1.6 \times 10^{-19}} = 25.875$ mV.

$$I_D = \frac{5 - 0.7}{4.3\ \text{k}} = \mathbf{1.000\ mA}, \qquad r_d = \frac{nV_T}{I_D} = \mathbf{25.9\ \Omega} \quad [1]$$

For the ac signal, the 5 V supply is a short and the diode is r_d. That gives a divider:

$$v_d = 20\ \text{mV} \times \frac{25.9}{4300 + 25.9} = \mathbf{0.120\ mV\ peak} \quad [1]$$

(b)(ii) $\Delta T = 50$ °C, so $V_D = 0.7 - 0.0025 \times 50 = 0.575$ V.

$I_D = (5 - 0.575)/4.3\ \text{k} = 1.029$ mA. $V_T = 1.38 \times 10^{-23} \times 350/1.6 \times 10^{-19} = 30.19$ mV. So $r_d = 30.19/1.029 = \mathbf{29.3\ \Omega}$ and $v_d = 20 \times 29.3/4329 = \mathbf{0.136\ mV}$ peak.

r_d rises because $V_T \propto T$ rises by 17 %, while I_D rises by only 3 % (the 4.3 kΩ resistor nearly fixes the current). [1]

COMMON MISTAKES: using $k = 1.38 \times 10^{-23}$ with E_g in eV in the exponent; forgetting °C → K; using L in mm and A in cm²; leaving out the 4.3 kΩ divider for v_d (the answer is not 20 mV).

REREAD: E1 (carrier concentrations, $n_i(T)$), E2 (small-signal r_d, temperature rules).

Question 3 (5 marks)

(a) Find each diode's turn-on condition from the node voltage.

- D1 (anode at v_o, cathode at +4 V) conducts when $v_o \geq 4 + 0.7 =$ **+4.7 V**. Then v_o is held at +4.7 V. [1]
- D2 (cathode at v_o, anode at −6 V) conducts when $v_o \leq -6 - 0.7 =$ **−6.7 V**. Then v_o is held at −6.7 V. [1]
- In between, both diodes are off, no current flows in the 1 kΩ resistor, and $v_o = v_i$.

Transition angles from $12 \sin \omega t$:

- Positive clipping: $\sin^{-1}(4.7/12) = 23.1^\circ$. Clipped from **23.1° to 156.9°**.
- Negative clipping: $\sin^{-1}(6.7/12) = 33.9^\circ$. Clipped from $180^\circ + 33.9^\circ =$ **213.9° to 326.1°**.

SKETCH DESCRIPTION: the output follows the sine from 0 V up to +4.7 V at 23.1°. It stays flat at +4.7 V until 156.9°, then follows the sine down through 0 V at 180° to −6.7 V at 213.9°. It stays flat at −6.7 V until 326.1°, then follows the sine back up to 0 V at 360°. The output is an asymmetric flat-topped sine. Its top is at +4.7 V, its bottom is at −6.7 V, and it is 11.4 V peak-to-peak. [1]

(b) Start in the half-cycle where the diode conducts. In the +8 V half, the output tries to rise, so the diode turns on and holds $v_o = 2 + 0.7 =$ **+2.7 V**. The capacitor charges to $8 - 2.7 = 5.3$ V (input side positive).

In the −8 V half, the diode turns off, and the capacitor's 5.3 V adds to the input: $v_o = -8 - 5.3 =$ **−13.3 V**. The diode is reverse biased (−13.3 V < 2.7 V), which confirms it is off. [1]

SKETCH DESCRIPTION: a square wave of the same shape and frequency (1 kHz). It sits at +2.7 V for the first half-period and at −13.3 V for the second. It is **16 V peak-to-peak**, the same as the input, and its average is −5.3 V. The whole input waveform has been shifted down by 5.3 V.

$RC = 100\ \text{k} \times 1\ \mu\text{F} = 100$ ms, which is 200 times the 0.5 ms half-period. The capacitor barely discharges, so the output is flat (tilt under 0.5 %). [1]

> **COMMON MISTAKES:** placing the clip at the battery voltage without the 0.7 V; getting the sign of the D2 level wrong (read which battery terminal the diode touches); for the clamper, starting the analysis in the half-cycle where the diode is off; giving an output peak-to-peak value different from the input's.

REREAD: E3 (clippers and clampers, playbooks E and F).

Question 4 (5 marks)

1. Secondary peak $V_{s,p} = 12\sqrt{2} = 16.97$ V. Two diodes conduct in series in a bridge: $V_P = 16.97 - 1.4 =$ **15.57 V**. [1]
2. Full-wave, so use $2f$: $C = \dfrac{V_P}{2fRV_r} = \dfrac{15.57}{2 \times 50 \times 150 \times 0.8} =$ **1297 µF**. [1]
3. With 1500 µF: $V_r = \dfrac{15.57}{2 \times 50 \times 150 \times 1500 \times 10^{-6}} =$ **0.692 V** p-p. $V_{dc} = 15.57 - 0.346 =$ **15.22 V**. [1]
4. $I_L \approx V_P/R = 103.8$ mA, and $\sqrt{V_P/(2V_r)} = \sqrt{15.57/1.384} = 3.354$. So $i_{Dmax} = 103.8(1 + 2\pi \times 3.354) =$ **2.29 A**. (The average during conduction is $i_{Dav} = 103.8(1 + \pi \times 3.354) = 1.20$ A.) The conduction angle is $\omega\Delta t = \sqrt{2 \times 0.692/15.57} = 0.298$ rad = 17.1°. [1]
5. In a bridge, the reverse-biased diode sees the secondary peak minus one conducting drop: PIV $= 16.97 - 0.7 =$ **16.3 V**. [1]

If you used $I_L = V_{dc}/R = 101.5$ mA in part 4, you get $i_{Dmax} = 2.24$ A. Give the mark if the method is stated.

> **COMMON MISTAKES:** using f instead of $2f$ (that doubles C); subtracting only one 0.7 V for a bridge; using the half-wave i_{Dmax} form; quoting the centre-tap PIV ($2V_m$).

REREAD: E3 (filter design, playbook D).

Question 5 (8 marks)

(a) Thevenin at the base:

$$V_{Th} = 12 \times \frac{47}{15+47} = 9.097\ \text{V}, \qquad R_{Th} = 15\ \text{k} \parallel 47\ \text{k} = 11.37\ \text{k}\Omega \quad [1]$$

pnp loop from the +12 V rail, through R_E, across E–B, through R_{Th} to V_{Th}. Base current flows **out** of the base:

$$12 - (\beta+1)I_B R_E - 0.7 - I_B R_{Th} = V_{Th}$$

$$I_B = \frac{12 - 0.7 - 9.097}{11.37\ \text{k} + 101 \times 1.5\ \text{k}} = \frac{2.203}{162.9\ \text{k}} = 13.53\ \mu\text{A} \quad [1]$$

$I_C = 100 I_B =$ **1.353 mA**, $I_E = 1.366$ mA. [1]

$V_E = 12 - 1.366 \times 1.5 = 9.95$ V. $V_C = I_C R_C = 1.353 \times 2.7 =$ **3.65 V**.
$V_{EC} = 9.95 - 3.65 =$ **6.30 V**. Since 6.30 V > 0.2 V, the transistor is **active**. [1]

(b)

β	I_C (mA)	Change from β = 100
50	1.254	−7.3 %
100	1.353	—
200	1.408	+4.1 %

A fourfold spread in β moves I_C by only about 11 % in total, so the bias is **stable**. The reason is that $(\beta+1)R_E = 151.5$ kΩ is about 13 times R_{Th} at β = 100. Even at β = 50 it is still about 6.7 times R_{Th}, so $I_C \approx (V_{CC} - V_{Th} - 0.7)/R_E$ almost independently of β. [2]

(c) Assume active: $V_C = 1.353 \times 8.2 = 11.09$ V, which is **above** $V_E = 9.95$ V. That would make $V_{EC} = -1.14$ V, which is impossible, so the transistor is **saturated**. [1]

Saturated model: $V_{EC} = 0.2$ V and $I_C \neq \beta I_B$. Two unknowns, I_B and I_C:

- Base loop: $12 - (I_B + I_C)(1.5\ \text{k}) - 0.7 = 9.097 + I_B(11.37\ \text{k})$
- Collector loop: $12 - (I_B + I_C)(1.5\ \text{k}) - 0.2 = I_C(8.2\ \text{k})$

Solving: $I_B = 29.9\ \mu$A and $I_C =$ **1.21 mA**. Check: $I_C/I_B = 40.5 < \beta = 100$, so it is saturated. $V_E = 10.14$ V and $V_C = 9.94$ V. [1]

(The largest R_C for active operation at β = 100 is $(9.95 - 0.2)/1.353\ \text{mA} = 7.21$ kΩ.)

COMMON MISTAKES: npn sign habits on a pnp (base current direction, V_{EB} not V_{BE}); using V_{CE} and getting a negative "active" answer without spotting it; in (c), keeping $I_C = \beta I_B$ after deciding the transistor is saturated.

REREAD: E4 (Thevenin two-loop method), E5 (pnp, β sensitivity, guess–analyse–check).

Question 6 (8 marks)

(a) DC:

$$I_B = \frac{V_{CC} - V_{BE}}{R_F + \beta R_C} = \frac{10 - 0.7}{180\ \text{k} + 150 \times 2.2\ \text{k}} = 18.24\ \mu\text{A}, \quad I_E = 151 I_B = \mathbf{2.754\ mA} \quad [1]$$

$r_e = 26\ \text{mV}/2.754\ \text{mA} = \mathbf{9.44\ \Omega}$. ($V_{CE} = 10 - 2.754 \times 2.2 = 3.94$ V, so it is active.) [1]

AC:

$$Z_o \approx R_C \parallel R_F = 2.2\ \text{k} \parallel 180\ \text{k} = \mathbf{2.17\ k\Omega}$$

$$A_v = -\frac{R_C \parallel R_F}{r_e} = -\frac{2173}{9.44} = \mathbf{-230} \quad [1]$$

$$Z_i = \frac{r_e}{\frac{1}{\beta} + \frac{R_C}{R_C + R_F}} = \frac{9.44}{0.00667 + 0.01207} = \mathbf{504\ \Omega} \quad [1]$$

Miller check: $\beta r_e \parallel R_F/(1 - A_v) = 1416 \parallel 779 = 502\ \Omega$. An exact nodal solution of the r_e model gives $A_v = -230.2$ and $Z_i = 502\ \Omega$.

(b) DC: the base is at 0 V, so $V_E = -0.7$ V and

$$I_E = \frac{-0.7 - (-4)}{1.2\ \text{k}} = \mathbf{2.75\ mA}, \qquad r_e = \frac{26\ \text{mV}}{2.75\ \text{mA}} = \mathbf{9.45\ \Omega} \quad [1]$$

($V_C = 12 - 0.99 \times 2.75 \times 3.3 = 3.02$ V, so $V_{CB} > 0$ and it is active.)

$Z_i = R_E \parallel r_e = 1200 \parallel 9.45 = \mathbf{9.38\ \Omega}$. [1]

$$A_{v_L} = \frac{\alpha(R_C \parallel R_L)}{r_e} = \frac{0.99 \times (3.3\ \text{k} \parallel 6.8\ \text{k})}{9.45} = \mathbf{232.6} \quad [1]$$

$$A_{v_s} = \frac{Z_i}{Z_i + R_s} A_{v_L} = \frac{9.38}{59.38} \times 232.6 = \mathbf{36.8}$$

v_o is **in phase** with v_s (CB does not invert). The 50 Ω source throws away 84 % of the signal, because Z_i is only 9.4 Ω. [1]

> **COMMON MISTAKES:** in (a), using $R_F + \beta(R_C + R_E)$ with an R_E that does not exist, or using $A_v = -R_C/r_e$ without $\parallel R_F$ (this gives −233, which is close enough to score only if the formula is shown); in (b), putting a minus sign on the CB gain, or forgetting the source divider.

REREAD: E6 (r_e model, DC→AC), E7 (sections 1.5 and 1.6).

Question 7 (5 marks)

1. KCL at Q1's collector: I_{REF} supplies I_{C1} plus both base currents. With matched transistors, $I_{C1} = I_{C2} = I_o$, so $I_{REF} = I_o(1 + 2/\beta) = 2.00 \times 1.02 = 2.04$ mA. [1] Q1's collector is at $V_{B1} = -9 + 0.7 = -8.3$ V, so $R = \dfrac{0 - (-8.3)}{2.04\ \text{mA}} =$ **4.07 kΩ**. [1]
2. Q2 stays active while $V_{C2} \geq V_{E2} + 0.2 = -8.8$ V. Since $V_{C2} = 0 - I_o R_L$, the condition is $R_L \leq 8.8/2\ \text{mA} =$ **4.40 kΩ**. [2]
3. I_{REF} is set by R alone: still 2.04 mA. Then $I_o = 2.04/(1 + 2/50) =$ **1.96 mA**. Halving β changes I_o by only 1.9 %. [1]

> **COMMON MISTAKES:** using $1/\beta$ instead of $2/\beta$; measuring R's voltage from the −9 V rail instead of from Q1's collector (−8.3 V); forgetting that the compliance limit is set by $V_{CE,\text{sat}}$, not by V_{BE}.

REREAD: E8 (playbook F, current mirror).

Question 8 (8 marks)

(a) No gate current flows, so $V_G = 18 \times \dfrac{20}{120} = \mathbf{3.00\ V}$. The bias line is $V_{GS} = V_G - I_D R_S = 3 - 300 I_D$. [1]

Substitute into Shockley:

$$I_D = 8\ \text{mA}\left(1 - \frac{3 - 300 I_D}{-4}\right)^2 = 8\ \text{mA}\left(\frac{7 - 300 I_D}{4}\right)^2$$

This gives the quadratic $45\, I_D^2 - 3.1\, I_D + 0.0245 = 0$ (with I_D in A), with roots $I_D = 59.8$ mA and **9.11 mA**. [1]

Reject 59.8 mA: it gives $V_{GS} = 3 - 17.9 = -14.9$ V, which is beyond $V_P = -4$ V (cut off), so it is not a valid operating point. Keep

$$I_{DQ} = \mathbf{9.11\ mA}, \qquad V_{GSQ} = 3 - 0.300 \times 9.11 = \mathbf{+0.268\ V} \quad [1]$$

V_{GS} is positive and $I_D > I_{DSS}$: the device is in enhancement mode.

(b) $V_S = 9.11 \times 0.3 = 2.73$ V. $V_D = 18 - 9.11 \times 1.0 = 8.89$ V. $V_{DS} = 6.16$ V. Saturation requires $V_{DS} \geq V_{GS} - V_P = 0.268 + 4 = 4.27$ V, and 6.16 V meets this. [1]

(c) $g_{m0} = 2(8)/4 = \mathbf{4.00\ mS}$. $g_m = 4\left(1 - \dfrac{0.268}{-4}\right) = \mathbf{4.27\ mS}$. $g_m > g_{m0}$ because the Q-point is above I_{DSS} on the transfer curve, where the curve is steeper than at $V_{GS} = 0$. [1]

(d) $r_d = 1/10\ \mu\text{S} = 100$ kΩ.

- $Z_i = R_1 \parallel R_2 = 100\ \text{M} \parallel 20\ \text{M} = \mathbf{16.7\ M\Omega}$ [1]
- $Z_o = r_d \parallel R_D = 100\ \text{k} \parallel 1\ \text{k} = \mathbf{990\ \Omega}$ [1]
- $A_v = -g_m(r_d \parallel R_D) = -4.27\ \text{mS} \times 990 = \mathbf{-4.23}$; with $R_L = 3.3$ kΩ: $A_{v_L} = -4.27\ \text{mS} \times (100\ \text{k} \parallel 1\ \text{k} \parallel 3.3\ \text{k}) = \mathbf{-3.25}$ [1]

> **COMMON MISTAKES:** taking the larger root because it "looks like more current"; dropping the minus on V_P when it is substituted ($1 - V_{GS}/V_P$ with $V_P = -4$ becomes $1 + V_{GS}/4$); assuming V_{GS} must be negative and forcing a sign; including R_S in the AC gain although it is bypassed.

REREAD: E10 (D-MOSFET bias, quadratic method), E11 (g_m, CS amplifier).

Question 9 (12 marks)

(a)(i) Self-bias line: $V_{GS} = -I_D R_S = -2000 I_D$. Shockley:

$$I_D = 10\ \text{mA}\left(1 - \frac{2000 I_D}{4}\right)^2$$

The roots are 3.12 mA and **1.28 mA**. 3.12 mA gives $V_{GS} = -6.23$ V, which is beyond V_P, so reject it. [1]

$I_{DQ} =$ **1.283 mA** and $V_{GSQ} =$ **−2.57 V**. ($V_{DS} = 15 - 2.57 = 12.4$ V, so it is saturated.) [1]

(ii) $g_{m0} = 2(10)/4 = 5$ mS. $g_m = 5\left(1 - \dfrac{-2.567}{-4}\right) =$ **1.79 mS**. [1]

(iii) $Z_i = R_G =$ **1 MΩ**. $r_d = 50$ kΩ. $Z_o = r_d \parallel R_S \parallel \dfrac{1}{g_m} = 50\ \text{k} \parallel 2\ \text{k} \parallel 558 =$ **433 Ω**. [1]

(iv) No load: $r_d \parallel R_S = 1923\ \Omega$, so $A_v = \dfrac{1.791 \times 1.923}{1 + 1.791 \times 1.923} =$ **0.775**.

With $R_L = 1$ kΩ: $r_d \parallel R_S \parallel R_L = 657.9\ \Omega$, so $A_{v_L} = \dfrac{1.791 \times 0.6579}{1 + 1.791 \times 0.6579} =$ **0.541**. [1]

(v) $v_o = 0.541 \times 0.5 =$ **0.270 V peak**, in phase with v_i. $v_{gs} = v_i - v_o =$ **0.230 V peak**. [1]

(b)(i) No gate current flows, so R_F carries no DC current and $V_{GS} = V_{DS}$. Then

$$V_{GS} = 12 - 2\ \text{k} \times 0.5\ \text{mA/V}^2\,(V_{GS} - 2)^2$$

so $(V_{GS} - 2)^2 + V_{GS} - 12 = 0$. The roots are $V_{GS} = 4.70$ V and -1.70 V. Reject −1.70 V because it is below $V_{GS(Th)}$. [1]

$V_{GSQ} = V_{DSQ} =$ **4.70 V** and $I_{DQ} = 0.5(2.70)^2 =$ **3.65 mA**.

Saturation needs $V_{DS} \geq V_{GS} - V_{GS(Th)}$. Here $V_{DS} = V_{GS}$, which is always greater than $V_{GS} - 2$, so saturation is automatic. [1]

(ii) $g_m = 2k(V_{GSQ} - V_{GS(Th)}) = 2(0.5)(2.70) =$ **2.70 mS**. [1]

(iii) $r_d \parallel R_D = 50\ \text{k} \parallel 2\ \text{k} = 1923\ \Omega$.

$$Z_i = \frac{R_F + r_d \parallel R_D}{1 + g_m(r_d \parallel R_D)} = \frac{10\ \text{M} + 1.92\ \text{k}}{1 + 2.70\ \text{mS} \times 1923} = \mathbf{1.61\ M\Omega} \quad [1]$$

$$Z_o = R_F \parallel r_d \parallel R_D = \mathbf{1.92\ k\Omega} \quad [1]$$

$$A_v = -g_m(R_F \parallel r_d \parallel R_D) = -2.70\ \text{mS} \times 1923 = \mathbf{-5.19} \quad [1]$$

An exact nodal solution gives the same values. The approximation $Z_i \approx R_F/(1 + g_m R_D) = 1.56$ MΩ is 3 % low, because it ignores r_d. Give the mark if the

formula is shown.

> **COMMON MISTAKES:** in (a), using $Z_o = R_S$ (the follower's Z_o is dominated by $1/g_m$); putting R_L in parallel with R_S for the DC bias. In (b), writing $Z_i = R_F = 10$ MΩ (the Miller effect divides R_F by roughly $1 + |A_v|$); taking the negative root.

REREAD: E11 (source follower), E12 (drain-feedback configuration).

Question 10 (7 marks)

1. Self-bias: $V_{GS} = -1000 I_D$. The quadratic gives roots of 9.37 mA (which gives $V_{GS} = -9.37$ V, beyond V_P, so reject it) and

$I_{DQ} =$ **2.16 mA** and $V_{GSQ} =$ **−2.16 V**. [1]

$V_{DSQ} = 18 - 2.161(3.3 + 1.0) =$ **8.71 V**, so both stages are saturated. [1]

2. $g_{m0} = 2(8)/4.5 = 3.556$ mS. $g_m = 3.556\left(1 - \dfrac{2.161}{4.5}\right) =$ **1.85 mS**. [1]

3. Stage 1 is loaded by $Z_{i2} = R_G = 2.2$ MΩ:

$$A_{v1} = -g_m(r_d \parallel R_D \parallel R_G) = -1.848\ \text{mS} \times 3044\ \Omega = \mathbf{-5.63} \quad [1]$$

Stage 2 is loaded by R_L:

$$A_{v2} = -g_m(r_d \parallel R_D \parallel R_L) = -1.848\ \text{mS} \times 2336\ \Omega = \mathbf{-4.32} \quad [1]$$

4. $A_{vT} = (-5.63)(-4.32) =$ **+24.3**. The input divider is $2.2\ \text{M}/(2.2\ \text{M} + 10\ \text{k}) = 0.9955$, so $v_o/v_{sig} =$ **24.2** and $V_o = 24.2 \times 20\ \text{mV} =$ **0.484 V rms**, in phase with the source. [1]

$Z_i = R_G =$ **2.2 MΩ** and $Z_o = r_d \parallel R_D =$ **3.05 kΩ**. [1]

Without r_d, you would get $A_{v1} = -6.09$ and $A_{v2} = -4.59$. Since r_d is given, it must be used.

> **COMMON MISTAKES:** multiplying the unloaded gains (−5.63 × −5.63 = 31.7); forgetting that two inverting stages give a positive overall gain; leaving out r_d when it is given.

REREAD: E12 (cascade), E8 (loading method).

Question 11 (7 marks)

(a)(i) $R'_L = V_{CC}/I_{CQ} = 12/0.15 = 80\ \Omega$. The secondary load is $8 \parallel 8 = 4\ \Omega$. [1]

$$a = \sqrt{\frac{R'_L}{R_L}} = \sqrt{\frac{80}{4}} = \mathbf{4.47}, \quad N_1 : N_2 \approx \mathbf{4.47 : 1} \quad [1]$$

(a)(ii) $P_i = V_{CC} I_{CQ} = 1.80$ W. The maximum ideal $P_o = \frac{1}{2} V_{CC} I_{CQ} =$ **0.90 W**, so $\eta =$ **50 %**. [1]

(b)(i) $V_{L(p)}$ can reach V_{CC}:

$$P_{o(max)} = \frac{V_{CC}^2}{2R_L} = \frac{400}{16} = \mathbf{25.0\ W}, \quad P_i = \frac{2V_{CC}^2}{\pi R_L} = 31.8\ \text{W}, \quad \eta = \mathbf{78.5\%} \quad [2]$$

(At this point $P_{2Q} = 6.83$ W.)

(b)(ii) Dissipation is largest at $V_{L(p)} = 2V_{CC}/\pi =$ **12.7 V**, not at full output:

$$P_{2Q(max)} = \frac{2V_{CC}^2}{\pi^2 R_L} = \frac{800}{\pi^2 \times 8} = \mathbf{10.1\ W}\ \text{total}, \quad \mathbf{5.07\ W}\ \text{per transistor} \quad [1]$$

At that point $P_o = 10.1$ W and $P_i = 20.3$ W, so $\eta =$ **50 %**. (A numerical sweep of $P_{2Q}(V_{L(p)})$ confirms the maximum at 12.73 V.) [1]

> **COMMON MISTAKES:** using $a = R'_L/R_L$ instead of its square root; using a single speaker (8 Ω) instead of the parallel pair (4 Ω); reporting the dissipation at maximum output (6.83 W) as the maximum dissipation; giving the total dissipation when the question asks per transistor.

REREAD: E13 (power amplifiers).

After marking

Q	Topic	Your mark	Max	If under 60 %, reread
1	MCQ		8	Pack named in the table under Q1
2	Semiconductors, diode r_d and T		7	E1, E2
3	Clipper and clamper		5	E3
4	Bridge and C filter		5	E3
5	pnp bias, β, saturation		8	E4, E5
6	Collector feedback, CB		8	E6, E7
7	Current mirror		5	E8
8	D-MOSFET		8	E10, E11
9	Source follower, drain feedback		12	E11, E12
10	JFET cascade		7	E12, E8
11	Power amps		7	E13
	Total		**80**	

Diodes and BJT total (Q2–Q6 plus Q1 I–III) out of 36: ____. FETs, multistage, op-amp and power total (Q7–Q11 plus Q1 IV–VIII) out of 44: ____. Spend tonight's remaining time on the weaker half, using the packs above. Do not start new material.

Night-before checklist

Pack the bag

- ☐ Calculator (with its model written down for the cover sheet) and spare batteries
- ☐ Printed open-book reference sheet, with the formula tables from this mock and from REFE
- ☐ Lecture notes for Weeks 6–8, WS7–10, and your Test 1 reference (Weeks 1–5), all tabbed
- ☐ Pens, pencil, ruler, eraser (you will be sketching waveforms and load lines)

Ten reflexes to have ready (read once, then sleep)

1. **Units first.** Convert °C to K. Put semiconductor lengths in cm. Use k in eV/K inside $e^{-E_g/2kT}$.
2. **Diodes:** write the state you assume, find where it changes (v_i at the transition), then sketch v_o under v_i and label the levels and angles in words.
3. **Full-wave means** $2f$, and a bridge loses two diode drops while its PIV is only one secondary peak.
4. **BJT DC:** use Thevenin and the exact I_B formula. For a pnp, flip every arrow. Always check V_{CE} (or V_{EC}) against 0.2 V. If it fails, redo the analysis with the saturation model; do not keep βI_B.
5. **Small-signal:** do DC first, then r_e or g_m, then Z_i, Z_o, A_v, then the load, then the source divider. Watch the sign: CE, CS and drain feedback invert; CB and followers do not.
6. **FET quadratics:** keep the root with V_{GS} between V_P (or $V_{GS(Th)}$) and the right side of the curve. A D-MOSFET may have a positive V_{GS} and $I_D > I_{DSS}$, and that is fine.
7. $g_m = g_{m0}(1 - V_{GSQ}/V_P)$. Use $r_d = 1/y_{os}$ whenever it is given.
8. **Cascades:** load each stage with the next stage's Z_i, then multiply. Two inverting stages give a positive gain.
9. **Power:** $R'_L = a^2 R_L$. Class A transformer-coupled η_{max} is 50 %; class B is 78.5 %. Class B dissipation peaks at $V_{L(p)} = 2V_{CC}/\pi$.
10. **Digital and op-amp:** $NM_L = V_{IL} - V_{OL}$, $NM_H = V_{OH} - V_{IH}$, and CMRR(dB) $= 20\log(A_d/A_c)$.

In the exam room

- ☐ Read the whole paper in the first 5 minutes. Mark the questions you can bank quickly.
- ☐ Do the MCQs and the short calculations first, then the long FET and BJT questions.
- ☐ Write the formula, then the substitution, then the answer with units, for every part. Method marks carry you when a number slips.
- ☐ If a question needs a figure value you cannot read, state your assumption in one line and carry on.
- ☐ Leave 5 minutes at the end to check signs, units and root choices.

Sleep by a sensible hour. A rested brain is worth more than one more pack tonight.